Groups, Rings and Fields

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Contents

1 Groups

Definition 1.1 (Group). *A group is a set S together with a binary operation $\circ$ such that the following properties hold:*

1. *Identity: There exists an element $e \in S$, satisfying*

$$e \circ a = a \circ e = a$$

for every $a \in S$.

2. *Inverses: For every $a \in S$ there exists $b \in S$ such that*

$$a \circ b = b \circ a = e, \text{ the identity element}$$

We usually denote this element as a^{-1} or $-a$, depending on context.

3. *Associativity: For any $a, b, c \in S$, we require*

$$a \circ (b \circ c) = (a \circ b) \circ c$$

A group is then the tupple $(S, \circ)$. We will often just write the set to refer to the group, e.g. refering to the group $(\mathbb{Z}, +)$ as just $\mathbb{Z}$.

1.1 Basic Examples

Example 1.1 (Integers under addition). *The set of integers $\mathbb{Z}$ with the operation $+$ forms a group:*

- *Identity: 0 since $0 + n = n + 0 = n$ for all $n \in \mathbb{Z}$.*
- *Inverses: For $n \in \mathbb{Z}$, the inverse is $-n$.*
- *Associativity: Addition is associative.*

Hence $(\mathbb{Z}, +)$ is a group.

Example 1.2 (Non-example: Natural numbers under addition). *The set $\mathbb{N}$ under $+$ is not a group since there is no inverse for $n > 0$.*

1.2 Basic Properties

Theorem 1.1 (Uniqueness of identity). *The identity element in a group is unique.*

Proof. Suppose e and e' are both identities. Then

$$e = e \circ e' = e',$$

so the identity is unique. □

Theorem 1.2 (Uniqueness of inverses). *Each element in a group has a unique inverse.*

Proof. Suppose b and c are inverses of a. Then

$$b = b \circ e = b \circ (a \circ c) = (b \circ a) \circ c = e \circ c = c.$$

□

1.3 Example Problem

Problem 1.1. *Determine whether the set*

$$G = \{1, -1, i, -i\} \subset \mathbb{C}$$

with multiplication is a group.

Solution. We check the group axioms:

1. **Closure:** Multiplying any two elements of G yields another element in G. True.
2. **Identity:** The element 1 acts as identity. True.
3. **Inverses:** Each element has an inverse in G: $1^{-1} = 1$, $(-1)^{-1} = -1$, $i^{-1} = -i$, $(-i)^{-1} = i$. True.
4. **Associativity:** Multiplication of complex numbers is associative. True.

Hence $(G, \cdot)$ is a group, in fact it is an abelian group which we shall describe below in definition 1.2. □

1.4 Abelian Groups

Definition 1.2 (Abelian Group). *A group* $(G, \circ)$ *is* abelian *(or commutative) if*

$$a \circ b = b \circ a \quad \forall a, b \in G.$$

Example 1.3. *The group* $(\mathbb{Z}, +)$ *is abelian because* $m + n = n + m$*.*

2 Important Groups

First let's define some often used notation.

For $g \in G$, define:

- $g^n = gg\dots g (n > 0)$
- $g^0 = e$
- $g^{-n} = (g^{-1})^n$

and recognize the following identities (which are provable by induction):

$$g^m g^n = g^{m+n}, \quad (g^m)^n = g^{mn}$$

Definition 2.1 (Roots of Unity).

$$U_n = \{e^{2\pi ik/n} : 0 \le k < n\}$$

are called the n-th roots of unity.

Definition 2.2. *An subset $\{g_1, \dots, g_k\} \subseteq G$ is a generating set if every $g \in G$ can be expressed as*

$$g_1^{n_1} \cdots g_k^{n_k}$$

If a singleton g' alone generates a group then it is called a generator of G and G is said to be cyclic.

The n-th roots of unity are finite, with exactly n elements. They are cyclic, with generator $e^{2i\pi/n}$.

Definition 2.3 (Symmetric Group). *The symmetric group S_n is the set of all permutations of n elements under composition. In other words it is the set of all $\sigma : [n] \to [n]$ with composition as its operation.*

The symmetric group is non-abelian for $n \ge 3$. The identity corresponds to the identity permutation (doing nothing) and inverses are undoing a permutation. We have the following notation for permutations in the symmetric Group (assume $n = 3$ for this example):

$$\begin{pmatrix} 1 & 2 & 3 \\ 2 & 3 & 1 \end{pmatrix}$$

corresponds to $1 \mapsto 2, 2 \mapsto 3, 3 \mapsto 1$.

Another way to write the same permutation is:

$$(1\,2\,3)$$

3 Subgroups

Definition 3.1 (Subgroup). *Let G be a group under $\star$. A subgroup $H \leq G$ is a subset $H \subseteq G$ that is itself a group under $\star$.*

Having to mechanically check if $H \subseteq G$ satisfies all the required axioms can be a bit tedious so next we introduce a powerful theorem which let's us easily determine whether some subset is a subgroup or not.

Theorem 3.1 (Subgroup Test). *A nonempty subset $H \subseteq G$ is a subgroup if and only if*

$$\forall x, y \in H,\ xy^{-1} \in H.$$

Proof Idea. If H is a subgroup the required conditions follows.

Conversely:

- Nonempty so some $h \in H$.
- $hh^{-1} = e \in H$.
- Closure under xy^{-1} will give inverses and closure.

□

We get some nice corollaries from this, as well as a nice and tidy way to prove whether or not a subset is a subgroup.

Corollary 3.1. *The intersection of any collection of subgroups is a subgroup.*
Given any subset $S \subseteq G$, there is a smallest subgroup containing it.

Definition 3.2 (Cyclic Subgroups). *Let G be a group.*
For any $g \in G$, define:

$$\langle g \rangle = \{g^n : n \in \mathbb{Z}\}.$$

$\langle g \rangle$ is called the cyclic subgroup generated by g.

It is very straightforward to show that the cyclic subgroup generated by some $g \in G$ is indeed a subgroup with the use of theorem 3.1.

On page 7 we say that a group is cyclic if it has a signle element which generates it, but now we can simply say that a group G is cyclic if $G = \langle g \rangle$ for some $g \in G$.

Theorem 3.2. *Suppose a group G is cyclic, i.e. $G = \langle g \rangle$ for some $g \in G$.*
Then it is isomorphic to one of the following:

- $(\mathbb{Z}, +)$, *or*
- $(\mathbb{Z}/n\mathbb{Z}, +)$ *for some* $n \geq 1$.

3.1 Structure of Cyclic Subgroups

Definition 3.3 (Order). *The order of an element $g \in G$ is*

$$\text{ord}(g) = |\langle g \rangle|$$

Furthermore, if $\text{ord}(g) = n < \infty$, then $g^n = e$ and n is minimal, otherwise we say g has infinite order.

If $\text{ord}(g) = n$, then:

$$\langle g \rangle = \{e, g, g^2, \dots, g^{n-1}\}$$

Moreover:

- $g^k = e \Leftrightarrow n \mid k$
- $\langle g^k \rangle = \langle g \rangle \Leftrightarrow \text{gcd}(k, n) = 1$

Theorem 3.3. *Every subgroup of a cyclic group is cyclic.*
More precisely:

- *Subgroups of $\mathbb{Z}$ are exactly $n\mathbb{Z}$.*
- *Subgroups of $\mathbb{Z}/n\mathbb{Z}$ correspond to the divisors of n.*

With all of this machinery we can approach group theory with geometric and algebraic intuition. Cyclic groups are like repeated motion, finite cyclic groups corresponding to rotations by rational angles and infinite ones corresponding to translations.

4 Cosets and Lagrange's Theorem

Definition 4.1 (Left and Right Cosets). *Let $H \leq G$, and let $g \in G$. The left- and right cosets are:*

$$gH = \{gh : h \in H\} \quad Hg = \{hg : h \in H\}$$

respectively.

A key intuition is that a coset is like a copy of H shifted by g. We may also recognize that the cosets are the same size as H by identifying $h \mapsto gh$ to be a bijection and so forth.

The next natural question is when are two cosets equal? Like with showing when a subset is a subgroup we have a little trick for this problem.

Theorem 4.1. *Let G be a group.*
For $g_1, g_2 \in G$, the following are equivalent:

$$\begin{gathered} g_1 H = g_2 H \\ \Updownarrow \\ g_2^{-1} g_1 \in H \\ \Updownarrow \\ g_1 \in g_2 H \end{gathered}$$

Theorem 4.2. *The set of all left cosets of $H \leq G$ forms a partition of G.*
That is,

- *Every element of G is in exactly on coset.*
- *Two cosets are either the same or disjoint.*

Definition 4.2. *Let G be a group with $H \leq G$.*
The index of H in G, written $[G : H]$, is the number of left cosets of H.

If G is finite we get $|G| = [G : H] \cdot |H|$.

Theorem 4.3 (Lagrange's Theorem). *Let G be a group with $H \leq G$. Then:*

$$|H| \mid |G|$$

Proof. Let G be a group with $H \leq G$.

Recall that by theorem 4.2 we have that the set of left cosets of H, $g_1H, g_2H, \ldots, g_kH$ (this set is finite since G is finite) are pairwise disjoint, satisfying

$$G = \bigcup_{i=1}^{k} g_iH$$

meaning

$$|G| = \sum_{i=1}^{k} |g_iH|$$

Recall also that $|H| = |g_iH|, \forall i \in [k]$. Thus

$$|G| = \sum_{i=1}^{k} |H| = k|H|$$

In other words $|G|$ is $|H|$ times some integer k, therefore $|H|$ divides $|G|$. □

Theorem 4.3 has the following immediate consequences:

- The order of any element divides $|G|$.
- $g^{|G|} = e$ for every $g \in G$.

Beware that it does not follow from Lagrange's Theorem that there exists a subgroup with the order of a divisor of $|G|$ for every divisor of G. The alternating group of order 12, A_4, has no subgroup of order 6 for example.

Proposition 4.1. *A group of order p where p is prime, is cyclic.*

Proof Sketch. Let G be a group with $|G| = p$ (prime).

Let $g \in G \setminus \{1\}$. Consider it's generated subgroup $\langle g \rangle$. By Lagrange's Theorem the order of this subgroup divides p so $|\langle g \rangle|$ is 1 or p, but it can't be 1 as $g \neq 1$ so it's order is p, i.e. g generates the entirety of G. □

4.1 Problems relating to subgroups

Problem 4.1. *Let G be a group and $g \in G$ with ord$(g) = 12$.*

- *List all distinct subgroups of $\langle g \rangle$.*
- *For which integers k does g^k generate $\langle g \rangle$?*

Solution. Recalling that $\langle g \rangle = \{g^k : k \in \mathbb{Z}\}$, we want to find all the distinct subgroups of $\langle g \rangle$. In other words, which subsets of $\langle g \rangle$ are also groups with respect to the operation of G?

ord$(g) = |\langle g \rangle| = 12$ hence $\langle g \rangle = \{g^0, g^1, \dots, g^{11}\}$. We know that there are two trivial subgroups of $\langle g \rangle$, namely $\{g^0\}$ and $\langle g \rangle$. Otherwise, we know that every cyclic subgroub of a cyclic group (which $\langle g \rangle$ is) is itself cyclic, so we can find the subgroups by looking at the generated groups of each element of $\langle g \rangle$. Otherwise, we could use the fact that $\langle g \rangle$ is isomorphic to $\mathbb{Z}/12\mathbb{Z}$ together with realizing that the number of subgroups is then the same as the number of positive divisors of 12.

Using the first approach we get:

$$\begin{aligned}
\langle g^2 \rangle &= \{g^n : n = 2k,\ k \in \mathbb{Z}\} \\
&= \{e, g^2, \dots, g^10\} = \langle g^10 \rangle \\
\langle g^3 \rangle &= \{e, g^3, g^6, g^9\} = \langle g^9 \rangle \\
\langle g^4 \rangle &= \{e, g^4, g^8\} = \langle g^8 \rangle \\
\langle g^6 \rangle &= \{e, g^6\}
\end{aligned}$$

Those integers which are coprime to 12, like $1, 5, 7, 11$, will generate $\langle g \rangle$. Thus we have 6 distinct subgroups, exactly the same as the number of positive divisors of 12, as expected. $\square$

Problem 4.2. *Let G be any group $H = \langle g \rangle$ where ord$(g) < \infty$. Prove that if $g^m \in H$, then $\langle g^m \rangle \leq H$.*

Proof. Assume, for contradiction, that $g^m \in H$, but $\langle g^m \rangle$ is not a subgroup of H.

Let $s = m \mod \text{ord}(g)$. Then $\langle g^m \rangle = \{g^{sk} : k \in \mathbb{Z}\} \neq \emptyset$. It is clear that this is a subset of $\langle g \rangle = H$, since $\langle g \rangle$ is closed under the group operation and $g^m \in H$. Then the only way our assumption is true is if there exists some $x, y \in \langle g^m \rangle$ such that $xy^{-1} \notin \langle g^m \rangle$.

Every element $x, y \in \langle g^m \rangle$ is of the form $x = g^{sn}, y = g^{st}$ for some $n, l \in \mathbb{N}$. By assumption, $xy^{-1} = g^{s(n-t)} \notin \langle g^m \rangle$ which would necessarily mean that $n - t \notin \mathbb{Z}$, impossible. $\square$

The above argument can be made more rigorous by considering the isomorphism between $\langle g^m \rangle$ and $\mathbb{Z}/s\mathbb{Z}$, or by proving the claim directly (I am an idiot, and this was a bad approach).

5 Normal subgroups

Definition 5.1 (Normal subgroup). *A subgroup $H \leq G$ is normal, written $H \trianglelefteq G$, if:*

$$gH = Hg, \forall g \in G$$

with the equivalent characterization:

$$H \trianglelefteq G \Leftrightarrow gHg^{-1} = H, \forall g \in G$$

A few things to tick off immediately;

1. Every subgroup of an abelian group is normal.
2. {e} and G are always normal.
3. The alternating group $A_n \trianglelefteq S_n$.
4. Subgroups of index 2 are always normal.

If you consider the map (conjugation)

$$x \mapsto gxg^{-1},$$

which constitutes an automorphism of G we can say the following.

Normal subgroups are precisely those subgroups which are fixed under every conjugation.

Definition 5.2 (Quotient Group). *If $H \trianglelefteq G$, define:*

$$G/H := \{gH : g \in G\}$$

with an operation we'll call multiplication by:

$$(gH)(kH) = (gk)H$$

Theorem 5.1. *The quotient group with multiplication, as defined above, is a well-defined group.*

Definition 5.3. *We define the* ***canonical projection*** *as*

$$\pi : G \to G/H,\ \pi(g) = gH.$$

Notably, the projection π constitutes a homomorphism with kernel $\ker \pi = H$. Notice that this claim of the kernel being H itself should be immediately obvious as hH for some $h \in H$ is itself H as it's closed.

6 Exercises for Week 1

Problem 6.1 (29, p.48). *Show that if G is a finite group with identity e and an even number of elements, then there is $a \neq e$ in the group such that $a * a = e$.*

Proof. We are being asked to show that in a finite group with even order, there exists some element other than the identity which is it's own inverse.

Recall uniqueness of inverses and that for $g \in G$, $(g^{-1})^{-1} = g$. Thus for every element there exists one and only one element which is it's inverse. $e^{-1} = e$ is covered. In $G \setminus \{e\}$ there are an odd number of elements. Each of these have exactly one unique inverse meaning that there is one element ”left” over which then necessarily has itself as it's inverse. $\square$

Problem 6.2 (33, p.49). *Let G be an abelian group and let $c^n = c * c * \cdots * c$ for n factors c, where $c \in G$ and $n \in \mathbb{Z}^+$. Give a proof by mathematical induction that $(a * b)^n = a^n * b^n$ for all $a, b \in G$.*

Proof. Let $a, b \in G$. The base case where $n = 1$ is trivial so we proceed to our assumption that for any $k \in \mathbb{Z}^+$ we have $(a * b)^k = a^k * b^k$.

Let $n = k + 1$.

$$\begin{aligned}
(a * b)^n &= (a * b)^{k+1} \\
&= (a * b)^k * (a * b) \\
&= a^k * b^k * (a * b) && \text{by hypothesis} \\
&= a^k * (b^k * a) * b && \text{by associativity} \\
&= a^k * (a * b^k) * b && \text{by commutativity} \\
&= (a^k * a) * (b^k * b) \\
&= (a^{k+1}) * (b^{k+1}) \\
&= a^n * b^n
\end{aligned}$$

By the principle of mathematical induction we have that

$$(a * b)^n = a^n * b^n, \ \forall n \in \mathbb{Z}^+$$

$\square$

Problem 6.3 (35, p. 49). *Show that if $(a*b)^2 = a^2 * b^2$ for a, b in a group, then $a*b = b*a$.*

Proof.

$$\begin{aligned} (a*b)^2 &= (a*b)*(a*b) \\ a^2*b^2 &= (a*a)*(b*b) \\ \Rightarrow (a*b)*(a*b) &= (a*a)*(b*b) \\ a*(b*a)*b &= a*(a*b)*b \end{aligned}$$

By one application each of left- and right cancellation we have

$$a*b = b*a$$

□

Problem 6.4 (41, p. 49). *Let G be a group and fix $g \in G$. Show that the map i_g, s.t. $i_g(x) = gxg^{-1}$ for $x \in G$. Show that i_g is an automorphism on G.*

Proof. Let $i_g : G \to G$ be defined as above. We need to show that it satisfies all the properties of an isomorphism.

It is clear, through left- and right cancellation, that i_g is injective.

Notice also that for any $x \in G$ we have the pre-image $g^{-1}xg$ since $i_g(g^{-1}xg) = gg^{-1}xgg^{-1} = x$, hence i_g is surjective.

Lastly we verify the homomorphism property:

$$\begin{aligned} i_g(xy) &= gxyg^{-1} \\ &= gxeyg^{-1} \\ &= gxg^{-1}gyg^{-1} \\ &= i_g(x)i_g(y) \end{aligned}$$

completing the proof. □

7 Problems for week of Jan 26

Problem 7.1 (Problem 44 Fraleigh's p.67). *Let $G = \langle a \rangle$ be a cyclic group isomorphic to a group G'. If $\phi : G \to G'$ is an isomorphism, show that for every $x \in G$, $\phi(x)$ is uniquely determined by $\phi(a)$. That is, if ϕ and ψ constitutes isomorphisms from $G \to G'$ with $\phi(a) = \psi(a)$, then $\phi(x) = \psi(x)$ for every $x \in G$.*

Proof. Let G be generated by a, isomorphic to G' via ϕ and ψ, satisfying

$$\phi(a) = \psi(a)$$

Let $x \in G$. Then $x = a^n$ for some $n \in \mathbb{Z}$, as $G = \langle a \rangle$. Observe the following:

$$\begin{aligned}\phi(x) &= \phi(a^n)\\ &= \phi(a)^n\\ &= \psi(a)^n\\ &= \psi(a^n) = \psi(x)\end{aligned}$$

This completes the proof. □

Problem 7.2 (Problem 6 Fraleigh's p.83). *Compute $|\sigma|$ for*

$$\sigma = \begin{pmatrix} 1 & 2 & 3 & 4 & 5 & 6 \\ 3 & 1 & 4 & 5 & 6 & 2 \end{pmatrix}$$

Solution. Observe that σ is the permutation which sends

$$\begin{gathered}1 \to 3\\ 2 \to 1\\ \vdots\\ 6 \to 2\end{gathered}$$

so σ^2 is

$$\begin{gathered}1 \to 4\\ 2 \to 3\\ \vdots\\ 6 \to 1\end{gathered}$$

We want to find $n \in \mathbb{N}$ such that $\sigma^n =$ the identity permutation. It's easy to see that performing the permutation 6 times gives $1 \to 1$ which we would require. We also see that $2 \to 2$, $3 \to 3$ and so on.

We could also write the permutation in compact form and see that σ is a cycle of length 6, so the order is 6. □

Problem 7.3 (Problem 7 Fraleigh's p.83). *Compute $|\tau|$ for*

$$\tau = \begin{pmatrix} 1 & 2 & 3 & 4 & 5 & 6 \\ 2 & 4 & 1 & 3 & 6 & 5 \end{pmatrix}$$

Solution. Note that τ in compact notation is

$$\tau = (1\ \ 2\ \ 4\ \ 3)(5\ \ 6)$$

so τ is the composition of a 2- and 4-cycle.

$$|\tau| = \mathtt{lcm}(4, 2) = 4$$

□

Problem 7.4 (Problem 49 Fraleigh's p.86). *If A is a set, then a subgroup $H \leq S_A$ is transitive on A if for each $a, b \in A$ there exists $\sigma \in H$ such that $\sigma(a) = b$. Show that if A is a nonempty finite set, then there exists a finite cylic subgroup H of S_A with $|H| = |A|$ that is transitive on A.*

Proof. Let $A = \{a_1, a_2, \dots, a_n\}$ with S_A being the permutations on A. Let $\sigma \in S_A$ be the permutation which sends $a_i \to a_{i+1}$ (take i modulo n). It is clear that σ^k is the permutation which sends a_i to a_{i+k} and furthermore $|\sigma| = n = |A|$. It is not hard to see that $\langle\sigma\rangle$ forms a subgroup where each permutation shifts all elements of A.

It is clear that for any $a_i, a_j \in A$ we can find a permutation which sends one to the other. Suppose, without loss of generality, $i < j$. Then $\sigma^{j-i}(a_i) = a_j$, and the inverse of said permutation sends a_j to a_i.

A bit inelegantly written, but I digress. □

Problem 7.5 (Problem 1 Fraleigh's p.94). *Find the orbit of*

$$\sigma = \begin{pmatrix} 1 & 2 & 3 & 4 & 5 & 6 \\ 5 & 1 & 3 & 6 & 2 & 4 \end{pmatrix}$$

Solution. We are looking at $\sigma \in S_6$ acting on $[6]$ (I assume) so we want to find

$$\{\mathtt{Orb}_\sigma(n) = \{\sigma^k(x) : k \in \mathbb{Z}\} \mid n = 1, 2, \dots, 6\}$$

Let us rewrite σ more compactly.

$$1 \to 5 \to 2 \to 1$$

$$3 \to 3$$

$$4 \to 6 \to 4$$

so

$$\sigma = (1\ \ 5\ \ 2)(4\ \ 6)$$

Thus we have the following orbits:

$$\begin{aligned} \texttt{Orb}_\sigma(1) = \texttt{Orb}_\sigma(2) = \texttt{Orb}_\sigma(5) = \{1, 2, 5\} \\ \texttt{Orb}_\sigma(4) = \texttt{Orb}_\sigma(6) = \{4, 6\} \\ \texttt{Orb}_\sigma(3) = \{3\} \end{aligned}$$

□

Problem 7.6 (Problem 7 Fraleigh's p.94). *Compute:*

$$(1\ \ 4\ \ 5)(7\ \ 8)(2\ \ 5\ \ 7)$$

Solution.

$$\begin{aligned} (1\ \ 4\ \ 5)(7\ \ 8)(2\ \ 5\ \ 7) &= (1\ \ 4\ \ 5) \circ ((7\ \ 8)(2\ \ 5\ \ 7)) \\ &= (1\ \ 4\ \ 5)(2\ \ 5\ \ 8\ \ 7) \\ &= (2\ \ 1\ \ 4\ \ 5\ \ 8\ \ 7) \end{aligned}$$

□

Problem 7.7 (Problem 29 Fraleigh's p.96). *Show that for every subgroup H of S_n for $n \geq 2$, either all permutations in H are even or exactly half are.*

Proof. Recall that an even permutation is a permutation which can be expressed as an even number of transpositions (swaps of 2 elements).

Let $\texttt{sgn}(\sigma) = 1$ if σ is even, $\texttt{sgn}(\sigma) = -1$ otherwise for all $\sigma \in S_n$. $\texttt{sgn} : S_n \to \{1, -1\}$ is a homomorphism. Restrict this map to H. Then, the image:

$$\texttt{sgn}[H] = \begin{cases} \{+1\} \\ \{+1, -1\} \end{cases}$$

In the first case it is clear that every element of H must be even.

In the other case where sgn is surjective, the kernel

$$\texttt{ker}(\texttt{sgn}[H]) = H \cap A_n,$$

the set of even permutations in H. By the first isomorphism theorem,

$$H/(H \cap A_n) \cong \{+1, -1\}$$

so

$$[H : H \cap A_n] = 2$$

hence

$$|H \cap A_n| = \frac{1}{2}H$$

i.e. half of H is even. $\square$

8 External- and Internal Direct Products

Definition 8.1. *Let* $H, K \leq G$,

$$G' = H \times K = \{(h, k) \mid h \in H, \ k \in K\}$$
$$G = H \oplus K = \{hk \mid h \in H, \ k \in K\}$$

The internal direct product $H \oplus K$ *presupposes that* $H, K \trianglelefteq G$, *and* $H \cap K = \{e\}$.

We really want to be able to say that $H \times K \cong H \oplus K$.

Example 8.1. *Consider the dihedral group* $D_3 = \{1, x, x^2, y, xy, x^y\}$. *Let* $H = \langle x \rangle$ *and* $K = \langle y \rangle$. *It's easy to see that* $H \cap K = \{e\} = \{1\}$, *and* $H \trianglelefteq D_3$. *However* $xK = \{x, xy\}$, *but* $Kx = \{x, yx\} = \{x, x^2 y\}$. *Hence*

$$D_3 \neq H \oplus K$$

Lemma 8.1. $H, K \trianglelefteq G$ *and* $H \cap K = \{e\}$.

1. *if* $ab = a'b'$ *where* $a, a' \in H$ *and* $b, b' \in K$ *then* $a = a'$ *and* $b = b'$.
2. *if* $a \in H$ *and* $b \in K$ *then* $ab = ba$.

Theorem 8.1. *Assuming* H, K *normal subgroups of* G *with trivial intersection, then*

$$H \times K \cong H \oplus K$$

Proof. Define a map

$$\varphi : H \times K \to HK, \qquad \varphi(h, k) = hk.$$

This map is well-defined since $h \in H$, $k \in K$ implies $hk \in HK$.

Define a second map

$$\psi : HK \to H \times K$$

by sending $g \in HK$ to the unique pair (h, k) such that $g = hk$. Such a decomposition exists since $HK = G$, and is unique because $H \cap K = \{e\}$.

We now verify that these maps are inverse to one another. For all $(h, k) \in H \times K$,

$$(\psi \circ \varphi)(h, k) = \psi(hk) = (h, k),$$

and for all $g \in HK$,

$$(\varphi \circ \psi)(g) = g.$$

Thus φ is bijective with inverse ψ.

Finally, φ is a homomorphism, since for $(h_1, k_1), (h_2, k_2) \in H \times K$,

$$\begin{aligned}\varphi((h_1, k_1)(h_2, k_2)) &= \varphi(h_1 h_2, k_1 k_2)\\ &= h_1 h_2 k_1 k_2\\ &= h_1 k_1 h_2 k_2\\ &= \varphi(h_1, k_1)\varphi(h_2, k_2),\end{aligned}$$

where we use that elements of H and K commute.

Therefore φ is an isomorphism, and $HK \cong H \times K$. □

Theorem 8.2. *If* $|x|, |y| < \infty$ *and* $(x, y) \in G \times H$ *then* $|(x, y)| =$ *lcm*$(|x|, |y|)$*.*

9 Problems from week of 8 Feb

Problem 9.1 (Problem 47 Fraleigh's p.113). *Let G be an abelian group. Let $H \subseteq G$ consisting of the identity and all elements of H with order 2. Show that $H \leq G$.*

Proof. Let $H = \{e\} \cup \{g \in G \mid |g| = 2\} \subseteq G$. Let $x, y \in H$, and for the sake of simplicity assume they are nontrivial, i.e. $x \neq e \neq y$. To show that H constitutes a subgroup of G it suffices to show that $xy^{-1} \in H$ (we already know it is nonempty).

$$\begin{aligned}
(xy^{-1})^2 &= xy^{-1}xy^{-1} \\
&= xxy^{-1}y^{-1} && (G \text{ abelian}) \\
&= x^2(y^2)^{-1} \\
&= e(e) \\
&= e
\end{aligned}$$

Hence $|xy^{-1}| = 2$, so $xy^{-1} \in H$. Thus H is a subgroup of G. □

Problem 9.2 (Problem 52 Fraleigh's p.113). *Show that a finie abelian group is not cyclic if and only if it contains a subgroup isomorphic to $\mathbb{Z}_p \times \mathbb{Z}_p$ for some prime p.*

Proof. Let G be a finite abelian group.

We begin by proving the right-to-left implication. Assume then that there is some $H \leq G$ with $H \cong \mathbb{Z}_p \times \mathbb{Z}_p$. Suppose, for a contradiction, that $G = \langle g \rangle$ (cyclic) for some $g \in G$. Let $\phi : H \to \mathbb{Z}_p \times \mathbb{Z}_p$ be an isomorphism.

Recall that as G is cyclic, generated by g, we have that there is some $k \in \mathbb{Z}$, for every $h \in H$, such that $h = k \cdot g$. As G is finite, $k \cdot g = (|G| - k) \cdot g$, so we can safely assume $k \geq 0$ without any loss of generality.

Since ϕ is bijective we know that, for every $y \in \mathbb{Z}_p \times \mathbb{Z}_p$, there is some $h \in H$ such that

$$\phi(h) = y$$

Using the homomorphism property, and the fact that $h = kg$, $k \in \mathbb{N}$, we get that:

$$\begin{aligned} y &= \phi(h) \\ &= \phi(k \cdot g) \\ &= \phi\left(\sum_{i=1}^{k} g\right) \\ &= \sum_{i=1}^{k} \phi(g) \\ &= k \cdot \phi(g) \end{aligned}$$

As y was chosen arbitrarily we now see that $\mathbb{Z}_p \times \mathbb{Z}_p = \langle \phi(g) \rangle$, i.e. cyclic. Being cyclic, together with the fact that $|\mathbb{Z}_p \times \mathbb{Z}_p| = p^2$, it is the case that there must be some element with order p^2, but no such element exists so $\mathbb{Z}_p \times \mathbb{Z}_p$ is not cyclic. We have arrived at a contradiction so our assumption that G is cyclic must be false.

Now we prove the other direction. We now assume that we have a finite abelian group G which is not cyclic. As G is finite we know that G is finitely generated. Thus by the fundamental theorem of finitely generated abelian groups we know that G is isomorphic to a direct product of cyclic groups of the form

$$\mathbb{Z}_{(p_1)^{\gamma_1}} \times \mathbb{Z}_{(p_2)^{\gamma_2}} \times \cdots \times \mathbb{Z}_{(p_n)^{\gamma_n}} \times \mathbb{Z} \times \mathbb{Z} \times \cdots \times \mathbb{Z}$$

As G is finite, the direct product does not include $\mathbb{Z}$. Furthermore as G is not cyclic we know that

$$G \ncong \mathbb{Z}_{p^\gamma}$$

Hence G must be the direct product of at least 2 $\mathbb{Z}_{(p_i)^{\gamma_i}}$. As $\gamma_i \in \mathbb{N} \setminus \{0\}$ and distinct from γ_j for $j \neq i$ there must be some $\mathbb{Z}_{(p_i)^{\gamma_l}}$ with $\gamma_l \geq 2$. Notice that this group necessarily has

$$H = \{0\} \times \cdots \times \{0\} \times \mathbb{Z}_{p_i} \times \{0\} \times \cdots \times \{0\} \times \mathbb{Z}_{p_i} \times \{0\} \times \cdots \times \{0\}$$

as a subgroup, which itself is isomorphic to $\mathbb{Z}_{p_i} \times \mathbb{Z}_{p_i}$. Then $\phi^{-1}[H]$ is a subgroup of G, which is isomorphic to $\mathbb{Z}_{p_i} \times \mathbb{Z}_{p_i}$. $\square$

> **Problem 9.3** (Problem 46 Fraleigh's p.135). *Let a group G be generated by $\{a_i \mid i \in I\}$, where I is some indexing set and $a_i \in G$ for all $i \in I$. Let ϕ, μ be two homomorphisms from G into a group G', such that $\phi(a_i) = \mu(a_i)$ for every $i \in I$. Prove that $\phi = \mu$.*

Proof. Let $G = \{a_i \mid i \in I\}$ with homomorphisms $\phi, \mu : G \to G'$ which coincide on a_i for all $i \in I$. Let $x \in G$.

$$\begin{aligned}
\phi(x) &= \phi\left(\prod_{i \in I} a_i^{k_i}\right), \, k_i \in \mathbb{Z} \\
&= \prod_{i \in I} \phi(a_i^{k_i}) \\
&= \prod_{i \in I} \underbrace{\phi(a_i) \cdots \phi(a_i)}_{k_i \text{ times}} \\
&= \prod_{i \in I} \underbrace{\mu(a_i) \cdots \mu(a_i)}_{k_i \text{ times}} \\
&= \prod_{i \in I} \mu(a_i^{k_i}) \\
&= \mu\left(\prod_{i \in I} a_i^{k_i}\right) = \mu(x)
\end{aligned}$$

x was chosen arbitrarily hence ϕ and μ coincide on all $x \in G$. In other words, $\phi = \mu$. □

> **Problem 9.4** (52 Fraleigh's p.135). *Let $\phi : G \to G'$ be a homomorphism with kernel H and let $a \in G$. Prove the set equality $\{x \in G \mid \phi(x) = \phi(a)\} = Ha$.*

Proof. We begin by proving that $Ha \subseteq \{x \in G \mid \phi(x) = \phi(a)\}$.

Suppose $ha \in Ha$. Then $h \in H$ so

$$\phi(ha) = \underbrace{\phi(h)}_{\in H} \phi(a) = e' \phi(a) = \phi(a)$$

Thus $ha \in \{x \in G \mid \phi(x) = \phi(a)\}$. As x was arbitrarily chosen, we conclude that

$$Ha \subseteq \{x \in G \mid \phi(x) = \phi(a)\}$$

Let $x \in \{x \in G \mid \phi(x) = \phi(a)\}$. Then

$$\begin{aligned}
\phi(x) &= \phi(a) \\
\phi(x)\phi(a)^{-1} &= e' \\
\phi(xa^{-1}) &= e'
\end{aligned}$$

hence $xa^{-1} \in H$ which implies $x \in Ha$. Once again x was arbitrarily chosen so

$$\{x \in G \mid \phi(x) = \phi(a)\} \subseteq Ha$$

Summing up we conclude that

$$Ha = \{x \in G \mid \phi(x) = \phi(a)\}$$

10 Problems week of 16 Feb, Quotient grps, simple grps

Problem 10.1 (Problem 3 Fraleigh's p.142). *Find the order of the given factor group:*

$$(\mathbb{Z}_4 \times \mathbb{Z}_2)/\langle(2,1)\rangle$$

Solution. We first compute the subgroup generated by $(2,1)$:

$$\langle(2,1)\rangle = \{(0,0),(2,1)\},$$

since $(2,1)+(2,1)=(0,0)$.

Because $\mathbb{Z}_4 \times \mathbb{Z}_2$ is abelian, every subgroup is normal, so the quotient group is well-defined.

The elements of the quotient

$$(\mathbb{Z}_4 \times \mathbb{Z}_2)/\langle(2,1)\rangle$$

are the cosets

$$(x,y)+\langle(2,1)\rangle = \{(x,y),(x+2,y+1)\}, \quad (x,y) \in \mathbb{Z}_4 \times \mathbb{Z}_2.$$

Each coset has exactly 2 elements. Since

$$|\mathbb{Z}_4 \times \mathbb{Z}_2| = 8,$$

the number of distinct cosets is

$$\frac{8}{2} = 4.$$

Hence the quotient group has order 4. □

Problem 10.2 (Problem 34 Fraleigh's p.143). *Show that if a finite group G has exactly one subgroup H of a given order, then H is a normal subgroup of G.*

Proof. Suppose G is a finite group with only one subgroup H of some order $k \in \mathbb{N}$. Recall that for the conjugacy of H is still a subgroup, furthermore

$$|gHg^{-1}| = |H| = k, \; g \in G$$

However H is the only subgroup with order k so

$$gHg^{-1} = H, \; \forall g \in G$$

which implies that H is normal.

□

Problem 10.3 (Problem 35 Fraleigh's p.143). *Show that if H and N are subgroups of a group G, and N is normal in G, then $H \cap N$ is normal in H.*

Proof. Let G be a group with $H \leq G$ and $N \trianglelefteq G$. We take it for granted that $H \cap N$ is indeed a group (this is not hard to show).

Fix some arbitrary $h \in H$. We must show that $h(H \cap N)h^{-1} \subseteq H \cap N$.

Let $x \in H \cap N$. As $x, h \in H$ we ceirtanly have $hxh^{-1} \in H$. As $x \in N$, N normal, we have that for any $h \in H \subseteq G$, $hxh^{-1} \in N$.

Hence
$$hxh^{-1} \in H \cap N$$
□

Problem 10.4 (Problem 3 Fraleigh's p.151). *Classify the groups according to the fundamental theorem of finitely generated abelian groups.*
$$(\mathbb{Z}_2 \times \mathbb{Z}_4)/\langle(1,2)\rangle$$
$$(\mathbb{Z}_4 \times \mathbb{Z}_8)/\langle(1,2)\rangle$$
$$(\mathbb{Z}_4 \times \mathbb{Z}_4 \times \mathbb{Z}_8)/\langle(1,2,4)\rangle$$

Solution. We begin with $(\mathbb{Z}_2 \times \mathbb{Z}_4)/\langle(1,2)\rangle$.

$$\begin{aligned}
\langle(1,2)\rangle &= \{(0,0),(1,2)\} \\
(0,0)\{(0,0),(1,2)\} &= \{(0,0),(1,2)\} = A \\
(1,0)\{(0,0),(1,2)\} &= \{(1,0),(0,2)\} = B \\
(0,1)\{(0,0),(1,2)\} &= \{(0,1),(1,3)\} = C \\
(1,1)\{(0,0),(1,2)\} &= \{(1,1),(0,3)\} = D \\
(0,2)\{(0,0),(1,2)\} &= \{(1,0),(0,2)\} = B \\
(1,2)\{(0,0),(1,2)\} &= \{(1,2),(0,0)\} = A \\
(0,3)\{(0,0),(1,2)\} &= \{(0,3),(1,1)\} = D \\
(1,3)\{(0,0),(1,2)\} &= \{(1,3),(0,1)\} = C
\end{aligned}$$

We get an abelian group with 4 distinct elements. Crucially, we can see that the coset of $(0,1)$ has order 4 and is thus a generator. We conclude that $(\mathbb{Z}_2 \times \mathbb{Z}_4)/\langle(1,2)\rangle \cong \mathbb{Z}_4$.

Now we look at $(\mathbb{Z}_4 \times \mathbb{Z}_8)/\langle(1,2)\rangle$.

Notice that
$$(\mathbb{Z}_4 \times \mathbb{Z}_8) = \langle a, b \mid 4a = 0,\ 8b = 0,\ ab = ba\rangle$$

so quotienting with $\langle(1,2)\rangle$ only imposes the extra relation that

$$a + 2b = 0 \Leftrightarrow a = -2b$$

then $\mathtt{ord}(1,2) = \mathtt{lcm}(4,4) = 4$ so $\langle(1,2)\rangle$ has 4 elements. $|\mathbb{Z}_4 \times \mathbb{Z}_8| = 32$ so the quotient group has

$$\frac{32}{4} = 8$$

elements. Up to isomorphism there exist 3 abelian groups of order 8. The relation imposed by $\langle(1,2)\rangle$ changes nothing so we require $8b = 0$. Clearly

$$(\mathbb{Z}_4 \times \mathbb{Z}_8)/\langle(1,2)\rangle \cong \mathbb{Z}_8$$

Lastly we look at

$$(\mathbb{Z}_4 \times \mathbb{Z}_4 \times \mathbb{Z}_8)/\langle(1,2,4)\rangle$$

Notice that

$$\mathbb{Z}_4 \times \mathbb{Z}_4 \times \mathbb{Z}_8 = \{a, b, c \mid 4a = 0,\ 4b = 0,\ 8c = 0,\ ab = ba,\ ac = ca,\ bc = ca\}$$

with

$$\langle(1,2,4)\rangle$$

imposing $a + 2b + 4c = 0$ which then means that $a = -2b - 4c$. The order of $\mathbb{Z}_4 \times \mathbb{Z}_4 \times \mathbb{Z}_8$ is 128. The order of the subgroup is $\mathtt{lcm}(4,2,2)$ which is 4. Thus the order of the quotient group is

$$\frac{128}{4} = 32 = 2^5$$

There are precisely 7 abelian groups unique up to isomorphism. b has order 4 and c has 8 so we conclude that the quotient group is isomorphic to

$$\mathbb{Z}_4 \times \mathbb{Z}_8$$

$\square$

Problem 10.5 (Problem 34 Fraleigh's p.153). *Let G be finite, containing a nontrivial subgroup of index 2 in G. Show that G is not simple.*

Proof. There exists a subgroup $H \leq G$ with 2 left cosets. Since H is always a left coset of H there must be one more left coset gH for any $g \notin H$. Since the number of right cosets is the same we have right cosets H and Hg. There are only 2 cosets so the non-H cosets must be the same

$$gH = Hg$$

Thus left and right cosets coincide. Therefore

$$H \trianglelefteq G$$

H is nontrivial, and crucially $H \neq G$ (since the index is 2). Hence G has a nontrivial proper subgroup. In other words, G is not simple. $\square$

11 Group Actions

Definition 11.1. *Let X be a set, and G a group. A (left) action of G on X is a map*

$$G \times X \to X$$
$$(g, x) \mapsto gx$$

such that $ex = x$ and $g_1(g_2x) = (g_1g_2)x$ for all $x \in X$ and $g_1, g_2 \in G$.

The right action is similarily defined, and in the case where G is abelian the actions coincide.

Example 11.1. *The following constitute group actions when considering matrix-vector multiplication.*

1. *$GL_2(\mathbb{R})$ and its subgroups on $\mathbb{R}^2$.*
2. *same for $GL_n(\mathbb{R})$ on $\mathbb{R}^n$.*

A very natural group action is D_n acting on the regular polygon with n sides, as D_n is the symmetries of the regular n-gon.
Furthermore S_n (which is often thought of as the bijections $f : [n] \to [n]$) can act on said $[n]$.
Another group action which we have also seen many times is conjugation where G acts on itself.

Definition 11.2. *Let G be a group and X a set.*
We say that a group action $G \times X \to X$ is faithful *if for any $g \in G$, $g \neq e$ implies there is some $x \in X$ such that $gx \neq x$.*
Alternatively it can be defined to be a group action for which any $g_1, g_2 \in G$ with $g_1 \neq g_2$ implies that there exists $x \in X$ such that $g_1x \neq g_2x$.

Proposition 11.1. *Consider a group action $G \times X \to X$. It defines a group homomorphism $\phi : G \to \mathrm{Sym}(X)$ for which $\ker \phi = \{e\}$ if and only if the group action is faithful.*

Problem 11.1 (Problem 11 Fraleigh's p.160). *Let X be a G-set. Show that G acts faithfully on X if and only if no two distinct elements of G have the same action on each element X.*
(I.e. prove the equivalent definition posed in definition 11.2)

Proof. Let G be a group, X a G-set.

Assume first that the action of G on X is faithful. Then $e \neq g \in G$ implies the existence of some $x \in X$ for which

$$gx \neq x$$

Suppose, for a contradiction, that there exist some distinct $g_1, g_2 \in G$ which coincide on every $x \in X$. Then

$$g_1 x = g_2 x$$
$$g_2^{-1} g_1 x = x$$

In other words there exists $g = g_2^{-1} g_1 \in G$ for which every $x \in X$ is a fixed-point, contradicting the assumption that G acts faithfully on X. Thus G acting faithfully on X must imply that any distinct pair of elements of G have some $x \in X$ for which they do not coincide.

Assume now that G acts on X such that for every distinct $g_1, g_2 \in G$ for which there exists $x \in X$ such that $g_1 x \neq g_2 x$. Let $g_2 = e \neq g_1$. They are clearly distinct and furthermore

$$g_1 x \neq g_2 x = x$$

This holds for any $g_1 \in G$ so G acts faithfully on X. $\square$

Definition 11.3 (Orbit). *Let G be a group acting on a G-set X. Let $x \in X$.*
The orbit of x, often denoted Orb(x) or Gx is the set

$$\text{Orb}(x) := \{gx \mid g \in G\}$$

Definition 11.4 (Stabilizer). *Let G be a group acting on a G-set X. Let $x \in X$.*
The stabilizer of x, often denoted Stab$_G(x)$ or G_x is the set

$$\{g \in G \mid gx = x\}$$

Theorem 11.1 (Orbit-Stabilizer). *For a finite group G acting on a set X, the following holds (conditions are equivalent).*

$$|\mathtt{Orb}(x)| = [G : \mathtt{Stab}_G(x)]$$
$$|G| = |\mathtt{Orb}(x)| \cdot |\mathtt{Stab}_G(x)|$$

Problem 11.2. *Let $\{X_i \mid i \in I\}$ be a disjoint collection of sets. Let each X_i be a G-set for the same group G.*
a) Show that $\bigcup_{i \in I} X_i$ can be viewed in a natural way as a G-set, the ***union*** *of the G-sets X_i.*
b) Show that every G-set X is the union of its orbits.

Proof of (a). Let $X = \bigcup_{i \in I} X_i$. Since the union is disjoint, each $x \in X$ lies in a unique X_i. Define

$$g \cdot x := g \cdot_{X_i} x$$

where the right-hand side denotes the given G-action on X_i.

This is well-defined because the union is disjoint. For $x \in X_i$ we have

$$e \cdot x = x$$

since X_i is a G-set, and

$$(gh) \cdot x = (gh) \cdot_{X_i} x = g \cdot_{X_i} (h \cdot_{X_i} x) = g \cdot (h \cdot x).$$

Thus the action axioms hold, and X becomes a G-set. □

Proof of (b). For $x \in X$, define the orbit

$$G \cdot x = \{g \cdot x \mid g \in G\}.$$

Since $x = e \cdot x$, every element lies in its orbit, so

$$X = \bigcup_{x \in X} G \cdot x.$$

If $G \cdot x \cap G \cdot y \neq \emptyset$, then $g \cdot x = h \cdot y$ for some $g, h \in G$. Then

$$x = g^{-1} h \cdot y,$$

so $x \in G \cdot y$, which implies $G \cdot x = G \cdot y$. Thus distinct orbits are disjoint, and X is the disjoint union of its orbits. □

Theorem 11.2 (Burnside's Formula). *Let G be a finite group and X a finite G-set. If r is the number of orbits in X under G, then*

$$r \cdot |G| = \sum_{g \in G} |X_g|$$

where X_g are the fixed points *of g, namely all $x \in X$ satisfying $gx = x$.*

Problem 11.3 (Problem 1 Fraileigh's p.164). *Find the number of orbits in $\{1, \ldots, 8\}$ under the cyclic subgroup $\langle (1\ 3\ 5\ 6) \rangle \leq S_8$.*

Solution. Both $[8]$ and the given subgroup of S_8 are finite so we use Burnside's formula. Denote $\langle \sigma \rangle := \langle 1\ 3\ 5\ 6 \rangle$. We know that $|\sigma| = 4$ so we must simply calculate the sum of sizes of fixed points of each permutation in $\langle \sigma \rangle$. Obviously $|X_e| = |X| = 8$. Next up σ has $2, 4, 7, 8$ as fixed points so $|X_\sigma| = 4$. Next up $\sigma^2 = (1\ 5)(3\ 6)$ and has the same fixed points. $\sigma^3 = (1\ 6\ 5\ 3)$ with the same fixed points once again. Thus

$$r = 8 + 4 + 4 + 4/|\sigma| = 5$$

$\square$

12 Rings, Fields & Integral domains

Definition 12.1 (Ring). *A ring R is a set together with two binary operations $+$ and $\cdot$, written $(R, +, \cdot)$, satisfying*

1. *$(R, +)$ is an abelian group (identity denoted 0),*
2. *$(R, \cdot)$ has closure and associativity,*
3. *with distributive laws, i.e.*

$$a \cdot (b + c) = ab + ac, \ \ (b + c) \cdot a = ba + ca.$$

If R has a multiplicative identity 1, we say R is a ring with identity, or R is a ring with unity or R is unital.
If $ab = ba$ for all $a, b \in R$, then R is a commutative ring.

So what, informally, a ring is, is a structure where you can add, multiply and subtract, but not necessarily divide.

Rings have a bunch of immediately nice properties such as

- $0 \in R$ is unique (follows from $(R, +)$ being abelian grp),
- $-a$ is unique (follows from the same logic),
- $a \cdot 0 = 0 = 0 \cdot a$ (very nice),
- $|R| \geq 2$, then $1 \neq 0$,
- $-(a + b) = (-a) + (-b)$,
- $-(-a) = a$,
- $-(ab) = (-a)b = a(-b)$,
- $ab = (-a)(-b)$.

Definition 12.2 (Unit). *Let $(R, +, \cdot)$ be a ring with identity, and let $u \in R$. u is called a unit if u has a multiplicative inverse u^{-1} such that*

$$u \cdot u^{-1} = u^{-1} \cdot u = 1$$

We denote the set of units

$$\{u \in R \mid u \text{ unit}\} =: R^x$$

(This set is a group under the multiplication of R)

Definition 12.3 (Zero divisor). *Let $(R, +, \cdot)$ be a ring. A zero divisor is a non-zero element $a \in R$ such that there exists non-zero $b \in R$ satisfying*

$$ab = 0 \quad \text{left zero divisor} ba = 0 \quad \text{right zero divisor}$$

Where these coincide in commutative rings.

Definition 12.4 (Field). *Let $(R, +, \cdot)$ be a ring.*
*R is a **field** if $(R \setminus \{0\}, \cdot)$ is an abelian group.*

Definition 12.5. *Let $(R, +, \cdot)$ be a ring.*
*R is a **division ring** if $(R \setminus \{0\}, \cdot)$ is a group.*

Crucially, in both a divison ring and a field we know that $\exists 1 \in R$, the multiplicative identity, meaning a field/division ring is unital.

The prototypical example of a commutative unital ring is $(\mathbb{Z}, +, \cdot)$. Importantly, $\mathbb{Z}$ is not a field. So a big area of focus will be to look at larger more powerful structures.

Definition 12.6 (Integral Domain). *Let $(R, +, \cdot)$ be a ring.*
*R is an **integral domain** if it is a commutative, unital ring.*

Proposition 12.1. *A ring R is an integral domain if and only if R has no zero-divisors and $|R| \geq 2$.*

The nice property we can gleam from integral domains is that $a \cdot b = 0$ if and only if either a or b is zero.

Usually, when we say integral domain, or field, we will be refering to said structure when it has 2 or more elements. The reason for this is that a ring or field with 1 element is not worth dwelling on.

Proposition 12.2. *A field is an integral domain.*

Proof sketch. If $ab = 0$ and $a \neq 0$, then $b = a^{-1}(ab) = a^{-1}0 = 0$. □

Theorem 12.1. *Every finite integral domain is a field.*

Proof. Let D be a finite integral domain.

$(D \setminus \{0\}, \cdot)$ is closed, because D has no zero divisors. Furthermore it is associative and $1 \in D \setminus \{0\}$ by definition. The last thing we need is inverses.

Let $r \in D$, $r \neq 0$. Consider $r, r^2, r^3, \ldots r^k, \ldots$

As D is finite there must be repeted powers in this list. Let $r^i = r^j$, $i < j$ wlog. Then

$$\underbrace{r \cdot r \cdots r}_{i} \cdot 1 = \underbrace{r \cdot r \cdots r}_{j}$$

Use left cancellation repeatedly to get

$$r^{j-i} = 1$$

Let $s = r^{j-i-1}$. Then $rs = sr = 1$, so $s = r^{-1}$. □

Theorem 12.2. *Every finite division ring is a field.*

13 Some problems pertaining to Rings, Fields, etc.

Problem 13.1 (Problem 3 & 5 Fraleigh's p.174). *Compute the products*

$$(11)(-4) \in \mathbb{Z}_{15}$$

$$(2,3)(3,5) \in \mathbb{Z}_5 \times \mathbb{Z}_9$$

Solution.

$$\begin{aligned} 11 \cdot -4 &= -(11 \cdot 4) \\ &= -(44) \\ &\equiv 1 \mod 15 \end{aligned}$$

$$\begin{aligned} (2,3)(3,5) &= (2 \cdot 3, 3 \cdot 5) \\ &= (6, 15) \\ &\equiv (1, 6) \end{aligned}$$

□

Problem 13.2 (Problem 11 Fraleigh's p.175). *Determine whether the following structure is a ring, then if it is a commutative ring, whether it is unital, and whether it is a field.*

$$\left\{ a + b\sqrt{2} \mid a, b \in \mathbb{Z} \right\}$$

Solution. Let's denote this structure as $\mathbb{Z}(\sqrt{2})$.

It is quite clear that $(\mathbb{Z}(\sqrt{2}), +)$ forms an abelian group as the integers is an abelian group and the identity here is $0 + 0\sqrt{2} \in \mathbb{Z}(\sqrt{2})$. Furthermore $(a+b\sqrt{2})+(c+d\sqrt{2}) = (a+c)+(b+d)\sqrt{2} \in \mathbb{Z}(\sqrt{2})$, so we're good.

The $\mathbb{Z} \setminus \{0\}$ with multiplication is closed, but not every element has an inverse (in fact only 1 and -1 do), so we'd expect something similar here.

Suppose $a + b\sqrt{2}, c + d\sqrt{2} \in \mathbb{Z}(\sqrt{2}) \setminus \{0\}$. Then

$$\begin{aligned} (a + b\sqrt{2})(c + d\sqrt{2}) &= ac + cb\sqrt{2} + ad\sqrt{2} + 2bd \\ &= (ac + 2bd) + (ad + cb)\sqrt{2} \in \mathbb{Z}(\sqrt{2}) \setminus \{0\} \end{aligned}$$

So the multiplicative structure is closed with identity, but there does not exist inverses for every element as for instance $2 \in \mathbb{Z}(\sqrt{2})^x$, but

2^{-1} does not exist. It is also not hard to see that distributive laws work fine.

Thus we conclude that $\mathbb{Z}(\sqrt{2})$ is a commutative unital ring. $\square$

Problem 13.3 (Problem 20 Fraleigh's p.175). *Consider the matrix ring $M_2(\mathbb{Z}_2)$.*
a. Find the order (number of elements) of the ring.
b. List all units in the ring.

Solution of (a). Recall that

$$M_2(\mathbb{Z}_2) = \left\{ \begin{pmatrix} a & b \\ c & d \end{pmatrix} \middle| \, a, b, c, d \in \mathbb{Z}_2 \right\}$$

There are 4 entries in each matrix in the ring, all of which can take on one of two values from $\mathbb{Z}_4$. Thus there are $2^4 = 16$ elements. $\square$

Problem 13.4 (Problem 37 Fraleigh's p.176). *Show that if U is the collection of all units in a ring $(R, +, \cdot)$ with unity, then $(U, \cdot)$ is a group.*

Proof. We conisder U the collection of the units of a ring R. $1 \in R$ (as R has unity), $(R, \cdot)$ has associativity by definition and the set of units U is precisely those $u \in R$ such that

$$u \cdot u^{-1} = u^{-1} \cdot u = 1$$

so it remains to show that U is closed. Let $u, v \in U$. Then, as said, there exists inverses $u^{-1}, v^{-1} \in U$. Then

$$\begin{aligned}(uv)(v^{-1}u^{-1}) &= u(vv^{-1})u^{-1} \\ &= u(1)u^{-1} \\ &= uu^{-1} \\ &= 1\end{aligned}$$

so $(uv)^{-1} = (v^{-1}u^{-1}) \in U$. Thus the product of two units is itself a unit. We have shown $(U, \cdot)$ to have all the properties of a group. $\square$

Problem 13.5 (Problem 46 Fraleigh's p.176). *An element a of a ring R is nilpotent if $a^n = 0$ for some $n \in \mathbb{Z}^+$. Show that if a and b are nilpotent in some ring commutative ring R, then $a + b \in R$ is also nilpotent.*

Proof. Suppose $a, b \in R$ (commutative) such that $a^n = 0$ and $b^m = 0$ for some $n, m \in \mathbb{Z}^+$.

Let $k \in \mathbb{Z}^+$. Then

$$\begin{aligned}(a+b)^k &= \sum_{i=0}^{k} \binom{k}{i} a^{k-i} b^i \\ &= a^k + ka^{k-1}b + \cdots + kab^{k-1} + b^k\end{aligned}$$

in particular, if $k := n + m$ then we observe the following:

$$\begin{aligned}(a+b)^{n+m} &= \sum_{k=0}^{n+m} \binom{n+m}{k} a^{n+m-k} b^k \\ &= \sum_{k=m+1}^{n+m} \binom{n+m}{k} a^{n+m-k} b^k + \sum_{k=0}^{m} \binom{n+m}{k} a^{n+m-k} b^k \\ &= \underbrace{\binom{n+m}{m+1} a^{n+m-m-1} b^{m+1}}_{\cdots\, a^{n-1} \cdot 0 \cdot b = 0} + \ldots + \underbrace{b^{n+m}}_{(b^n) \cdot 0 = 0} \\ &+ \underbrace{a^{n+m}}_{0 \cdot a^m = 0} + \cdots + \underbrace{\binom{n+m}{m} a^{n+m-m} b^m}_{\cdots\, 0 \cdot 0 = 0} \\ &= 0\end{aligned}$$

since every term is 0. □

Problem 13.6 (Problem 1 Fraleigh's p.182). *Find all solutions of* $x^3 - 2x^2 - 3x = 0$ *in* $\mathbb{Z}_{12}$.

Solution.

$$\begin{aligned}f(x) = x^3 - 2x^2 - 3x &= 0 \\ x(x^2 - 2x - 3) &= 0 \\ x(x-3)(x+1) &= 0\end{aligned}$$

Since we are in a ring with zero-divisors we are really asking for which $x \in \mathbb{Z}_{12}$ the following holds

$$12 \mid x(x-3)(x+1)$$

We begin by solving $\mod 3$ where

$$f(x) = x^2(x+1) \in \mathbb{Z}_3[x]$$

with solutions $0, 2 \in \mathbb{Z}_3$. $f(x)$ reduces to

$$f(x) = x(x-1)^2 \in \mathbb{Z}_4[x]$$

with solutions $0, 1, 2 \in \mathbb{Z}_4$. We get the solutions

$$\{0, 2, 5, 6, 8, 9\} \subset \mathbb{Z}_{12}$$

$\square$

Problem 13.7 (Problem 11 Fraleigh's p.182). *Let R be a commutative unital ring with characteristic 4. Compute and simplify $(a+b)^4$ for $a, b \in R$.*

Solution.

$$\begin{aligned}(a+b)^4 &= a^4 + \binom{4}{1}a^3b + \binom{4}{2}a^2b^2 + \binom{4}{3}ab^3 + b^4 \\ &= a^4 + 4a^3b + 6a^2b^2 + 4ab^3 + b^4\end{aligned}$$

As $4 \equiv 0 \mod 4$ and $6 \equiv 2 \mod 4$ we get $(a+b)^4 \equiv a^4 + 2a^2b^2 + b^4$ in integral domain R with $\mathrm{char}(R) = 4$. $\square$

Problem 13.8 (Problem 23 Fraleigh's p.183). *Show that a division ring has exactly two idempotent elements.*

Proof. Recall that we define division rings as non-trivial so $|R| \geq 2$ crucially. The candidates for idempotent elements are $0, 1 \in R$ as $0 \cdot 0 = 0$ as $0 \cdot a = 0\ \forall a \in R$. Then since $1 \cdot a = a\ \forall a \in R$ we conclude that $1 \cdot 1 = 1$. Now we need to show that there do not exist any other elements which satisfy this property. Assume, for a contradiction, that there exists $a \in R \setminus \{0, 1\}$ such that

$$a^2 = a$$

As R is a division ring we know there exists $a^{-1} \in R$ such that

$$\begin{aligned}a^2a^{-1} &= aa^{-1} \\ a(aa^{-1}) &= 1 \\ a &= 1\end{aligned}$$

contradiction our assumption that $a \neq 1$. $\square$

Problem 13.9 (Problem 29 Fraleigh's p.183). *Show that the characteristic of an integral domain D is either 0 or a prime p.*

Proof. Let D be an integral domain, i.e. a commutative ring with unity and no zero-divisors. Assume, for a contradiction, that $\mathrm{char}(D) = k \in$

$\mathbb{Z}^+$ (not prime). Then char$(D) = k = nm,\ n, m \in \mathbb{Z}^+$. Then we can conclude that

$$k \cdot 1 = \underbrace{1 + 1 + \cdots + 1 + 1}_{k=nm \text{ times}} = 0$$

but then

$$(n \cdot 1)(m \cdot 1) = 0$$

so $n \cdot 1 = 0$ or $m \cdot 1 = 0$ so k is not the characteristic of D. $\square$

14 Fermat's and Euler's theorems, Quotient Fields

Theorem 14.1 (Fermat's Little Theorem). *If p is prime and* gcd$(a,p)=1$, *then*

$$a^{p-1} \equiv 1 \mod p$$

Equivalently

$$a^p \equiv a \mod p$$

Proof idea. The key fact is that

$$(\mathbb{Z}/p\mathbb{Z})^\times$$

the nonzero elements mod p, form a group under multiplication. This group has $p-1$ elements (finite) and cyclic. By Lagrange's Theorem, if G is finite and $g \in G$ then

$$g^{|G|} = e$$

Applying this to $G = (\mathbb{Z}/p\mathbb{Z})^\times$ we get

$$a^{p-1} \equiv 1$$

□

Theorem 14.2. *Euler's Theorem If* gcd$(a,n)=1$, *then*

$$a^{\phi(n)} \equiv 1 \mod n,$$

where $\phi(n)$ is Euler's totient function.

Proof idea. We consider

$$(\mathbb{Z}/n\mathbb{Z})^\times$$

the group of units mod n. It has exactly $\phi(n)$ elements. Again by Lagrange:

$$a^{\phi(n)} \equiv 1$$

□

So Fermat's Little Theorem is just Euler's Theorem for n prime.

14.1 Quotient Fields

In $\mathbb{Z}$, you generally cannot divide by 2. But in $\mathbb{Q}$, you can. The questions we want to answer is:

"Can we build a field from a ring?"

As it turns out the answer is yes (conditions may apply...).

Recall that an integral domain is

- a commutative ring,
- with identity,
- with no zero divisors

Suppose D is an integral domain. We construct Frac(D) like rational numbers.

Definition 14.1 (Quotient Field). *The* ***field of fractions*** *or* ***quotient field*** *of an integral domain D is the smallest field K (often denoted* Frac(D)*) containing D as a subring, constructed by forming formal fractions $\frac{a}{b}$ where $a, b \in K$ $(b \neq 0)$.*
The operations on K are defined as

$$\frac{a}{b} + \frac{c}{d} = \frac{ad + bd}{bd}$$

and

$$\frac{a}{b} \cdot \frac{c}{d} = \frac{ac}{bd}$$

We equip K with an equivalence relation $\sim$ where we say

$$\frac{a}{b} \sim \frac{c}{d}$$

if $ad = bc$.

As we see we get a field Frac(D) and we get an injective embedding

$$D \hookrightarrow \text{Frac}(D)$$

via

$$a \mapsto \frac{a}{1}$$

Problem 14.1 (Problem 20.5 Fraleigh's p.189). *Use Fermat's theorem to find the remainder of 37^{49} when it is divided by 7.*

Solution. $\gcd(37, 7) = 1$ and $p = 7$ is ceirtainly prime so Fermat's Little Theorem applies.

We can rewrite

$$37^{49} = 37^{7^2} \equiv 2^{7^2}$$

From FLT we have $2^6 \equiv 1$.

$$2^{7^2} = 2^{6 \cdot 8 + 1} \equiv 1^8 \cdot 2 \equiv 2 \mod 7$$

□

Problem 14.2 (Problem 20.12 Fraleigh's p.189). *Describe all solutions to*

$$22x \equiv 5 \mod 15$$

Solution. The greatest common divisor $\gcd(22, 15) = 1$ which is a divisor of 5 so there does exist solutions (in fact, as $\gcd(22, 15) = 1$ there exist exactly one solution). We can simplify to get

$$7x \equiv 5 \mod 15$$

As the greates common divisor is 1 we know that 7 is a unit in $\mathbb{Z}/15\mathbb{Z}$. We want to find 7^{-1}. Notice that $7 \cdot 13 = 91 \equiv 1$.

$$\begin{aligned} x \equiv 13 \cdot 5 &\mod 15 \\ 13 \cdot 5 &= 65 \\ 65 \equiv 5 &\mod 15 \end{aligned}$$

Thus

$$x \equiv 5 \mod 15$$

□

Problem 14.3 (Problem 20.15 Fraleigh's p.189). *As in the task above for*

$$39x \equiv 125 \mod 9$$

Solution. Notice that

$$39 \equiv 3 \mod 9$$

and

$$125 \equiv -1 \equiv 8 \mod 9$$

so we can transform the problem into

$$3x \equiv 8 \mod 9$$

Notice that $\gcd(3, 9) = 3$, but 3 is not a divisor of 8 thus the solution DNE. □

Problem 14.4 (Problem 20.27 Fraleigh's p.190). *Show that 1 and $p-1$ are the only elements of the field $\mathbb{Z}_p$ that are their own multiplicative inverse. [Hint: Consider the equation $x^2 - 1 = 0$].*

Proof. Let p be prime and let $x \in \mathbb{Z}_p$. Notice that $x = x^{-1}$ iff

$$x^2 \equiv 1 \mod p \Leftrightarrow x^2 - 1 \equiv 0 \mod p$$

It is not hard to see that 1 and $p-1$ are solutions to these equations, but we have to show that they are the only ones. Notice that x^2-1 can be rewritten so we get

$$(x-1)(x+1) \equiv 0 \mod p$$

Recall that since $\mathbb{Z}_p$ is a field (no zero divisors) it is the case that $x-1 \equiv 0$ or $x+1 \equiv 0$. I.e.

$$x \equiv 1 \mod p, \quad x \equiv -1 \mod p.$$

Therefore

$$x = \begin{cases} 1 \\ p-1 \end{cases}$$

$\square$

Problem 14.5 (Problem 21.1 Fraleigh's p.196). *Describe the field F of quotients of the integral subdomain*

$$D = \{n + mi \mid n, m \in \mathbb{Z}\}$$

of $\mathbb{C}$. "Describe" means give the elements of $\mathbb{C}$ that make up the field of quotients of D in $\mathbb{C}$.

Solution. Recall that

$$\text{Frac}(D) = \left\{ \frac{a}{b} \,\middle|\, a, b \in D \ (b \neq 0) \right\}$$

$a, b \in D$ means that a, b can be written as

$$a = n + mi, \quad b = k + li, \quad n, m, k, l \in \mathbb{Z}$$

(note: $k + li \neq 0 \Rightarrow k \neq 0 \neq li$) so

$$\text{Frac}(D)$$

are elements of the form

$$\begin{aligned} \frac{n+mi}{k+li} &= \frac{n+mi}{k+li} \cdot \frac{k-li}{k-li} \\ &= \frac{(n+mi)(k-li)}{k^2+l^2} \\ &= \frac{nk + kmi - nli + ml}{k^2+l^2} \\ &= \frac{nk + ml + (km - nl)i}{k^2+l^2} \\ &= \frac{nk+ml}{k^2+l^2} + \frac{km-nl}{k^2+l^2} i \end{aligned}$$

Now we know Frac$(D) \subseteq \mathbb{Q}(i)$, but is this subset relation strict? Let $a = r/s$, $b = t/s$. Then

$$a + bi = \frac{r}{s} + \frac{t}{s}i$$

and since

$$r + ti \in \mathbb{Z}(i) = D, \text{ and } s \in \mathbb{Z} \subset \mathbb{Z}(i) = D$$

we conclude that $\mathbb{Q}(i) \subseteq D$. I.e.

$$D = \text{Frac}(\mathbb{Z}(i)) = \mathbb{Q}(i)$$

□

Problem 14.6 (Problem 21.12 Fraleigh's p.197). *Let R be a nonzero commutative ring, and let T be a nonempty subset of R closed under multiplication and containing neither 0 nor divisors of 0. Starting with $R \times T$ and otherwise exactly following the construction in this section, we can show that the ring R can be enlarged to a partial ring of quotients $Q(R,T)$. Think about this for 15 minutes or so; look back over the construction and see why things still work. In particular, show the following:*
a. $Q(R,T)$ has unity even if R does not.
b. In $Q(R,T)$, every nonzero element of T is a unit.

Proof of (a). We let R be some nonzero commutative ring and take a closed subset T not containing 0 or any divisors of it and construct $Q(R,T)$ to be

$$\frac{r}{t}, \; r \in R \text{ and } t \in T$$

Let $x \in Q(R,T)$ then

$$x = \frac{r}{t}$$

for some $r \in R$, $t \in T$. As $T \subseteq R$ we know that

$$\frac{t}{t} \in Q(R,T)$$

Notice that

$$\frac{t}{t} \cdot \frac{r}{t} = \frac{tr}{t^2} = \frac{r}{t} \cdot \frac{t}{t}$$

but

$$\frac{tr}{t^2} = \frac{r}{t}$$

as

$$t^2 r = trt$$

so $t/t = 1_{Q(R,T)}$. □

Proof. Suppose $t, s \in T$, nonzero, with

$$\frac{t}{s} \in Q(R, T)$$

We claim this arbitrary nonzero element is invertible. Consider

$$\frac{s}{t}$$

Observe that

$$\frac{t}{s}\frac{s}{t} = \frac{t}{t} = \frac{s}{t}\frac{t}{s}$$

and as we showed in (a), and following from $st = ts$, we get

$$1 = \frac{s}{t}\frac{t}{s}$$

$\square$

Problem 14.7 (Problem 22.2 Fraleigh's p.207). *Find the sum and product of the given polynomial in the polynomial ring:*

$$f(x) = x + 1,\ g(x) = 3x^2 + 2x + 3, \text{ in } \mathbb{Z}_2[x].$$

Solution.

$$\begin{aligned} f(x) + g(x) &= 3x^2 + 3x + 4 \\ &\equiv x^2 + x \mod 2 \end{aligned}$$

$$\begin{aligned} f(x)g(x) &= (x + 1)(3x^2 + 2x + 3) \\ &\equiv (x + 1)(x^2 + 1) \mod 2 \\ &= x^3 + x^2 + x + 1 \end{aligned}$$

$\square$

Problem 14.8 (Problem 22.5 Fraleigh's p.207). *How many polynomials are there of degree ≤ 3 in $\mathbb{Z}_2[x]$? (Include 0)*

Solution. The polynomials in $\mathbb{Z}_2$ are precisely

$$ax^3 + bx^2 + cx + d,\ \ a, b, c, d \in \mathbb{Z}_2$$

So there are 4 independent choices of coefficients which each can take on one of two values. Thus there are

$$2^4 = 16$$

possible polynomials in $\mathbb{Z}_2$. $\square$

Problem 14.9 (Problem 22.12/13 Fraleigh's p.207). *Find all zeros of the indicated finite field of the given polynomial in said field.*

$$x^2 + 1, \text{ in } \mathbb{Z}_2$$

and

$$x^3 + 2x + 2, \text{ in } \mathbb{Z}_7$$

Solution.

$$\begin{aligned} x^2 + 1 &= 0 \\ x^2 &= -1 \equiv 1 \mod 2 \end{aligned}$$

so the only zero is $1 \in \mathbb{Z}_2$.

Next we work with

$$x^3 + 2x + 2 = 0$$

Let us begin by trying some values. Clearly 0 does not work. If we plug in 1 we get

$$1 + 2 + 2 = 5 \neq 0$$

Next 2 gives us

$$8 + 4 + 2 = 14 \equiv 0$$

so 2 is a solution.

$$27 + 6 + 2 = 35 \equiv 0$$

$$64 + 8 + 2 = 74 \equiv 4 \neq 0$$

and so on to find the rest of the zeros. □

Problem 14.10 (Problem 27 Fraleigh's p.208). *Let F be a characteristic zero field and let D be the formal polynomial differentiation map, so that*

$$D\left(\sum_{k=0}^{n} a_k x^k\right) = \sum_{k=1}^{n} k a_k x^{k-1}$$

a) Show that $D : F[x] \to F[x]$ is a group homomorphism of $\langle F[x], +\rangle$ onto itself. Is D a ring homomorphism?
b) Find the kernel of D.
c) Find the image of $F[x]$ under D.

Proof of (a). Let $f, g \in F[x]$, i.e.

$$f(x) := \sum_{k=0}^{n} a_k x^k, \quad g(x) := \sum_{k=0}^{n} b_k x^k$$

Then

$$
\begin{aligned}
D(f+g) &= D\left(\left[\sum_{k=0}^{n} a_k x^k\right] + \left[\sum_{k=0}^{n} b_k x^k\right]\right) \\
&= D\left(\sum_{k=0}^{n} (a_k + b_k) x^k\right) \\
&= \sum_{k=1}^{n} k(a_k + b_k) x^{k-1} \\
&= \sum_{k=1}^{n} k(a_k) x^{k-1} + \sum_{k=1}^{n} k(b_k) x^{k-1} \\
&= D(f) + D(g)
\end{aligned}
$$

It is not a ring homomorphism. Consider

$$x(x^2+1)$$

Then

$$D(x(x^2+1)) = D(x^3+x) = 3x^2+1$$

but

$$D(x)D(x^2+1) = 2x$$

$\square$

Solution of (b). The preimage of the zero polynomial under differentiation is simply all the constant polynomials i.e.

$$c \in F$$

$\square$

Solution of (c). Differentiation is surjective so the image of $F[x]$ is simply $F[x]$. Take any $f \in F[x]$, i.e. $f = \sum_{k=0}^{n} a_k x^k$. Then

$$D^{-1}(f) = c + \sum_{k=0}^{n} \frac{a_k}{k+1} x^{k+1}$$

for any $c \in F$. So $f \in F[x]$ has precisely $|F|$ elements in the preimage.

$\square$

15 Polynomial Rings

Definition 15.1. *Let R be a ring.*
A polynomial over R is

$$\sum_{k=0}^{n} a_k x^k$$

where $a_k \in R$ and $n \geq 0$.
The set of all such polynomials is denoted $R[x]$, and is called the polynomial ring over R.

Two polynomials $\sum_{k=0}^{n} a_k x^k$ and $\sum_{k=0}^{n} b_k x^k$ are equal iff $a_k = b_k, \forall k \in [n]$.

With addition, they are very nicely behaved:

$$\sum_{k=0}^{n} a_k x^k + \sum_{k=0}^{n} b_k x^k = \sum_{k=0}^{n} (a_k + b_k) x^k$$

while multiplication looks like:

$$\sum_{k=0}^{n} a_k x^k \cdot \sum_{k=0}^{n} b_k x^k = \sum_{k} \left(\sum_{i+j=k} a_i b_j \right) x^k$$

Notably, if R is a ring then $R[x]$ is a ring. Furthermore commutativity, identity and being and integral domain carry over. Note that not all properties transfer, e.g. being a field does not always carry over.

15.1 Degree of a Polynomial

Definition 15.2. *Let*

$$f(x) = a_n x^n + \cdots + a_0$$

*with $a_n \neq 0$. The **degree** of f is*

$$\text{deg}(f) = n.$$

The degree of the zero polynomial is defined to be $-\infty$.

If R is an integral domain then

$$\text{deg}(fg) = \text{deg}(f) + \text{deg}(g).$$

15.2 Units in Polynomial Ring

If R is an integral domain, the only units in $R[x]$ are the constant polynomials whose coefficients are units in R.

For example, the only units in $\mathbb{Z}[x]$ are ± 1.

15.3 Evaluation Homomorphisms

For $c \in R$ define

$$\varphi_c : R[x] \to R$$

by

$$\varphi_c(f(x)) = f(c).$$

This map is a ring homomorphism and is called the **evaluation homomorphism**.

15.4 Roots

Definition 15.3. *Let $f(x) \in R[x]$ and $c \in R$. We say c is a **root** of f if*

$$f(c) = 0.$$

Theorem 15.1 (Factor Theorem). *If c is a root of $f(x)$ in R, then*

$$(x - c) \mid f(x).$$

If R is an integral domain and $\deg(f) = n$, then f has at most n roots in R.

16 Polynomial Factorisation

Polynomial arithmetic behaves especially well over fields.

Theorem 16.1 (Division Algorithm). *Let F be a field and $f, g \in F[x]$ with $g \neq 0$. Then there exist unique polynomials $q, r \in F[x]$ such that*

$$f(x) = q(x)g(x) + r(x)$$

where either $r = 0$ or

$$\deg(r) < \deg(g).$$

This allows us to compute greatest common divisors of polynomials using the Euclidean algorithm.

Definition 16.1. *A polynomial* $f(x) \in F[x]$ *is* ***irreducible*** *if*

- deg$(f) \geq 1$
- *whenever* $f = gh$ *with* $g, h \in F[x]$*, one of* g, h *is a unit.*

Theorem 16.2. *If* F *is a field then* $F[x]$ *is a unique factorization domain. In particular, every polynomial can be written as a product of irreducible polynomials, uniquely up to order and multiplication by units.*

16.1 Useful Irreducibility Criterion

If $f(x) \in F[x]$ has degree 2 or 3, then

$$f \text{ is irreducible} \quad \Longleftrightarrow \quad f \text{ has no root in } F.$$

Thus to test irreducibility of such polynomials it suffices to check whether they have roots.

17 Problems from week of March 22

Problem 17.1 (Problem 23.9 Fraleigh's p.218). *The polynomial x^4+4 can be factored into linear factors in $\mathbb{Z}_5[x]$. Find this factorization.*

Solution. We work with

$$f(x) = x^4 + 4 = (x^2 + 2x + 2)(x^2 - 2x + 2)$$

We see that

$$x^2 + 2x + 2 \equiv x^2 + 2x - 3 = (x + 3)(x - 1) \equiv (x + 3)(x + 4)$$

and

$$x^2 - 2x + 2 \equiv x^2 + 3x + 2 = (x + 2)(x + 1)$$

so

$$f(x) \equiv (x + 4)(x + 3)(x + 2)(x + 1)$$

□

Problem 17.2 (Problem 23.14 Fraleigh's p.218). *Show that $f(x) = x^2 + 8x - 2$ is irreducible over $\mathbb{Q}$. Is $f(x)$ irreducible over $\mathbb{R}$?*

Proof. We consider $f(x) \in \mathbb{Q}[x]$.

$$f(x) = x^2 + 8x - 2$$

Since $f(x)$ is degree 2 it is irreducible in $\mathbb{Q}$ if and only if it has no root in $\mathbb{Q}$.

$$\begin{aligned} f(x) &= 0 \\ x^2 + 8x - 2 &= 0 \\ (x + 4 - 3\sqrt{2})(x + 4 + 4\sqrt{2}) &= 0 \end{aligned}$$

$\sqrt{2} \notin \mathbb{Q}[x]$ so $f(x)$ is irreducible in $\mathbb{Q}$.

As $\sqrt{2} \in \mathbb{R}$ and $f(x)$ is degree 2 we conclude that $f(x)$ is reducible in $\mathbb{R}[x]$. □

Problem 17.3 (Problems 18 & 19 Fraleigh's p.219). *Determine whether the given polynomials satisfy the Eisenstein criterion for irreducibility over $\mathbb{Q}$.*

$$f(x) = x^2 - 12$$

$$g(x) = 8x^3 + 6x^2 - 9x + 24$$

Solution. Recall that for some polynomial

$$\sum_{k=0}^{n} a_k x^k$$

If there exists a prime p which divdes $a_0, \dots, a_{n-1}$ and doesn't divide a_n and p^2 doesn't divide a_0, then it is irreducible over the rationals.

For

$$f(x) = x^2 - 12$$

we have $a_0 = -12$, $a_1 = a_n = 1$. No prime number divides 1 so that condition is fulfilled. Then we see that $3|12$, but 3^2 does not divide -12. We conclude that $f(x)$ is irreducible in $\mathbb{Q}[x]$.

Next look at

$$g(x)$$

Observe that for $p = 3$ we have

$$p \mid 24$$
$$p \mid -9$$
$$p \mid 6$$

but p does not divide 8 and $p^2 = 9$ does not divide 24 so $g(x)$ is irreducible over the rationals. □

Problem 17.4 (Problem 23.34 Fraleigh's p.219). *Show that for a prime p, the polynomial $x^p + a$ in $\mathbb{Z}_p[x]$ is not irreducible for any $a \in \mathbb{Z}_p$*

Proof. Let p be prime and

$$x^p + a \in \mathbb{Z}_p[x]$$

By Fermat's Little Theorem $a = a^p$. Furthermore observe that

$$x^p + a = x^p + a^p = (x + a)^p$$

So $x^p + a$ is reducible to

$$\prod_{[p]} (x + a) \in \mathbb{Z}_p[x]$$

□

Problem 17.5 (Problem 23.35 Fraleigh's p.219). *If F is a field and $a \neq 0$ is a zero of $f(x) = a_0 + a_1 x + \cdots x_n x^n$ in $F[x]$, show that*

$1/a$ *is a zero of* $a_n + a_{n-1}x + \cdots + a_0x^n$*.*

"Thinking out loud" proof. Let F be a field and let

$$f(x) := \sum_{i=0}^{n} a_i x^i$$

be a polynomial in $F[x]$. Assume $a \neq 0$ is a root of f, i.e.

$$f(a) = \sum_{i=0}^{n} a_i a^i = 0.$$

Consider the expression

$$\frac{f(x)}{x^n} = \sum_{i=0}^{n} a_i x^{i-n} = a_0x^{-n} + a_1x^{1-n} + \cdots + a_{n-1}x^{-1} + a_n.$$

This is not a polynomial in $F[x]$, but it suggests reversing the powers of x.

To obtain a genuine polynomial, define

$$g(x) := x^n f\left(\frac{1}{x}\right).$$

Then

$$\begin{aligned} g(x) &= x^n \sum_{i=0}^{n} a_i \left(\frac{1}{x}\right)^i \\ &= \sum_{i=0}^{n} a_i x^{n-i} \\ &= a_0x^n + a_1x^{n-1} + \cdots + a_{n-1}x + a_n. \end{aligned}$$

Thus $g(x) \in F[x]$ and it is the polynomial obtained by reversing the coefficients of f.

Now evaluate g at $1/a$:

$$\begin{aligned} g\left(\frac{1}{a}\right) &= \left(\frac{1}{a}\right)^n f(a) \\ &= \frac{f(a)}{a^n} \\ &= 0. \end{aligned}$$

Hence $1/a$ is a root of g whenever $a \neq 0$ is a root of f. $\square$

18 Homomorphisms of rings, Prime/maximal ideals

Definition 18.1. *A* ***ring homomorphism***

$$\phi : R \to S$$

between rings satisfies

$$\phi(a+b) = \phi(a) + \phi(b)$$
$$\phi(ab) = \phi(a)\phi(b)$$

and typically

$$\phi(1_R) = 1_S$$

Theorem 18.1. *Let R, S be rings and $\phi : R \to S$ a homomorphism. Then* $\ker \phi$ *is an ideal of R.*

Proof. Let $a, b \in \ker \phi$.

$$\phi(a-b) = 0$$

so closed under subtraction. For any $r \in R$ we have

$$\phi(ra) = \phi(r)0 = 0$$

so $ra \in \ker \phi$.

Therefore kernels behave like normal subgroups. □

Theorem 18.2 (First Isomorphism Theorem for Rings)**.** *If*

$$\phi : R \to S$$

then

$$R/\ker \phi \cong \operatorname{Im} \phi$$

meaning every homomorphic image of a ring is a quotient by an ideal.

Example 18.1. *Take*

$$\phi : \mathbb{Z} \to \mathbb{Z}_n$$

as

$$\phi(k) = k \mod n$$

The kernel is then

$$\ker \phi = n\mathbb{Z}$$

so

$$\mathbb{Z}/n\mathbb{Z} \cong \mathbb{Z}_n$$

Definition 18.2. *If I is an ideal of a ring R we define*

$$R/I$$

just like quotient groups, with operations

$$(a+I)+(b+I)=(a+b)+I$$
$$(a+I)(b+I)=ab+I$$

18.1 Prime Ideals

Definition 18.3. *An ideal $P \subset R$ is **prime** if*

$$ab \in P \Rightarrow a \in P \text{ or } b \in P$$

Example 18.2. *In $\mathbb{Z}$ all ideals are (n) and the prime ideals are (p) where p is a prime number. If $a, b \in (p)$ then $p|ab$ so $p|a$ or $p|b$.*

Theorem 18.3. *Suppose $P \subset R$ is an ideal. Then*

$$P \text{ prime} \Leftrightarrow R/P \text{ is an integral domain}$$

18.2 Maximal Ideals

Definition 18.4. *An ideal $M \subset R$ is **maximal** if $M \neq R$ and there is no ideal strictly between M and R, meaning*

$$M \subset I \subset R$$

implies $M = I$ or $R = I$.

Theorem 18.4. *Let $M \subset R$ be an ideal.*
Then

$$M \text{ maximal} \Leftrightarrow R/M \text{ is a field}$$

An important note is that maximal $\Rightarrow$ prime, but not necessarily the other way around, so being maximal is a strictly stronger condition. This is easy to see since a field is also an integral domain, but not vice versa.

19 Problems problems and more problems

Problem 19.1 (Problem 26.12 Fraleigh's p.244). *Give an example to show that a factor ring of an integral domain may be a field.*

Solution. Recalling that an integral domain is a nonzero, commutative, unital ring, we give the following example.

Consider the integral domain $\mathbb{Z}$. Take some prime $p \in \mathbb{Z}$ and examine the factor ring $\mathbb{Z}/\langle p \rangle$. It is known that

$$\mathbb{Z}/\langle p \rangle \cong \mathbb{Z}_p$$

The integers modulo a prime is a field. □

Problem 19.2 (Problem 26.13 Fraleigh's p.244). *Give an example to show that a factor ring of an integral domain may have* 0 *divisors.*

Solution. Consider again the integers $\mathbb{Z}$. Take the ideal generated by 6

$$\langle 6 \rangle = \{\dots, -6, 0, 6, \dots\}$$

Notice that

$$\mathbb{Z}/\langle 6 \rangle \cong \mathbb{Z}_6$$

and $2, 3 \in \mathbb{Z}_6$, but

$$2 \cdot 3 \equiv 0$$

□

Problem 19.3 (Problem 26.25 Fraleigh's p.244). *Show that if* R *is a ring with unity and* N *is an ideal of* R *such that* $N \neq R$, *then* R/N *is a ring with unity.*

Proof. Let R be a unital ring with ideal $N \neq R$.

We study the quotient ring

$$R/N \neq \{N\}$$

Let $(a + N) \in R/N$. Consider $(1 + N) \in R/N$ Then

$$(a + N)(1 + N) = (a \cdot 1) + N = (1 \cdot a) + N = (1 + N)(a + N)$$

□

Problem 19.4 (Problem 26.26 Fraleigh's p.245). *Let* R *be a commutative ring and let* $a \in R$. *Show that* $I_a = \{x \in R \mid ax = 0\}$ *is an* *ideal of R.*

Proof. We consider, for some arbitrary element of a commutative ring R, the set of right (and correspondingly left) zero divisors of a

$$I_a = \{x \in R \mid ax = 0\}$$

We know $I_a \neq \emptyset$ as $a \cdot 0 = 0$. Let $b, c \in I_a$. Then $ab = ac = 0$, so

$$a(b - c) = ab - ac = 0 - 0 = 0$$

so $b - c \in I_a$. Therefore $(I_a, +) \leq (R, +)$. Let $r \in R$. As stated $ab = 0$ and furthermore

$$a(rb) = a(br) = (ab)r = 0r = 0$$

so $rb \in I_a$. Thus I_a is an ideal of R. □

Problem 19.5 (Problem 26.30 Fraleigh's p.245). *An element a of a ring R is nilpotent if $a^n = 0$ for some $n \in \mathbb{Z}^+$. Show that the collection of all nilpotent elements in a commutative ring R is an ideal, the nilradical of R.*

Proof. Let R be a commutative ring and let $A \subseteq R$ be the set

$$A = \{a \in R \mid \exists n \in \mathbb{Z}^+ : a^n = 0\}$$

$A \neq \emptyset$ as $0^1 = 0$. Let $a, b \in A$. Then there is some $m, n \in \mathbb{Z}^+$ such that

$$a^m = 0 \text{ and } b^n = 0$$

Let $k = m + n \in \mathbb{Z}^+$.

$$\begin{aligned}(a - b)^k &= \sum_{i=0}^{k} \binom{k}{i} a^i (-b)^{k-i} \\ &= \sum_{i=0}^{m+n} \binom{m+n}{i} a^i (-b)^{k-i} \\ &= \sum_{i=m+1}^{m+n} \binom{m+n}{i} a^i (-b)^{k-i} \\ &\quad + \sum_{i=0}^{m} \binom{m+n}{i} a^i (-b)^{k-i}\end{aligned}$$

Notice that $\sum_{i=m+1}^{m+n} \binom{m+n}{i} a^i (-b)^{k-i}$ has $i \geq m + 1$ for every term so $a^i = 0$ for all terms. Hence every term is 0. For $\sum_{i=0}^{m} \binom{m+n}{i} a^i (-b)^{k-i}$ we have $k - i \geq n$ so $(-b)^{k-i} = 0$ for every term. Therefore

$$\sum_{i=m+1}^{m+n} \binom{m+n}{i} a^i (-b)^{k-i} + \sum_{i=0}^{m} \binom{m+n}{i} a^i (-b)^{k-i} = 0 + 0 = 0$$

Therefore $(a-b) \in A$. Now let $r \in R$ and observe that

$$(ra)^m = r^m a^m = r^m \cdot 0 = 0$$

so $ra \in A$. Thus A constitutes an ideal of R. $\square$

Problem 19.6 (Problem 26.34 Fraleigh's p.245). *Let R be a commutative ring and N an ideal of R. Show that the set $\sqrt{N}$ of all $a \in R$, such that $a^n \in N$ for some $n \geq \mathbb{Z}^+$, is an ideal of R, the **radical ideal** of N.*

Proof. Let N be an ideal of some commutative ring R.

Let

$$\sqrt{N} = \{a \in R \mid \exists n \in \mathbb{Z}^+ \text{ s.t. } a^n \in N\}$$

First, observe that $\sqrt{N} \neq \emptyset$ as $0 = 0^1 \in N$. Next, let $a, b \in \sqrt{N}$. We show that $(a-b)^k \in N$ for some k, i.e. $a - b \in \sqrt{N}$. By assumption there exists $m, n \in \mathbb{Z}^+$ so that $a^m, b^n \in N$. Let $k = m + n$. Then

$$\begin{aligned}
(a-b)^k &= \sum_{i=0}^{k} \binom{k}{i} a^i (-b)^{k-i} \\
&= \sum_{i=0}^{m+n} \binom{m+n}{i} a^i (-b)^{m+n-i} \\
&= \sum_{i=m+1}^{m+n} \binom{m+n}{i} a^i (-b)^{m+n-i} \\
&\quad + \sum_{i=0}^{m} \binom{m+n}{i} a^i (-b)^{m+n-i}
\end{aligned}$$

The sum $\sum_{i=m+1}^{m+n} \binom{m+n}{i} a^i (-b)^{m+n-i}$ has $i \geq m+1$ so each term is of the form $a^m (a \cdots a)(-b)^{m+n-i}$. By assumption $a^m \in N$ and as R is commutative we can rearange terms and see that since N is an ideal of R and $a \cdots a(-b)^{m+n-i} \in R$ we have that each term is in N by the multiplication property of the ideal. As N is a group under the addition, the sum of these terms is in N. Apply this same argument to the sum from $i = 0$ to m, observing that $m+n-i \geq n$, to find that $\sum_{i=0}^{m} \binom{m+n}{i} a^i (-b)^{m+n-i} \in N$. Both sums are in N so

$$(a-b)^k \in N$$

i.e. $a - b \in \sqrt{N}$.

Lastly let $r \in R$ and see that

$$(ra)^m = r^m a^m \in N$$

as N is an ideal. Thus $ra \in \sqrt{N}$.

Therefore $\sqrt{N}$ is an ideal of R. $\square$

Problem 19.7 (Problem 27.5 Fraleigh's p.252). *Find all $c \in \mathbb{Z}_3$ such that*

$$\mathbb{Z}_3[x]/\langle x^2 + c\rangle$$

is a field.

Proof. Since $\mathbb{Z}_3$ is a field, the quotient

$$\mathbb{Z}_3[x]/\langle x^2 + c\rangle$$

is a field if and only if $x^2 + c$ is irreducible in $\mathbb{Z}_3[x]$.

The elements of $\mathbb{Z}_3$ are $0, 1, 2$. We check each case.

If $c = 0$, then x^2 is reducible since $x^2 = x \cdot x$.

If $c = 1$, we check whether $x^2 + 1$ has a root in $\mathbb{Z}_3$. The squares in $\mathbb{Z}_3$ are

$$0^2 = 0, \quad 1^2 = 1, \quad 2^2 = 1.$$

Thus x^2 never equals 2, so $x^2 + 1$ has no root and is irreducible.

If $c = 2$, then

$$x^2 + 2 = x^2 - 1 = (x - 1)(x + 1),$$

so it is reducible.

Therefore the quotient ring is a field only when $c = 1$. □

Problem 19.8 (Problem 27.15/16 Fraleigh's p.253). *Find a maximal ideal of $\mathbb{Z} \times \mathbb{Z}$.*
Find a prime ideal of $\mathbb{Z}^2$ which is not maximal.

Solution. We want to find the the proper ideal $M \subsetneq \mathbb{Z} \times \mathbb{Z}$ such that for any ideal I, if $M \subseteq I \subseteq \mathbb{Z} \times \mathbb{Z}$ then $I = \mathbb{Z} \times \mathbb{Z}$ or $I = M$.

The maximal ideals of $\mathbb{Z}^2$ is $\langle p\rangle \times \mathbb{Z}$ and $\mathbb{Z} \times \langle p\rangle$ where p is some prime. $\langle p\rangle^2$ does not work as the quotient would be $\mathbb{Z}_p^2$. Both factors are nontrivial so this is not a field. □

Problem 19.9 (Problem 18 Fraleigh's p.253). *Is $\mathbb{Q}[x]/\langle x^2 - 5x + 6\rangle$ a field? Why?*

Proof. Recall that the ideal generated by $x^2 - 5x + 6$ is

$$\langle x^2 - 5x + 6\rangle = \{f(x)(x^2 - 5x + 6) : f(x) \in \mathbb{Q}[x]\}$$

We apply a theorem

$$\mathbb{Q}/\langle p\rangle \text{ is a field} \Leftrightarrow p \text{ is irreducible over } \mathbb{Q}$$

Notice that $x^2-5x+6=(x-2)(x-3)$ and $x-2, x-3 \in \mathbb{Q}[x]$, so

$$\mathbb{Q}[x]/\langle x^2-5x+6\rangle$$

is **not** a field. $\square$

Problem 19.10 (Problem 19 Fraleigh's p.253). *Is* $\mathbb{Q}[x]/\langle x^2-6x+6\rangle$ *a field? Why?*

Proof. Notice that

$$x^2-6x+6=(x-(3-\sqrt{3}))(x-(3+\sqrt{3}))$$

Notice that $\sqrt{3} \notin \mathbb{Q}$, i.e. irreducible since the polynomial is a quadratic, so this quotient ring is a field. ($\cong \mathbb{Q}(\sqrt{3})$) $\square$

20 Tackling some extension field problems early

Problem 20.1 (Problem 29.1/3/4 Fraleigh's p.272). *For each following α, show that the given number $\alpha \in \mathbb{C}$ is algebraic over $\mathbb{Q}$ by finding $f(x) \in \mathbb{Q}$ such that $f(\alpha) = 0$.*

$$\alpha = 1 + \sqrt{2}$$
$$\alpha = 1 + i$$
$$\alpha = \sqrt{1 + \sqrt[3]{2}}$$

Solutions. We begin by considering $\alpha = 1 + \sqrt{2}$. Let $x = 1 + \sqrt{2}$. Then

$$\begin{aligned} x &= 1 + \sqrt{2} \\ \sqrt{2} &= x - 1 \\ 2 &= (x-1)^2 \\ x^2 - 2x + 1 &= 2 \\ x^2 - 2x - 1 &= 0 \end{aligned}$$

Thus $x^2 - 2x - 1 \in \mathbb{Q}$ is zero for $x = \alpha = 1 + \sqrt{2}$, hence $1 + \sqrt{2}$ is algebraic over $\mathbb{Q}$.

Next we look at $\alpha = 1 + i$. Set $x = 1 + i$.

$$\begin{aligned} i &= x - 1 \\ -1 &= (x-1)^2 \\ -1 &= x^2 - 2x + 1 \\ x^2 - 2x + 2 &= 0 \end{aligned}$$

$f(x) = x^2 - 2x + 2 \in \mathbb{Q}[x]$ and $f(\alpha) = 0$ so $\alpha = 1 + i$ is algebraic over $\mathbb{Q}$.

Lastly let $\alpha = \sqrt{1 + \sqrt[3]{2}}$.

$$\begin{aligned} x &= \sqrt{1 + \sqrt[3]{2}} \\ x^2 &= 1 + 2^{1/3} \\ x^2 - 1 &= 2^{1/3} \\ (x^2 - 1)^3 &= 2 \\ (x^2 - 1)^3 - 2 &= 0 \end{aligned}$$

As above ... α is algebraic over $\mathbb{Q}$. □

Problem 20.2 (Problem 29.18 Fraleigh's p.272). *a. Show that x^2+1 is irreducible in $\mathbb{Z}_3[x]$.*
b. Let α be a zero of x^2+1 in an extension field of $\mathbb{Z}_3$. As in Example 29.19, give the multiplication and additions tables for the nine elements of $\mathbb{Z}_3[\alpha]$, written in the order $0, 1, 2, \alpha, 2\alpha, 3\alpha, 1+\alpha, 1+2\alpha, 2+\alpha$ and $2+2\alpha$.

Proof of (a). Let $f(x) = x^2+1$. $f(x)$ is a classical introductory example to motivate the complex numbers, having $\alpha = \pm i \in \mathbb{C} \setminus \mathbb{R}$ as zero.

As f is a degree 2 polynomial with no zeros in $\mathbb{Z}_3$ we conclude that it is irreducible in $\mathbb{Z}_3[x]$. $\square$

21 Trying an earlier exam

Problem 21.1 (Problem 1a, Exam 2023). *Prove that the set*

$$S = \{2^n \mid n \in \mathbb{Z}\}$$

equipped with the usual operation of multiplication is a group.

Proof. It suffices to show that $(S, \cdot)$ is isomorphic to a known group. Let $\phi : \mathbb{Z} \to S$ be defined by

$$\phi(n) = 2^n$$

It is clear that ϕ is a bijection as the preimage of 2^n is n and $2^n = 2^m$ iff $n = m$.

Lastly we show ϕ to be a homomorphism.

$$\begin{aligned} \phi(n+m) &= 2^{n+m} \\ &= 2^n \cdot 2^m \\ &= \phi(n)\phi(m) \end{aligned}$$

Thus $(S, \cdot)$ is isomorphic to $(\mathbb{Z}, +)$ which is a known group. □

Problem 21.2 (Problem 1b, Exam 2023). *List all finite abelian groups of order* 20*, up to isomorphism.*

Solution. The finite abelian groups of order 20 are trivially finitely generated so the fundamental theorem of finitely generated abelian groups applies.

I.e.

$$G \cong \left(\prod_{i=1}^{n} \mathbb{Z}_{p_i^{e_i}} \right) \times \mathbb{Z}^k$$

We are asked to find those of order 20 (finite) so

$$G \cong \prod_{i=1}^{n} \mathbb{Z}_{p_i^{e_i}}$$

where $\prod_{i=1}^{n} p_i^{e_i} = 20$.

$20 = 4 \cdot 5 = 2 \cdot 2 \cdot 5$. Thus there exists exactly two abelian groups of order 20, namely

$$\mathbb{Z}_4 \times \mathbb{Z}_5$$

and

$$\mathbb{Z}_2 \times \mathbb{Z}_2 \times \mathbb{Z}_5$$

□

Problem 21.3 (Problem 1c, Exam 2023). *Let A_4 be the alternating group on 4 letters, that is the subgroup of S_4 consisting of even permutations. Find elements $\sigma_1, \sigma_2, \sigma_3 \in A_4$ such that σ_i has order i (for $i = 1, 2, 3$).*

Solution. σ_1 is the trivial permutation, i.e. the one which sends

$$1 \mapsto 1, 2 \mapsto 2, \text{ etc.}$$

Next a candidate for σ_2 is any permutation which is the composition of two transpositions, e.g.

$$(1\ \ 2)(3\ \ 4)$$

Notice that

$$(1\ \ 2)(3\ \ 4) \circ (1\ \ 2)(3\ \ 4)$$

sends

$$1 \to 2 \to 1, 2 \to 1 \to 2, 3 \to 4 \to 3, 4 \to 3 \to 4$$

So $(1\ \ 2)(3\ \ 4)$ has order 2.

An exampele of σ_3 is something like

$$(1\ \ 2\ \ 3) = (1\ \ 3)(1\ \ 2)$$

□

Problem 21.4 (Problem 1d, Exam 2023). *Prove that A_4 is not abelian.*

Proof. Observe that $(1\ \ 2)(3\ \ 4)$ and $(1\ \ 3)(1\ \ 2)$ do not commute.

$$(1\ \ 2)(3\ \ 4) \circ (1\ \ 3)(1\ \ 2)$$

sends

$$\begin{gathered} 1 \mapsto 2 \mapsto 1 \\ 2 \mapsto 3 \mapsto 4 \\ 3 \mapsto 1 \mapsto 2 \\ 4 \mapsto 4 \mapsto 3 \end{gathered}$$

So

$$(1\ \ 2)(3\ \ 4) \circ (1\ \ 3)(1\ \ 2) = (2\ \ 4\ \ 3)$$

while

$$(1\ \ 3)(1\ \ 2) \circ (1\ \ 2)(3\ \ 4)$$

sends

$$3 \mapsto 4 \mapsto 4$$

So we can already see that

$$(1\ \ 3)(1\ \ 2) \circ (1\ \ 2)(3\ \ 4) \neq (1\ \ 2)(3\ \ 4) \circ (1\ \ 3)(1\ \ 2)$$

Hence A_4 is not abelian. □

Problem 21.5 (Problem 2a, Exam 2023). *Determine the 0-divisors in the ring $\mathbb{Z}_{10}$.*

Solution. $\mathbb{Z}_{10}$ is commutative so left- and right divisors coincide.

$$
\begin{aligned}
(2)(5) &= 10 \equiv 0 \\
(4)(5) &= 20 \equiv 0 \\
(6)(5) &= 30 \equiv 0 \\
(8)(5) &= 40 \equiv 0
\end{aligned}
$$

so the 0-divisrors in $\mathbb{Z}_{10}$ are 5 with any power of 2, which just so happens to be the nonzero elements of $\mathbb{Z}_{10}$ which are not coprime to $\mathbb{Z}_{10}$. □

Problem 21.6 (Problem 2b, Exam 2023). *Let R be the ring $\mathbb{Z}\times\mathbb{Z}$, where addition and multiplication is given by, for $(a, b), (c, d) \in \mathbb{Z} \times \mathbb{Z}$, setting*

$$
\begin{aligned}
(a, b) + (c, d) &= (a + c, b + d) \\
(a, b)(c, d) &= (ac, bd)
\end{aligned}
$$

Let

$$I = \{(a, 0) | a \in \mathbb{Z}\} \subset R$$

Prove that I is an ideal of R.

Proof. We show that I is a subgroup of $(R, +)$ with the proper behavior w.r.t left-multiplication.

First we begin by observing that $(0, 0) \in I \neq \emptyset$. Let $(a, 0), (b, 0) \in I$. Then we see that

$$
\begin{aligned}
(a, 0) - (b, 0) &= (a, 0) + (-b, 0) \\
&= (a - b, 0 + 0) \\
&= (a - b, 0) \in I
\end{aligned}
$$

So I is a subgroup of the additive group of R.

Lastly let $(c, d) \in R$. Then

$$
\begin{aligned}
(c, d)(a, 0) &= (ca, d0) \\
&= (ca, 0) \in I
\end{aligned}
$$

Thus I is an ideal of R. □

Problem 21.7 (Problem 2c, Exam 2023). *Prove that I is a prime ideal and that I is not a maximal ideal.*

Proof. It is clear that $I \neq R$ as $(1,1) \in R \setminus I$.

Now suppose $(ab, 0) \in I$. Then

$$(ab, 0) = (a, c)(b, d)$$

with $a, b, c, d \in \mathbb{Z}$ and $cd = 0$. In $\mathbb{Z}$ there are no zero-divisors, thus

$$c = 0 \text{ or } d = 0$$

and hence $(a, c) \in I$ or $(b, d) \in I$.

We show that I is not maximal by giving some ideal J so that

$$I \subseteq J \subseteq R$$

with $I \neq J$ and $J \neq R$.

Let J be the set of $(a, b) \in R$ so that b is even. Formally

$$J = \{(a, b) \in R \mid b = 2n \text{ for some } n \in \mathbb{Z}\}$$

It is trival to check that this is an ideal and that

$$I \subsetneq J \subsetneq R$$

$\square$

Problem 21.8 (Problem 3a, Exam 2023). *Let p be a prime number, and let $f(x) = x^p - x \in \mathbb{Z}_p[x]$. Determine (with justification) the zeros of $f(x)$ in $\mathbb{Z}_p[x]$.*

Solution. The zeros are all numbers in $\mathbb{Z}_p$. We can apply Fermat's little theorem

$$a^p \equiv a \mod p$$

to get that

$$f(x) = x^p - x^p = x - x = 0$$

□

Problem 21.9 (Problem 3b, Exam 2023). *Consider $x^2+1 \in \mathbb{Z}_3[x]$. Prove that the ring $\mathbb{Z}_3[x]/\langle x^2+1\rangle$ is a field, and determine its number of elements.*

Proof. We use the following theorem:

For some field F and $f \in F[x]$

$$F[x]/\langle f(x)\rangle$$

is a field if and only if $f(x)$ is irreducible in $F[x]$. Since $f(x) = x^2+1$ is a degree 2 polynomial we see that f is irreducible in $\mathbb{Z}_3[x]$ if and only if f has no zeros in $\mathbb{Z}_3[x]$.

It is known that $f(x)$ is irreducible since it is the motivating introductory example for the complex numbers with solution $x = \pm i$.

We quickly check that $f(0) \neq 0$, $f(1) \neq 0$ and $f(2) \neq 0$ to confirm this.

As 3 is prime we see that $\mathbb{Z}_3$ is a field.

We conclude therefore that

$$\mathbb{Z}_3[x]/\langle x^2+1\rangle$$

is a field. □

22 Extension fields & Vector spaces

Definition 22.1. *A* ***field extension*** *of a field F is a field K with $F \subseteq K$.*
It is common to write

$$K/F$$

Some basic examples of the above include $\mathbb{Q} \subseteq \mathbb{R}$, $\mathbb{Q} \subseteq \mathbb{C}$.

As we have seen earlier,

$$\mathbb{Z}_3 \subseteq \mathbb{Z}_3[x]/\langle x^2+1\rangle \cong \mathbb{Z}_3(i)$$

Definition 22.2. *Given a field F and an element $\alpha \notin F$, define*

$$F(\alpha)$$

to be the smallest field containing both F and α.

Theorem 22.1. *If α satisfies a polynomial over F, then*

$$F(\alpha) \cong F[x]/\langle f(x)\rangle$$

where f is the minimal polynomial.
In other words, if α is algebraic over F then this holds.

Definition 22.3. *The minimal polynomial of α over F is the monic polynomial of smallest degree in $F[x]$ such that α is a zero of said polynomial.*

Definition 22.4. *The degree of an extension $[K : F]$ is the dimension of K as a vector space over F.*

Example 22.1. $[\mathbb{Q}(\sqrt{2}) : \mathbb{Q}] = 2$ *with basis* $\{1, \sqrt{2}\}$.

Theorem 22.2. *If K/F is a field extension then K is a vector space over F.*

Theorem 22.3. *If minimal polynomial has degree n then*

$$[F(\alpha) : F] = n$$

Theorem 22.4 (Tower Law). *If*

$$F \subseteq K \subseteq L$$

then

$$[L : F] = [L : K][K : F]$$

Problem 22.1 (Problem 4a, Exam 2023). *Let* $f(x) = x^5 - 8x^4 + 2x + 2 \in \mathbb{Q}[x]$*. You may use w/o proof that* f *is irreducible.*
Let $\alpha \in \overline{\mathbb{Q}}$ *be a zero of* f*. Compute* $[\mathbb{Q}(\alpha) : \mathbb{Q}]$*, and give a basis for* $\mathbb{Q}(\alpha)$ *as a vector space over* $\mathbb{Q}$*.*

Proof. Since $f(x) \in \mathbb{Q}[x]$ is irreducible and α is a root, f is the minimal polynomial of α over $\mathbb{Q}$.

Therefore,

$$[\mathbb{Q}(\alpha) : \mathbb{Q}] = \deg f = 5.$$

A basis for $\mathbb{Q}(\alpha)$ over $\mathbb{Q}$ is

$$\{1, \alpha, \alpha^2, \alpha^3, \alpha^4\}.$$

□

Problem 22.2 (Problem 4b, Exam 2023). *Let* $\beta \in \mathbb{Q}(\alpha) \setminus \mathbb{Q}$*. Prove that* $\mathbb{Q}(\beta) = \mathbb{Q}(\alpha)$*.*

Proof. $\beta \in \mathbb{Q}(\alpha)$ and $\beta \notin \mathbb{Q}$, by assumption. So

$$\mathbb{Q} \subseteq \mathbb{Q}(\beta) \subseteq \mathbb{Q}(\alpha)$$

We know that

$$[\mathbb{Q}(\alpha) : \mathbb{Q}] = [\mathbb{Q}(\alpha) : \mathbb{Q}(\beta)] \cdot [\mathbb{Q}(\beta) : \mathbb{Q}]$$

5 is prime so one of the factors is 5 and the other is 1. It can't be the case that

$$[\mathbb{Q}(\alpha) : \mathbb{Q}(\beta)] = 5$$

since that would imply $\mathbb{Q}(\beta) = \mathbb{Q}$ which is not the case, by assumption. Hence $\mathbb{Q}(\alpha) = \mathbb{Q}(\beta)$. □

23 Some problems relevant to Fraleigh's sections 29-33

Problem 23.1. *Let $\alpha = \sqrt{2} + \sqrt{5}$. Find the minimal polynomial for α over $\mathbb{Q}$. Then compute degree and identify a basis.*

Solution.

$$\begin{aligned} x &= \sqrt{2} + \sqrt{5} \\ x^2 &= 7 + 2\sqrt{10} \\ x^2 - 7 &= 2\sqrt{10} \\ \frac{x^2-7}{2} &= \sqrt{10} \\ \frac{(x^2-7)(x^2-7)}{4} &= 10 \\ \frac{x^4 - 14x^2 + 49 - 40}{4} &= 0 \\ \frac{x^4 - 14x^2 + 9}{4} &= 0 \\ x^4 - 14x^2 + 9 &= 0 \end{aligned}$$

is the minimal polynomial of α. It is degree r and thus a basis of $Q(\alpha)$ over $\mathbb{Q}$ is $\{1, \alpha, \alpha^2, \alpha^3\}$. □

Problem 23.2. *Give an example of a field extension which is finitely generated and not finite.*

Solution. Consider $\mathbb{C}/\mathbb{R}$. $\mathbb{C}$ is a 2-dimensional vector space over $\mathbb{R}$, but as a field $|\mathbb{C}| = \aleph_1$ and

$$|\mathbb{C} \setminus \mathbb{R}| = \aleph_1$$

□

Problem 23.3. *Let α, β be algebraic over F. Prove*

$$[F(\alpha,\beta) : F] \leq [F(\alpha) : F][F(\beta) : F]$$

and explain when equality is achieved.

Proof. Suppose α, β algebraic over some field F. Then there exists minimal polynomials

$$f, g \in F[x]$$

so that $f(\alpha) = 0$ and $g(\beta) = 0$. With

$$F(\alpha) = \left\{ \sum_{i=0}^{\deg f - 1} c_i \alpha_i \,\middle|\, c_i \in F \right\}$$

and

$$F(\beta) = \left\{ \sum_{i=0}^{\deg g - 1} c_i \beta_i \,\middle|\, c_i \in F \right\}$$

Then

$$[F(\alpha) : F][F(\beta) : F] = (\deg f - 1)(\deg g - 1)$$

It is clear that the two expressions are equal when g is irreducible in $F(\alpha)$, recalling

$$F \subseteq F(\alpha) \subseteq F(\alpha, \beta)$$

A nice example is $\mathbb{Q}(\sqrt{2}, \sqrt{5})$.

$F(\alpha, \beta)$ is an extension field of $F(\alpha)$ so if β and α are independent in their minimal polynomials then we get equality. $\square$

Problem 23.4 (Problem 31.3/5/7 Fraleigh's p.291). *Find the degree and basis for the given field extensions. Be prepared to justify your answers.*

3. $\mathbb{Q}(\sqrt{2}, \sqrt{3}, \sqrt{18})$ *over* $\mathbb{Q}$

5. $\mathbb{Q}(\sqrt{2}, \sqrt[3]{2})$ *over* $\mathbb{Q}$

7. $\mathbb{Q}(\sqrt{2}, \sqrt{3})$ *over* $\mathbb{Q}$

Solution. Notice that

$$\mathbb{Q}(\sqrt{2}, \sqrt{3}, \sqrt{18}) = \mathbb{Q}(\sqrt{2}, \sqrt{3})$$

since $\sqrt{18} = \sqrt{9}\sqrt{2} = 3\sqrt{2} \in \mathbb{Q}(\sqrt{2})$. Now notice that

$$[\mathbb{Q}(\sqrt{2}) : \mathbb{Q}] = 2$$

since $x^2 - 2$ is irreducible over $\mathbb{Q}$. Lastly we conclude that $\sqrt{3} \notin \mathbb{Q}(\sqrt{2})$ since

$$\sqrt{3} = a + b\sqrt{2}$$
$$3 = a^2 + 2ab\sqrt{2} + 2b^2$$

Thus $2ab = 0$ so $a = 0$ or $b = 0$. Both cases work out to show that $a, b \notin \mathbb{Q}$. Therefore

$$[\mathbb{Q}(\sqrt{2}, \sqrt{3}) : \mathbb{Q}(\sqrt{2})] = 2$$

By the tower law:

$$[\mathbb{Q}(\sqrt{2}, \sqrt{3}) : \mathbb{Q}] = 4$$

with a standard basis being $\{1, \sqrt{2}, \sqrt{3}, \sqrt{6}\}$.

Next up we examine $\mathbb{Q}(\sqrt{2}, \sqrt[3]{2})$.

$$\begin{aligned}\sqrt[3]{2} &= a + b\sqrt{2} \\ 2 &= (a+b\sqrt{2})(a^2 + 2b^2 + 2ab\sqrt{2}) \\ 2 &= a^3 + 2ab^2 + 2a^2b\sqrt{2} + a^2b\sqrt{2} + 2b^3\sqrt{2} + 2ab^2\end{aligned}$$

The left side is rational so we require $b = 0$, but then $a = \sqrt[3]{2} \notin \mathbb{Q}(\sqrt{2})$. Hence we conclude that

$$[\mathbb{Q}(\sqrt{2}, \sqrt[3]{2}) : \mathbb{Q}] = 6$$

For problem 7 notice that we have already shown it for problem 3. $\square$

Problem 23.5 (Problem 31.23 Fraleigh's). *Show that if E is a finite extension of a field F and $[E : F]$ is prime, then E is a simple extension of F and, indeed $E = F(\alpha)$ for every $\alpha \in E \setminus F$.*

Proof. We assume E is a finite extension of F and in particular

$$[E : F] = p$$

for some prime p. Let $\alpha \in E \setminus F$ then α is algebraic over F so α has a minimal polynomial over F and

$$[F(\alpha) : F] = \deg m_\alpha(x)$$

Now note that

$$F \subseteq F(\alpha) \subseteq E$$

since $\alpha \in E$. By the tower law:

$$[E : F] = [E : F(\alpha)][F(\alpha) : F] = p$$

so one side of the product must be 1. Since $\alpha \notin F$ it must be the case that

$$[F(\alpha) : F] > 1$$

so $[F(\alpha) : F] = p$, i.e.

$$E = F(\alpha)$$

$\square$

Problem 23.6 (Problem 31.24 Fraleigh's p.292). *Prove that $x^2 - 3$ is irreducible over $\mathbb{Q}(\sqrt[3]{2})$.*

Proof. We can make the simple observation that $\sqrt{3} \notin \mathbb{Q}(\sqrt[3]{2})$. Let $\alpha = \sqrt[3]{2}$. Then

$$\mathbb{Q}(\alpha) = \{a + b\alpha + c\alpha^2 : a, b, c \in \mathbb{Q}\}$$

So if $\sqrt{3} \in \mathbb{Q}(\alpha)$ then

$$\sqrt{3} = a + b\alpha + c\alpha^2$$

Suppose this holds. Since $\mathbb{Q}(\sqrt{3}) \subseteq \mathbb{Q}(\alpha)$ then by the tower law

$$[\mathbb{Q}(\alpha)] = [\mathbb{Q}(\alpha) : \mathbb{Q}(\sqrt{3})] \cdot [\mathbb{Q}(\sqrt{3}) : \mathbb{Q}]$$

But then

$$3 = [\mathbb{Q}(\alpha) : \mathbb{Q}(\sqrt{3})] \cdot 2$$

but 2 does not divide 3 so this is impossible. $\square$

Problem 23.7 (Problem 31.29 Fraleigh's p.292). *Let E be a finite extension of F, and let $p(x) \in F[x]$ be irreducible over F and have degree that is not a divisor of $[E : F]$. Show that $p(x)$ has no zeros in E.*

Proof. Let $\alpha \in E$ be a hypothetical zero of $p(x)$. Since p is irreducible in F and $p(\alpha) = 0$, p must be the minimal polynomial of α.

Thus

$$[F(\alpha) : F] = \deg p(x)$$

Using the tower law

$$[E : F] = [E : F(\alpha)][F(\alpha) : F]$$

But then $\deg p$ divides $[E : F]$ which, by assumption, is not the case. Hence we have a contradiction. $\square$

Problem 23.8 (Problem 31.30 Fraleigh's p.292). *Let E be an extension of F. Let $\alpha \in E$ be algebraic of odd degree over F. Show that α^2 is algebraic of odd degree over F, and $F(\alpha) = F(\alpha^2)$.*

Proof. We fix some $\alpha \in E$ algebraic of odd degree over F. Then

$$[F(\alpha) : F] = 2n + 1, \quad n \in \mathbb{N}$$

Crucially $\alpha \notin F$, by assumption. Now suppose $\alpha^2 \in F$. Then $\alpha^2 \notin F(\alpha) \backslash F$ so $[F(\alpha) : F] = 2$ (since $x^2 - \alpha^2 \in F[x]$ would be a minmal polynomial of α), which contradicts our assumption. Thus $\alpha^2 \notin F$. Now since $\alpha^2 \in F(\alpha)$ we see that α satisfies

$$x^2 - \alpha^2 \in F(\alpha^2)[x]$$

Hence $[F(\alpha) : F(\alpha^2)] \leq 2$. By the tower law

$$[F(\alpha) : F] = [F(\alpha) : F(\alpha^2)][F(\alpha^2) : F]$$

We see that then $[F(\alpha) : F(\alpha^2)]$ must be 1, hence

$$F(\alpha) = F(\alpha^2)$$

$\square$

Problem 23.9. *Let* $\alpha = \sqrt{2} + \sqrt{3}$*.*
Show that α *is algebraic over* $\mathbb{Q}$*.*
Show that $\mathbb{Q}(\alpha) = \mathbb{Q}(\sqrt{2}, \sqrt{3})$*.*
Deduce that $\mathbb{Q}(\sqrt{2}, \sqrt{3})$ *is a simple extension.*

Proof. Clearly $\sqrt{2} + \sqrt{3} \notin \mathbb{Q}$. Now it remains to find a polynomial in $\mathbb{Q}[x]$ for which α is a zero.

$$\begin{aligned} x &= \sqrt{2} + \sqrt{3} \\ x^2 &= 2 + 2\sqrt{2}\sqrt{3} + 3 \\ x^2 - 5 &= 2\sqrt{2}\sqrt{3} \\ \frac{x^2 - 5}{2} &= \sqrt{6} \\ \frac{(x^2-5)(x^2-5)}{4} &= 6 \\ \frac{x^4 - 10x^2 + 25}{4} - 6 &= 0 \\ x^4 - 10x^2 + 1 &= 0 \end{aligned}$$

and $x^4 - 10x^2 + 1 \in \mathbb{Q}[x]$ so α is algebraic over $\mathbb{Q}$.

Thus

$$[\mathbb{Q}(\alpha) : \mathbb{Q}] = [\mathbb{Q}(\sqrt{2}, \sqrt{3}) : \mathbb{Q}] = 4$$

The standard basis for $\mathbb{Q}(\sqrt{2}, \sqrt{3})$ is $\{1, \sqrt{2}, \sqrt{3}, \sqrt{2}\sqrt{3}\}$ which just so happens to be the basis for $\mathbb{Q}(\alpha)$. We can also take some arbitrary element from either extension field and show how to express it in terms of the other to show that they are indeed equal.

Since $\mathbb{Q}(\alpha)$ is simple, it follows from the above observation that

$$\mathbb{Q}(\sqrt{2}, \sqrt{3})$$

is indeed simple. □

24 Some problems from Section 33

Problem 24.1 (Problem 33.4 Fraleigh's). *Find the number of primitive* 8*th roots of unity in* $GF(9)$ *(also known as* $\mathbb{F}_9$*).*

Proof. The multiplicative group $\mathbb{F}_9^x$ has order 8 since 0 is excluded. The multiplicative group of a finite field is cyclic so

$$\mathbb{F}_9^x \cong \mathcal{C}_8$$

The 8th roots of unity are the elements $x \in \mathcal{C}_8$ for which

$$x^8 = 1$$

Since the whole group has order 8, every nonzero element satisfies this. A primitive 8th root of unity has order exactly 8 and the Euler totient function gives

$$\phi(8) = 4$$

□

Problem 24.2 (Problem 33.9 Fraleigh's p.305). *Let $\overline{\mathbb{Z}}_2$ be the algebraic closure of $\mathbb{Z}_2$, and let $\alpha, \beta \in \overline{\mathbb{Z}}_2$ be zeros of $x^3 + x^2 + 1$ and of $x^3 + x + 1$, respectively. Using the results of this section, show that $\mathbb{Z}_2(\alpha) = \mathbb{Z}_2(\beta)$.*

Proof. We begin by showing that the polynomials $f(x) = x^3 + x^2 + 1$ and $g(x) = x^3 + x + 1$ are irreducible over $\mathbb{Z}_2$. These are degree 3 polynomials and $\mathbb{Z}_2$ is a field so they are irreducible if and only if they have no zeros in $\mathbb{Z}_2$.

$$f(0) = 1 \text{ and } f(1) = 1,$$

and

$$g(0) = 1 \text{ and } g(1) = 1.$$

Thus f, g are irreducible over $\mathbb{Z}_2$. Hence

$$[\mathbb{Z}_2(\alpha) : \mathbb{Z}_2] = [\mathbb{Z}_2(\beta) : \mathbb{Z}_2] = 3$$

So both $\mathbb{Z}_2(\alpha)$ and $\mathbb{Z}_2(\beta)$ are field with degree 2^3. Since 2 is prime we can use the finite-field theorem and conclude that these two fields are isomorphic. Furthermore in an algebraic closure of some field $\mathbb{F}_p$, there is exactly one subfield with p^n elements, namely the set of roots of

$$x^{p^n} - x.$$

Thus

$$\mathbb{Z}_2(\alpha) = \mathbb{Z}_2(\beta).$$

□

Problem 24.3 (Problem 33.10 Fraleigh's p.305). *Show that every irreducible polynomial in $\mathbb{Z}_p[x]$ is a dvisor of $x^{p^n} - x$ for some n.*

Proof. Let $\alpha \in \overline{\mathbb{Z}}_p$ be the root of some $f(x) \in \mathbb{Z}_p$, irreducible. Then

$$[\mathbb{Z}_p(\alpha) : \mathbb{Z}_p] = \deg f$$

So $\mathbb{Z}_p(\alpha)$ is a finite field with p^m elements. If a field has p^m elements then $a^{p^m} = a$ for all a in said field. Equivalently every elements is a root of

$$x^{p^m} - x.$$

Then

$$\alpha^{p^m} - \alpha = 0.$$

If a polynomial g is the minimal polynomial of α over $\mathbb{Z}_p$, and α is also a root of another polynomial f then

$$g \mid f$$

$\square$

Problem 24.4 (Problem 3a Exam 2016). *What is an integral domain? Show that $\mathbb{Z}[x]/(x^2 + 1)$ is an integral domain. Is $\mathbb{Z}[x]/(x^2 - 1)$ an integral domain? Give reasons for your answer.*

Solution and Proof. An integral domain is a non-zero ring with multiplicative identity ($1 \neq 0$) that has no zero divisors.

For $f(x) \in F[x]$, $F[x]/\langle f(x)\rangle$ is an integral domain iff $f(x)$ is irreducible in $F[x]$. Since x^2+1 is a degree 2 polynomial it is irreducible if and only if it has no zeros in $\mathbb{Z}$. $f(x) = x^2 + 1$ is famously a motivating example for complex numbers with zeros $\pm i \in \mathbb{C}$. A quick sanity check gives $f(0) = 1$, $f(\pm 1) = 2$ and generally $f(n) = f(-n) < f(n+1) = f(-(n+1))$ so it ceirtainly does not have any zeros in $\mathbb{Z}$ (can be proven with induction or standard techniques if necessary). Thus f is irreducible and so $\mathbb{Z}[x]/\langle f(x)\rangle$ is an integral domain.

Using this same procedure we see that $x^2 - 1$ has zeros $\pm 1 \in \mathbb{Z}$ and is therefore not irreducible so

$$\mathbb{Z}[x]/\langle x^2 - 1\rangle$$

is not an integral domain. $\square$

25 Automorphisms of Field Extensions & Isomorphism Extension

Definition 25.1 (Field Automorphism). *A **field automorphism** of E is a bijection*

$$\sigma : E \to E$$

such that $\forall x, y \in E$:

1. $\sigma(x+y) = \sigma(x) + \sigma(y)$
2. $\sigma(xy) = \sigma(x)\sigma(y)$
3. $\sigma(1) = 1$

If $F \subseteq E$ is a field extension we write

$$\mathrm{Aut}(E/F) := \{\sigma : E \to E \mid \sigma \text{ is a field automorphism and } \sigma|_F = \mathrm{id}_F\}.$$

Proposition 25.1. *A field automorphism σ preserves algebraic structure:*

1. $\sigma(0) = 0$
2. $\sigma(-x) = -\sigma(x)$
3. $\sigma(x^{-1}) = \sigma(x)^{-1}$ *for* $x \neq 0$

Proof Sketch. Straightforward from homomorphism properties. For instance:

$$1 = \sigma(1) = \sigma(xx^{-1}) = \sigma(x)\sigma(x^{-1})$$

and so on. □

Proposition 25.2. $\mathrm{Aut}(E/F)$ *is a group under composition.*

Theorem 25.1. *Let E/F be an extension, and let $\alpha \in E$ be algebraic over F. Let $m_\alpha(x) \in F[x]$ be a minimal polynomial.*
Then for any $\sigma \in \mathrm{Aut}(E/F)$,

$$m_\alpha(\sigma(\alpha)) = 0.$$

Proof. Write $m_\alpha(x) = \sum_{i=0}^{n} a_i x^i$, $a_i \in F$ $(a_n = 1)$.

Since σ fixes F, we have $\sigma(a_i) = a_i$. Then:

$$0 = \sigma(m_\alpha(\alpha)) = \sigma\left(\sum a_i \alpha^i\right) = \sum a_i \sigma(\alpha)^i = m_\alpha(\sigma(\alpha))$$

□

Proposition 25.3. *Let $E = F(\alpha)$, and let $\sigma \in \mathsf{Aut}(E/F)$. Then σ is completely determined by $\sigma(\alpha)$.*

Theorem 25.2. *Let $E = F(\alpha)$, where α is algebraic over F with minimal polynomial $m_\alpha(x)$.*
Then
$$|\mathsf{Aut}(E/F)| \leq \deg m_\alpha.$$

Definition 25.2. *A field embedding*
$$\sigma : F \to E$$
is an injective field homomorphism.

Theorem 25.3. *Let $F \subseteq E = F(\alpha)$, $\sigma : F \to K$ be a field homomorphism, $m_\alpha(x)$ the minimal polynomial of α. Then there exists an extension*
$$\tilde{\sigma} : E \to K$$
with $\tilde{\sigma}(\alpha) = \beta$ if and only if β is a root of $\sigma(m_\alpha)(x)$.

Theorem 25.4 (Isomorphism Extension Theorem)**.** *Let $\sigma : F \to K$ be a field isomorphism and α algebraic over F with minimal polynomial $m_\alpha(x)$.*
Let $\beta \in \overline{K}$ be a root of $\sigma(m_\alpha)(x)$.
Then σ is extends to an isomorphism
$$\tilde{\sigma} : F(\alpha) \to K(\beta)$$
with
$$\tilde{\sigma}(\alpha) = \beta.$$

Theorem 25.5 (Automorphisms of Finite Fields)**.** *Let $E = \mathbb{F}_{p^n}$. Then*
$$\mathsf{Aut}(E/\mathbb{F}_p) = \langle \phi \rangle$$
where
$$\phi(x) = x^p$$
and
$$|\mathsf{Aut}(E/\mathbb{F}_p)| = n.$$

26 Exam Practice

Henceforth these notes will be mostly revision, even though I haven't covered all topics yet. There will also probably be a lot of problems repeated.

Problem 26.1 (1 Exam 2019). *a) Is $\mathbb{Z}_{\geq 0}$ equipped with addition a group? Justify your answer.*

b) What are the abelian groups of order 18 up to isomorphism?

c) What are the cosets of the subgroup $\langle (0,3) \rangle$ in $\mathbb{Z}_2 \times \mathbb{Z}_9$? Describe the quotient

$$\frac{\mathbb{Z}_2 \times \mathbb{Z}_9}{\langle (0,3) \rangle}$$

as a product of finite cyclic groups.

Proof of (a). No it is not a group because there are additive inverses for all elements except 0.

A single counter-example is $1 \in \mathbb{Z}_{\geq 0}$ which has no inverse.

$$1 + x = 0 \Rightarrow x = -1 \notin \mathbb{Z}_{\geq 0}.$$

We can prove it rigorously as follows. 0 is obviously not the additive inverse of 1. Suppose $n \in \mathbb{Z}_{\geq 0}$ is not an inverse of 1. Assume, for a contradiction, that $n+1$ is an inverse of 1. Then

$$\begin{aligned} 1 + (n+1) &= 0 \\ n &= -2 \end{aligned}$$

From this we conclude that the inverse of 1 is the inverse of 2, i.e

$$-1 = -2$$

so $1 = 2$. Hence $n+1$ is not an inverse of 1. By induction no element of $\mathbb{Z}_{\geq 0}$ is an inverse of 1. □

Solution of (b). We use the fundamental theorem of finitely generated abelian groups. 18 is finite so we can simply find all ways of expressing 18 as products of prime powers and we'll be done. We see that

$$18 = 2 \cdot 9 = 2 \cdot 3^2 = 2 \cdot 3 \cdot 3$$

so there are two (up to isomprhism) abelian groups of order 18, namely

$$\mathbb{Z}_2 \times \mathbb{Z}_9 \text{ and } \mathbb{Z}_2 \times \mathbb{Z}_3 \times \mathbb{Z}_3.$$

□

Solution of (c). $\langle(0,3)\rangle = \{(0,0),(0,3),(0,6)\}$. The left and right cosets coincide since this group is abelian and they are precisely

$$\{x+y : y \in \langle(0,3)\rangle\}, \ \ x \in \mathbb{Z}_2 \times \mathbb{Z}_9.$$

Since for (a,b), b is taken mod 9 only b mod 3 matters. So the left cosets are

$$(a,b) + \langle(0,3)\rangle$$

where $a \in \mathbb{Z}_2$ and $b \in \mathbb{Z}_3$. Denote $H := \langle(0,3)\rangle$. That gives us cosets

$$\begin{gathered}(0,0)+H,\ \ (0,1)+H,\ \ (0,2)+H\\ (1,0)+H,\ \ (1,1)+H,\ \ (1,2)+H\end{gathered}$$

i.e. 6 cosets, which is what we expect because the order of $\mathbb{Z}_2 \times \mathbb{Z}_9$ is 18 and $18/|H| = 6$. The quotient groups is abelian and $6 = 2 \cdot 3$ so necessarily

$$\frac{\mathbb{Z}_2 \times \mathbb{Z}_9}{\langle(0,3)\rangle} \cong \mathbb{Z}_2 \times \mathbb{Z}_3.$$

We can also argue by observing the surjective homomorphism $\phi : \mathbb{Z}_2 \times \mathbb{Z}_9 \to \mathbb{Z}_2 \times \mathbb{Z}_3$, $\phi(a,b) = (a, b \mod 3)$ has kernel $\ker\phi = H$ so the result follows by the first isomorphism theorem. □

Problem 26.2 (2 Exam 2019). *a) Let* $\sigma = (2\ \ 6\ \ 1\ \ 4\ \ 3)$ *and* $\tau = (5\ \ 2\ \ 1\ \ 3)$ *be permutations in* S_6. *Express the composition* $\sigma\tau$ *in disjoint cycle notation. What is the order of* $\sigma\tau$*?*

b) Show that if $\sigma \in S_n$ *is a cycle of odd length* $2k+1$, *then* σ^2 *is also a cycle of length* $2k+1$.

c) Let G *be a subgroup of the symmetric group* S_n *such that there exist permutations* $\sigma_2, \dots, \sigma_n \in G$ *satisfying* $\sigma_i(1) = i$ *for* $2 \le i \le n$. *Show that* G *acts* ***transitively*** *on* $\{1, \dots, n\}$. *In other words, show that for each pair* $1 \le i, j \le n$ *there exists a permutation* $\sigma \in G$ *such that* $\sigma(i) = j$.

Solution of (a). σ is $2 \to 6 \to 1 \to 4 \to 3 \to 2$ while τ is $5 \to 2 \to 1 \to 3$. We work out $\sigma\tau$ like this:

$$\begin{gathered}5 \to 2 \to 6 \Rightarrow 5 \to 6\\ 2 \to 1 \to 4 \Rightarrow 2 \to 4\\ 1 \to 3 \to 2 \Rightarrow 1 \to 2\\ 3 \to 5 \to 5 \Rightarrow 3 \to 5\\ 4 \to 4 \to 3 \Rightarrow 4 \to 3\\ 6 \to 6 \to 1 \Rightarrow 6 \to 1\end{gathered}$$

so $\sigma\tau = (1\ \ 2\ \ 4\ \ 3\ \ 5\ \ 6)$. The order of an n-cycle is n so $|\sigma\tau| = 6$. □

Proof of (b). Suppose σ is a cycle of length $2k+1$, i.e.

$$\sigma = (a_1 \ a_2 \ \cdots \ a_{2k+1}).$$

Composing σ^2 sends $a_1 \to a_3$, $a_2 \to a_4$ and so on. In particular $a_i \to a_{(i+2)}$ where $i+2$ is taken mod $2k+1$. Then all odd elements are taken to odd elements, but the $2k+1$-th element is sent to a_2 which is sent to a_4 and so on. Even elements are sent to even elements until $a_{2k} \to a_1$ so

$$\sigma^2 = (a_1 \ a_3 \ \cdots \ a_{2k+1} \ a_2 \ a_4 \ \cdots \ a_{2k})$$

which has length $2k+1$. □

Proof of (c). Since G is a group, each permutation σ_k has an inverse σ_k^{-1} which sends k to 1. Also $e \in G$ so for $1 \leq i, j \leq n$ we can send i to j by sending i to 1 and then 1 to j so

$$(\sigma_j \circ \sigma_i^{-1})(i) = \sigma_j(1) = j.$$

This works well for the case where neither i, j are 1. If i is one then let $\sigma_i := e$ and likewise for j. □

Problem 26.3 (3 Exam 2019). *a) Show that if n is not prime, then $n\mathbb{Z}$ is not a prime ideal of $\mathbb{Z}$.*

b) Show that the quotient

$$\frac{\mathbb{Z}_2[x]}{\langle x^3 + x + 1 \rangle}$$

is a field. How many elements does it contain?

c) Find a polynomial $p(x) \in \mathbb{Z}_2[x]$ such that $\deg(p) \leq 2$ and

$$x^5 + \langle x^3 + x + 1 \rangle = p(x) + \langle x^3 + x + 1 \rangle$$

in $\frac{\mathbb{Z}_2[x]}{\langle x^3+x+1 \rangle}$.

Proof. Suppose n is not prime. Then $n = ab$ for some integers a, b with $1 < a < n$ and $1 < b < n$. Hence

$$ab = n \in n\mathbb{Z}.$$

However, $a \notin n\mathbb{Z}$ and $b \notin n\mathbb{Z}$, since neither a nor b is divisible by n. Thus there exist $a, b \in \mathbb{Z}$ such that $ab \in n\mathbb{Z}$ but $a, b \notin n\mathbb{Z}$. Therefore $n\mathbb{Z}$ is not a prime ideal of $\mathbb{Z}$. □

Proof of (b). We will use the fact that the quotient (in this case) is a field if and only if $f(x) = x^3 + x + 1$ is irreducible over $\mathbb{Z}_2$. Notice that

$$f(0) = f(1) = 1 \neq 0$$

and $f(x)$ is degree 3 hence irreducible over $\mathbb{Z}_2$. Thus the quotient is a field. There are $2^3 = 8$ elements. $\square$

Solution of (c). Since $x^3+x+1 = 0$ we have $x^3 = x+1$. Then $x^5 = x^2x^3 = x^2(x+1) = x^3 + x^2$. Using the relation again we get $x^5 = x^2 + x + 1$. So let $p(x) := x^2 + x + 1$. Then $x^5\langle x^3 + x + 1\rangle = p(x) + \langle x^3 + x + 1\rangle$. $\square$

Problem 26.4 (1 Exam 2022, (Group Theory)).

1. *Let G be a group of order 200. What are the sizes of Sylow p-subgroups of G?*
2. *Let $G = \mathbb{Z}_{12} \times \mathbb{Z}_8$. Find the order of the element $(4, 2) \in G$. Determine the quotient group $G/\langle(4, 2)\rangle$ up to isomorphism.*
3. *Let p be a prime number. Suppose G is a group of order p^n acting on a set X such that p does not divide $|X|$. Show that the action of G must have a fixed point, i.e. there exists an $x \in X$ such that $|\mathcal{O}| = 1$.*

Note: Problem 1 is not relevant to the current curriculum.

Solution of (2). 4 has order 3 in $\mathbb{Z}_{12}$ and 2 has order 4 in $\mathbb{Z}_8$ so the order of $(4, 2)$ in $\mathbb{Z}_{12} \times \mathbb{Z}_8$ is $\operatorname{lcm}(3, 4) = 12$.

Thus there are 12 elements in $\langle(4, 2)\rangle$ which we will use to characterize the quotient group.

G has $12 \cdot 8 = 96$ elements which by Lagrange's theorem gives $96/12 = 8$ elements in the quotient group. Now since the quotient group is a finite abelian group it must be isomorphic to $\mathbb{Z}_8$ or $\mathbb{Z}_2 \times \mathbb{Z}_2$ or $\mathbb{Z}_2 \times \mathbb{Z}_2 \times \mathbb{Z}_2$.

We see quickly that $(0, 1)$ has order 4 in the quotient so it can't be $\mathbb{Z}_2^3$. Furthermore $4(a, b) \equiv (a, b)$ when taken mod 8 so the maximum order of any element is 4 so we conclude that the quotient is (up to isomorphism)

$$\mathbb{Z}_4 \times \mathbb{Z}_2.$$

$\square$

Proof of (3). We particion X into disjoint orbits $\mathcal{O}_1, \ldots, \mathcal{O}_j$. Assume, for contradiction, that $|\mathcal{O}_x| > 1$ for all $x \in X$. Since $|\mathcal{O}_x| = [G : G_x] = |G|/|G_x|$ we have that $|\mathcal{O}_x| = p^{n_x}$ for some $n_x > 0$. Then since

$$|X| = \sum_{i=1}^{s} |O_i|,$$

we have that

$$|X| = \sum_{i=1}^{s} p^{n_i}.$$

Then since p divides every term we have

$$p \mid |X|$$

which contradicts our assumption that p does not divide $|X|$. □

Problem 26.5 (2 Exam 2022, Ring Theory).

1. *Show that $\mathbb{Z}_{12}$ is not an integral domain.*
2. *Let p be prime and $n > 1$. Find a maximal ideal M of $\mathbb{Z}_{p^n}$ and identify the quotient $\mathbb{Z}_{p^n}/M$.*
3. *Show that if an ideal I of the polynomial ring $\mathbb{Q}[x]$ contains both $f(x) = x^3 - x^2 + x$ and $g(x) = x^2 - 1$ then $I = \mathbb{Q}[x]$.*

Solution sketch of (1). $\mathbb{Z}_{12}$ contains zero-divisors $3 \cdot 4 = 12 \equiv_{12} 0$ and is therefore not an integral domain. □

Proof attempt of (2). We need to find an ideal M so that if $M \subseteq K \subseteq \mathbb{Z}_{p^n}$ then $K = M$ or $K = \mathbb{Z}_{p^n}$.

Let $M := \langle p \rangle$. Then M is a maximal ideal.

M absorbs any power of p so $|\mathbb{Z}_{p^n}|/M| < p^n - n$. Etc. □

Proof of (3). We notice that

$$\frac{x^3 - x^2 + x}{x^2 - 1} = x - 1 - \frac{1}{x^2 - 1} =: r(x).$$

Something, something use addition and multiplication with these polynomials to get scalars, x and such to be able to make all polynomials over $\mathbb{Q}$. □

Problem 26.6 (3, Exam 2022 (Finite Fields)). *Consider $F = \mathbb{Z}_3$ and the irreducible polynomial $f(x) = x^3 + 2x + 2 \in F[x]$. Let E be $F[x]/\langle f(x) \rangle$.*

1. *What is the size of E?*
2. *Let $\alpha \in \overline{F}$ denote a zero of $f(x)$ so that $E \cong F(\alpha)$. Let*

$$\sigma_3 : F(\alpha) \to F(\alpha)$$

denote the Frobenius automorphism defined by $\sigma_3(a) = a^3$ for all $a \in F(\alpha)$. Write $\sigma_3(\alpha)$ and $\sigma_3(\alpha)^2$ in the basis $\{1, \alpha, \alpha^2\}$ for $F(\alpha)$ over F.

3. *Assume* $g(x) = x^3 + 2x + 1 \in F[x]$ *is irreducible over* F *and let* $\beta \in \overline{F}$ *be a zero of* $g(x)$.

   Prove that $F(\alpha)$ *and* $F(\beta)$ *are isomorphic without giving an isomorphism. However explain why the map* $\psi_{\alpha,\beta} : F(\alpha) \to F(\beta)$ *which fixes* F *and sends* α *to* β *cannot be made into a field isomorphism.*

Solution of (1). $\deg f = 3$ so $|E| = 3^3 = 27$. □

Solution of (2). $f(\alpha) = \alpha^3 - 2\alpha + 2 = 0$ so $\alpha^3 = 2\alpha - 2 = \alpha + 1$.

Then

$$\sigma_3(\alpha) = \alpha^3 = \alpha + 1.$$

Next

$$\sigma_3^2(\alpha) = (\alpha + 1)^3 = \alpha^3 + 1 = \alpha + 2.$$

□

Proof of (3). Since both are fields of size $27 = 3^3$ they are isomorphic.

The map $\alpha \to \beta$ cannot be a field isomorphism because their respective minimal polynomials do not coincide. □

Problem 26.7. *Let*

$$f(x) = x^4 - 2 \in \mathbb{Q}[x].$$

Find

1. *the roots of* f,
2. *the splitting field* K,
3. *the degree* $[K : \mathbb{Q}]$,
4. *what the automorphism in* $\mathrm{Gal}(K/\mathbb{Q})$ *can do to* $\sqrt[4]{2}$ *and* i.

Solution.

$$\begin{aligned} f(x) &= (x^2 - \sqrt{2})(x^2 + \sqrt{2}) \\ &= (x - \sqrt[4]{2})(x + \sqrt[4]{2})(x^2 + \sqrt{2}) \end{aligned}$$

so we apply the quadratic formula to the latter term:

$$\begin{aligned} x &= \frac{\pm\sqrt{-4\sqrt{2}}}{2} \\ &= \pm\sqrt{-\sqrt{2}} \\ &= \pm i\sqrt[4]{2} \end{aligned}$$

yielding
$$f(x) = (x + \sqrt[4]{2})(x - \sqrt[4]{2})(x + i\sqrt[4]{2})(x - i\sqrt[4]{2}).$$
which has roots
$$\pm\sqrt[4]{2} \text{ and } \pm i\sqrt[4]{2}.$$
Thus the splitting field is
$$K = \mathbb{Q}(\sqrt[4]{2}, i).$$

As we can see f is irreducible over $\mathbb{Q}$ by Eisenstein with $p = 2$ and in fact it is the minimal polynomial of $\sqrt[4]{2}$ so
$$[\mathbb{Q}(\sqrt[4]{2}) : \mathbb{Q}] = \deg f = 4.$$
We can also see that $i \notin \mathbb{R} \supset \mathbb{Q}(\sqrt[4]{2})$ so K has nontrivial degree over $\mathbb{Q}(\sqrt[4]{2})$ and since the minimal polynomial is $x^2 + 1$ we conclude that
$$[K : \mathbb{Q}(\sqrt[4]{2})] = 2$$
so
$$[K : \mathbb{Q}] = 8.$$

Let $\alpha = \sqrt[4]{2}$. Then $K = \mathbb{Q}(\alpha, i)$ and every automorphism in the Galois group is determined by where it sends α and i. For the roots to satisfy their respective polynomials we have
$$\sigma(\alpha) \in \{\alpha, -\alpha, i\alpha, -i\alpha\}$$
and
$$\sigma(i) \in \{i, -i\}.$$
□

Problem 26.8. *Let*
$$R = \mathbb{F}_3[x]/\langle x^2 + x + 2\rangle.$$
Let
$$\alpha = x + \langle x^2 + x + 2\rangle.$$

a) *Show that $x^2 + x + 2$ is irreducible over $\mathbb{F}_3$. Conclude that R is a field.*

b) *How many elements does R have? Write them in the form*
$$a + b\alpha, \quad a, b \in \mathbb{F}_3.$$

c) *Use the quotient relation to compute*
$$\alpha^2, \alpha^3, \alpha^4.$$

d) *Find the inverse of*
$$1 + \alpha$$
in R.

Proof of (a). Let $f(x) := x^2 + x + 2$. Since $\deg f \in \{2, 3\}$ we have that f is irreducible if and only if f has no zeros in the ground field.

We quickly check that

$$f(0) = 2, \ \ f(1) \equiv 1, \ \ f(2) \equiv 2,$$

meaning f has no zeros in $\mathbb{F}_3$ and is thus irreducible.

Since $\mathbb{F}_3$ is a field we have that the quotient is a field if and only if f is irreducible over $\mathbb{F}_3$. Thus we conclude that R is a field. □

Solution to (b). We use the standard result to get

$$|R| = |\mathbb{F}_3|^{\deg f} = 9.$$

Conclusively

$$R = \{0, 1, 2, \alpha, 1 + \alpha, 2 + \alpha, 2\alpha, 2\alpha + 1, 2\alpha + 2\}.$$

□

Solution to (c). We use

$$\alpha^2 + \alpha + 2 = 0$$

so

$$\alpha^2 = 2\alpha + 1$$

and then

$$\alpha^3 = \alpha(2\alpha + 1) = 2\alpha^2 + \alpha = 2(2\alpha + 1) + \alpha = 2\alpha + 2$$

and lastly

$$\alpha^4 = \alpha(2\alpha + 2) = 2\alpha^2 + 2\alpha = 2(2\alpha + 1) + 2\alpha = 4\alpha + 2 + 2\alpha = 2.$$

□

Solution to (c). We must find $a + b\alpha$ such that

$$(1 + \alpha)(a + b\alpha) = 1.$$

Expanding we get

$$\begin{aligned} a + b\alpha + a\alpha + b\alpha^2 &= 1 \\ a + a\alpha + b\alpha + 2b\alpha + b &= 1 \\ a + 3b &= a \\ &\vdots \end{aligned}$$

We get $(1 + \alpha)^{-1} = \alpha$. □

Problem 26.9. *Let*
$$I = \langle x^2 + 1 \rangle \subseteq \mathbb{Q}[x].$$

a) Show that I is maximal.

b) Show that
$$\mathbb{Q}[x]/I \cong \mathbb{Q}(i).$$

c) Find the inverse of
$$x + 1 + I$$
in the quotient ring.

Proof of (a). $\mathbb{Q}[x]$ is a PID so every ideal is generated by a polynomial. Since $\mathbb{Q}$ is a field, $\langle f(x) \rangle$ is maximal if and only if f is irreducible over $\mathbb{Q}$, which $x^2 + 1$ obviously is. □

Proof of (b). We will use the isomorphism theorem for rings.

Let $\psi : \mathbb{Q}[x] \to \mathbb{Q}(i)$ be
$$\psi(p) := p(i).$$
Then very clearly $\langle x^2 + 1 \rangle \subseteq \ker \psi$. Now take any $p(x) \in \ker \psi$. Then
$$p(x) = q(x)(x^2 + 1) + r(x),$$
where
$$\deg r < 2.$$
So $r(x) = ax + b$ for some $a, b \in \mathbb{Q}$. Evaluate at i to get
$$0 = p(i) = q(i)(i^2 + 1) + r(i) = ai + b,$$
forcing $a = b = 0$. Thus
$$p(x) = q(x)(x^2 + 1) \in \langle x^2 + 1 \rangle.$$
Hence $\ker \psi = \langle x^2 + 1 \rangle$.

Lastly the map is surjective, i.e. $\operatorname{im} \psi = \mathbb{Q}(i)$, because take any $\gamma \in \mathbb{Q}(i)$. Then
$$\gamma = a + bi, \ \ a, b \in \mathbb{Q}.$$
Notice then that we can find the polynomial
$$f(x) := a + bx \in \mathbb{Q}[x],$$
for which we have
$$\psi(f) = a + bi.$$
Thus by the first isomorphism theorem for rings,
$$\frac{\mathbb{Q}[x]}{\langle x^2 + 1 \rangle} \cong \mathbb{Q}(i).$$
□

Problem 26.10 (Problem 3 Exam 2021).

a) *Determine if $\mathbb{Z}_2[x]/\langle x^4+1\rangle$ is an integral domain.*

b) *Show that $f(x) = x^4+x^3+1$ is irreducible in $\mathbb{Z}_2[x]$. Explain why $f(x)$ admits a zero θ in the field $\mathbb{Z}_{16}$, seen as an extension of $\mathbb{Z}_2$, with degree $[\mathbb{Z}_{16} : \mathbb{Z}_2] = 4$. Prove that $f(x)$ is the primitive polynomial in $\mathbb{Z}_2[x]$.*

Solution of (a). We notice that, in $\mathbb{Z}_2[x]$,

$$x^4+1 = x^4-1 = (x^2+1)(x^2-1)$$

and so is the product of two lower-degree (yet non-zero) polynomials in $\mathbb{Z}_2[x]$, and is thus not irreducible. From this we can very quickly see that $\langle x^4+1\rangle$ is not a prime ideal, meaning $\mathbb{Z}_2[x]/\langle x^4+1\rangle$ is not an integral domain. □

Proof of (b). We see that f has no zeros in $\mathbb{Z}_2$ and that any factorization into two degree-2 polynomials

$$f(x) = (x^2+ax+b)(x^2+cx+d)$$

requires $b = d = 1$, but then $1 = a+c = 0$. Therefore f is irreducible over the ground field. □

Problem 26.11. *Let*

$$f(x) = x^3+x+1 \in \mathbb{F}_2[x],$$

and let

$$K = \mathbb{F}_2[x]/\langle f\rangle.$$

Let

$$\theta = x + \langle f\rangle \in K.$$

a) *Show that f is irreducible over $\mathbb{F}_2$. Conclude K is a field.*

b) *Use*

$$f(\theta) = 0$$

to express θ^3 in terms of $1, \theta, \theta^2$.

c) *Determine whether θ is a generator of $K^\times$.*

d) *Write the factorization of f into linear factors over K using Frobenius conjugates.*

Proof of (a). Since f is degree-3 we have irreducible if and only if no zeros. We notice that

$$f(0) = f(1) = 1.$$

Thus f is irreducible over $\mathbb{F}_2$.

Since $\mathbb{F}_2$ is a field we have that the quotient

$$\mathbb{F}_2/\langle f\rangle$$

is a field if and only if f is irreducible over $\mathbb{F}_2$. Thus K is a field with

$$|K| = 2^3 = 8.$$

$\square$

Solution of (b). We have

$$\theta^3 + \theta + 1 = 0$$

meaning

$$\theta^3 = -\theta - 1 = \theta + 1.$$

$\square$

Solution of (c). $\langle f\rangle$ is maximal so

$$\frac{\mathbb{F}_2}{\langle f\rangle}$$

can be viewed as a field-extension of $\mathbb{F}_2$ of degree $\deg f = 3$, with the basis $\{1, \theta, \theta^2\}$. We know

$$|K| = 8$$

so

$$K \cong \mathbb{F}_8.$$

Now the question is whether

$$\langle\theta\rangle \cong K^\times \cong \mathbb{F}_7.$$

We compute the lowest j such that

$$\theta^j = 1.$$

We see that

$$\begin{aligned}
\theta^7 &= \theta^2\theta^2\theta^3 \\
&= \theta^4(\theta + 1) \\
&= \theta^5 + \theta^4 \\
&= \theta^2(\theta + 1) + \theta^4 \\
&= \theta^3 + \theta + 2\theta^2 \\
&= \theta^3 + \theta \\
&= -1 \\
&= 1.
\end{aligned}$$

Doing further computations for θ^j for $j < 7$ we see that this is indeed the order of θ. Thus

$$|\langle\theta\rangle| = 7 = |\mathbb{F}_7|$$

and we conclude that θ does indeed generate the multiplicative group.

□

Problem 26.12. *Let*

$$f(x) = x^4 + x + 1 \in \mathbb{F}_2[x],$$

and define

$$K = \mathbb{F}_2[x]/\langle f\rangle.$$

Let

$$\alpha = x + \langle f(x)\rangle \in K.$$

a) Show that $f(x)$ is irreducible over $\mathbb{F}_2$.

b) Conclude that K is a field. How many elements does K have? Give a basis for K as a vector space over $\mathbb{F}_2$.

c) Use $f(\alpha) = 0$ to compute

$$\alpha^4,\ \alpha^5,\ \alpha^6,\ \alpha^7.$$

Write each in the basis from $b)$.

d) Determine whether α generates $K^\times$.

Proof of (a). Suppose f is reducible. Then it factors into two degree-2 polynomials

$$f(x) = x^4 + x + 1 = (x^2 + ax + b)(x^2 + cx + d).$$

Then we require $bd = 1$ so $b = d = 1$, and

$$adx + bcx = (ad + bc)x = x,$$

so

$$ad + bc = 1$$

and since $b = d = 1$ we then have

$$a + c = 1.$$

Then either $a = 1$ or $c = 1$ while the other is 0. Suppose wlog $a = 1$ and $c = 0$. Then

$$x^4+x+1 = (x^2+x+1)(x^2+1) = x^4+x^2+x^3+x+x^2+1 = x^4+x^3+x+1$$

which is false. Our assumption was incorrect and f is irreducible. □

Solution of (b). As f is irreducible we conclude $\langle f \rangle$ is a maximal ideal of $\mathbb{F}_2[x]$. Since $\mathbb{F}_2$ is a field (and therefore a commutative ring), we have that

$$\frac{\mathbb{F}_2[x]}{\langle f \rangle}$$

is a field.

Since $\mathbb{F}_2$ is a finite field, and any non-trivial quotient of the polynomial ring over a finite field by an irreducible polynomial is a finite field extension, we conclude

$$\frac{\mathbb{F}_2[x]}{\langle f \rangle}$$

must be a (finite) field extension.

The standard result is that

$$\left| \frac{\mathbb{F}_2[x]}{\langle f \rangle} \right| = |\mathbb{F}_2|^{\deg f} = 16.$$

The standard basis is then

$$1, \alpha, \alpha^2, \alpha^3$$

since

$$\left[\frac{\mathbb{F}_2[x]}{\langle f \rangle} : \mathbb{F}_2 \right] = 4.$$

□

Solution of (c). We have that

$$\alpha^4 + \alpha + 1 = 0,$$

so

$$\alpha^4 = \alpha + 1.$$

Then

$$\alpha^5 = \alpha(\alpha + 1) = \alpha^2 + \alpha,$$

and

$$\alpha^6 = \alpha(\alpha^2 + \alpha) = \alpha^3 + \alpha^2.$$

Lastly

$$\alpha^7 = \alpha^4 + \alpha^3 = \alpha^3 + \alpha + 1.$$

□

Proof of (d). We have that

$$K := \frac{\mathbb{F}_2[x]}{\langle f \rangle} \cong \mathbb{F}_{16},$$

since the finite fields are classified up to isomorphism. Then the question is whether

$$\langle \alpha \rangle = K^\times \cong \mathbb{F}_{16}^\times \cong C_{15} \leq \mathbb{F}_{16}.$$

Then if we find that $|\langle\alpha\rangle| = 15$ we will be finished. First we can check whether $\alpha^{15} = 1$ as we would expect.

$$\begin{aligned} \alpha^{15} &= \alpha^8(\alpha^3 + \alpha + 1) \\ &= \alpha^{11} + \alpha^9 + \alpha^8 \\ &= (\alpha^5\alpha^6) + \alpha^3\alpha^3 + (\alpha + 1)^2 \\ &= (\alpha^2 + \alpha)(\alpha^3 + \alpha^2) + \alpha^3\alpha^3 + \alpha^2 + 1 \\ &= \alpha^5 + \alpha^4 + \cdots \end{aligned}$$

□

Problem 26.13. *Let $G = C_4 = \langle r\rangle$ act on the set*

$$X = \{0, 1\}^4$$

of binary strings of length 4 by cyclic rotation. Thus

$$r(a_1, a_2, a_3, a_4) = (a_4, a_1, a_2, a_3).$$

In other words, two binary strings are considered equivalent if one can be obtained from the other by rotating the positions.

a) *Write down the elements of G and describe how each acts on X.*

b) *For each $g \in G$, compute the number of fixed points*
$$|X^g| = \{x \in X : gx = x\}.$$

c) *Use Burnside's formula to determine the number of orbits.*

d) *Interpret the answer: how many binary of necklaces of length 4 are there, up to rotation?*

Solution and proof to all. $G = \{e, r, r^2, r^3\}$. Suppose $(a_1, a_2, a_3, a_4) \in X$. Then
$$r(a_1, a_2, a_3, a_4) = (a_4, a_1, a_2, a_3)$$
and
$$r^2(a_1, a_2, a_3, a_4) = r(a_4, a_1, a_2, a_3) = (a_3, a_4, a_1, a_2)$$
and
$$r^3(a_1, a_2, a_3, a_4) = (a_2, a_3, a_4, a_1)$$
with e acting on X by doing nothing.

$X^e = X$ as always.

$X^r = \{(a, a, a, a) : a \in \{0, 1\}\} = \{(0, 0, 0, 0), (1, 1, 1, 1)\}$. This is quickly seen since we require $a_1 = a_4$ and $a_2 = a_1$, but then $a_2 = a_4$ and since

$a_3 = a_2$ they must all be equal. By a similar argument we get the same result for X^{r^3}.

However, X^{r^2} is different since it just swaps entries 1 and 2 with 3 and 4, respectively.

Thus

$$X^{r^2} = \{(a,b,a,b) : a,b \in \{0,1\}\}$$

which means $|X^{r^2}| = 2^2 = 4$.

The number of orbits, by Burnside's lemma/formula is given by

$$|X/G| = \frac{1}{|G|}\sum_{g\in G}|X^g|.$$

Meaning we get

$$|X/G| = \frac{1}{4}(\underbrace{|X^e|}_{|X|} + \underbrace{|X^r|}_{2} + \underbrace{|X^{r^2}|}_{4} + \underbrace{|X^{r^3}|}_{2}) = \frac{1}{4}(16+2+4+2) = 24/4 = 6.$$

Thus there are 6 necklaces unique, up to rotation, when you have two types of pearls, which is quick to verify mentally. $\square$

Problem 26.14. *Let $G = D_4$ be the symmetries of the square, acting on the set*

$$X = \{\textit{colorings of the four vertices of a square using red/blue}\}.$$

We will say that two colorings are the same if one can be obtained from the other by rotating or reflecting the square.

a) List the elements of D_4, and describe their action on the vertices.

b) Compute the number of fixed colorings $|X^g|$ for each type of symmetry $g \in D_4$.

c) Use Burnside's formula to compute the number of orbits.

d) Pick the coloring

$$x = (R,R,B,B)$$

around the square. Find the orbit and stabilizer. Verify orbit-stabilizer:

$$|Gx| = \frac{|G|}{|G_x|}.$$

e) Explain in words how the orbit-stabilizer theorem and Burnside's formula are related, but different.

Proof and solution of all. D_4 is the set of rigid symmetries of the square. We can rotate the square r and reflect it s. Rotating 4

times is the same as doing something so $|r| = 4$. Then also $s^2 = e$ and $srs = r^{-1}$. Then

$$D_4 = \{r, s \mid r^4 = e, \;\; s^2 = e, \;\; srs = r^{-1}\} = \{e, r, r^2, r^3, s, rs, r^2s, r^3s\}.$$

Rotating a square $(1, 2, 3, 4)$ gives

$$r(1, 2, 3, 4) = (4, 1, 2, 3)$$

and reflecting gives

$$s(1, 2, 3, 4) = (2, 1, 4, 3).$$

So we can think of these actions as the bijections acting on the indices of the vertices $r : \mathbb{Z}_4 \to \mathbb{Z}_4$, $r(i) := i + 1 \mod 4$,

$$s(i) = \begin{cases} i + 1 \mod 4, & \texttt{if odd } i \\ i - 1 \mod 4, & \texttt{if even } i \end{cases}.$$

Then $r^n(i) = i + n \mod 4$ and

$$s^n(i) = \begin{cases} s(i), & \texttt{if } n \texttt{ odd} \\ i, & \texttt{if } n \texttt{ even} \end{cases}.$$

The number of colorings for e is $|X^e| = |X|$.

For $g = r$ we, by the same argument as in the necklace problem,

$$|X^r| = 2.$$

For $g = r^3 = r^{-1}$ we get the same result.

For r^2 we get

$$X^{r^2} = \{(a, b, a, b) : a, b \in \{R, B\}\},$$

so

$$|X^{r^2}| = 4.$$

For $g = s$ we once again get

$$|X^s| = 4.$$

For $g = rs$ we get

$$r \circ s(1, 2, 3, 4) = r(2, 1, 4, 3) = (3, 2, 1, 4)$$

which is fixed only if $1 = 3$. Meaning we have 2 choices for indices 1 and 3 and 2 for index 2 and likewise for index 4. Thus $|X^{rs}| = 2^3 = 8$.

For $g = r^2s$ we get

$$r^2 \circ s(1, 2, 3, 4) = r(3, 2, 1, 4) = (4, 3, 2, 1).$$

Here we have $1 \mapsto 4$ and vice versa, likewise for 2 and 3. Thus we get $|X^{r^2s}| = 4$.

Lastly for $g = r^3 s$ we have

$$r^3 s(1,2,3,4) = r(4,3,2,1) = (1,4,3,2)$$

for which 1 and 3 are fixed and 2 and 4 are swapped. Like with $g = rs$ we get $|X^{r^3 s}| = 8$.

Then by Burnside's lemma, the number of orbits is

$$\#(G/X) = \frac{1}{8}(16+2+2+4+4+8+4+8) = \frac{48}{8} = 6.$$

Let $x = (R,R,B,B)$. The orbit is

$$\{x, (B,R,R,B), (B,B,R,R), (R,B,B,R)\}$$

with $|\mathrm{Orb}_G(x)| = 4$. Then the stabilizer is

$$\{e, s\}.$$

Then we have

$$|G| = 8 = 2 \cdot 4 = |\mathrm{Orb}_G(x)| \cdot |\mathrm{Stab}_G(x)|.$$

Burnside's formula relates the number of orbits to the number of elements fixed by elements in G while the orbit-stabilizer theorem relates elements of G which fix $x \in X$, what elements can be reached by x through the action of the group and states that the product of those will be the size of the group (when finite). The easiest example is a group acting on itself and seeing that e can reach every element in the group ($|G|$ elements), but is only fixed by itself, $|\{e\}| = 1$. $\square$

Problem 26.15. *Let*

$$f(x) = x^3 + x + 1 \in \mathbb{F}_2[x],$$

and let

$$K = \mathbb{F}[x]/\langle f(x) \rangle.$$

Let

$$\alpha = x + \langle f(x) \rangle.$$

a) Show that f is irreducible over $\mathbb{F}_2$. Conclude that K is a field.

b) Use

$$f(\alpha) = 0$$

to compute

$$\alpha^3, \alpha^4, \alpha^5, \alpha^6, \alpha^7.$$

in the basis $\{1, \alpha, \alpha^2\}$.

c) The Frobenius automorphism is

$$\sigma_2 : K \to K, \quad \sigma_2(a) = a^2.$$

Compute

$$\sigma_2(\alpha), \ \ \sigma_2^2(\alpha), \ \ \sigma_2^3(\alpha).$$

d) *Use Frobenius conjugates to factor $f(x)$ into linear factors over K.*

Proof of (a). Since $\deg f \in \{2, 3\}$ we can use the fact that f is irreducible over the ground field if and only if it has no zeros in said field. Since

$$f(0) = 1 = f(1)$$

we have that f is irreducible over $\mathbb{F}_2$.

As $\mathbb{F}_2[x]$ is a field and $\langle f \rangle$ is a maximal ideal by f being reducible we have that

$$\mathbb{F}_2[x]/\langle f \rangle$$

is a field. □

Solution of (b).

$$\begin{aligned} f(\alpha) &= 0 \\ \alpha^3 + \alpha + 1 &= 0 \\ \alpha^3 &= \alpha + 1. \end{aligned}$$

Then

$$\begin{aligned} \alpha^4 &= \alpha^2 + \alpha \\ \alpha^5 &= \alpha^3 + \alpha^2 \\ &= \alpha^2 + \alpha + 1 \\ \alpha^6 &= \alpha(\alpha^2 + \alpha + 1) \\ &= \alpha + 1 + \alpha^2 + \alpha \\ &= \alpha^2 + 1 \\ \alpha^7 &= \alpha + 1 + \alpha \\ &= 1. \end{aligned}$$

□

Solution of (c).

$$\begin{aligned}
\sigma_2(\alpha) &= \alpha^2 \\
\sigma_2^2(\alpha) &= \sigma_2(\alpha^2) \\
&= \alpha^4 \\
&= \alpha^2 + \alpha \\
\sigma_2^3(\alpha) &= \sigma(\alpha^2 + \alpha) \\
&= (\alpha^2 + \alpha)(\alpha^2 + \alpha) \\
&= \alpha^4 + 2\alpha^3 + \alpha^2 \\
&= \alpha^2 + \alpha + \alpha^2 \\
&= \alpha.
\end{aligned}$$

□

Solution of (d). Since f has coefficients in $\mathbb{F}_2$, Frobenius sends roots to roots. Therefore the roots of K are

$$\alpha, \ \alpha^2, \ \alpha^4.$$

Writing this in the basis we get roots

$$\alpha, \ \alpha^2, \ \alpha^2 + \alpha.$$

Thus

$$f(x) = (x - \alpha)(x - \alpha^2)(x - \alpha^4) = (x + \alpha)(x + \alpha^2)(x + \alpha^4).$$

□

Problem 26.16. *Let*

$$f(x) = x^3 - 5 \in \mathbb{Q}[x].$$

Let

$$\alpha = \sqrt[3]{5}$$

be the real cube root of 5*, and let*

$$\zeta = \zeta_3 = e^{2i\pi/3}.$$

a) *Show that* $f(x)$ *is irreducible over the rationals.*

b) *Find all roots of* $f(x)$ *in* $\mathbb{C}$*, and describe the splitting field* K *of* f *over* $\mathbb{Q}$*.*

c) *Compute*

$$[K : \mathbb{Q}].$$

d) *Show that*

$$\mathtt{Gal}(K/\mathbb{Q}) \cong S_3.$$

e) *Describe the two automorphisms of* K *that generate the Galois group.*

Proof of (a). Since f is over $\mathbb{Q}$ with integer coefficients we can use Eistenstein's criterion. Let $p = 5$. Then p divides 5, but not the leading coefficient 1 and $p^2 = 25$ does not divide 5. Thus f is irreducible over $\mathbb{Q}$. □

Solution of (b) and (c). We have the obvious

$$f(\sqrt[3]{5}) = 0.$$

Then we notice that

$$f(\zeta_3 \sqrt[3]{5}) = e^{2\pi} \cdot 5 - 5 = 0.$$

Thus the roots in $\mathbb{C}$ are

$$\sqrt[3]{5} \text{ and } \zeta_3 \sqrt[3]{5}.$$

Thus the splitting field is

$$K = \mathbb{Q}(\sqrt[3]{5}, \zeta_3).$$

Noticing that f is actually the minimal polynomial of $\sqrt[3]{5}$ we have

$$[\mathbb{Q}(\sqrt[3]{5}) : \mathbb{Q}] = 3.$$

Furthermore, generally for an n-th root of unity the minimal polynomial is

$$\prod_{d|n} (x^{n/d})^{\mu}(d)$$

where μ is the Möbius function. So the minimal polynomial of ζ_3 over $\mathbb{Q}(\sqrt[3]{5})$ is

$$x^2 + x + 1$$

meaning, by the tower law,

$$[K : \mathbb{Q}] = 2 \cdot 3 = 6.$$

□

Solution of (d). For the splitting field for a degree n polynomial we have

$$\mathtt{Gal}(K/F) \leq S_n$$

so

$$\mathtt{Gal}(K/\mathbb{Q}) \leq S_3.$$

Since $K/\mathbb{Q}$ is Galois we have

$$|\,\mathtt{Gal}(K/\mathbb{Q})| = [K : \mathbb{Q}] = 6 = |S_3|.$$

The only subgroup of S_3 with $|S_3|$ elements is S_3 itself. Hence

$$\mathtt{Gal}(K/\mathbb{Q}) \cong S_3.$$

□

Solution of (e). Define $\sigma(\alpha) = \zeta\alpha$ and $\sigma(\zeta) = \zeta$. Then

$$\sigma(\alpha) = \zeta\alpha$$

and

$$\sigma(\zeta\alpha) = \zeta^2\alpha$$

and

$$\sigma(\zeta^2\alpha) = \alpha.$$

So σ permutes the three roots, and has order 3.

Define $\tau(\alpha) = \alpha$, and $\tau(\sigma) = \sigma^2$. Then

$$\tau(\alpha) = \alpha,$$

and

$$\tau(\alpha\sigma) = \alpha\sigma^2$$

and

$$\tau(\alpha\sigma^2) = \alpha\sigma.$$

Thus τ is complex conjugation and since α is fixed it has order 2.

If we label $\alpha = 1, \alpha\zeta = 2, \alpha\zeta^2 = 3$ we get

$$\sigma = (1\ \ 2\ \ 3),\ \ \tau = (2\ \ 3)$$

so a 3-cycle and a transposition, which generate S_3. $\square$

Problem 26.17. *Let*

$$f(x) = x^3 + x + 1 \in \mathbb{F}_2[x],$$

and let

$$K = \frac{\mathbb{F}_2[x]}{\langle f(x) \rangle}.$$

Write

$$\alpha = x + \langle f(x) \rangle \in K.$$

a) *Show that $f(x)$ is irreducible over $\mathbb{F}_2$.*

b) *Conclude that K is a field. How many elements does K have? Give a basis for K as a vector space over $\mathbb{F}_2$.*

c) *Use $f(\alpha) = 0$ to compute*

$$\alpha^3,\ \ \alpha^4,\ \ \alpha^5,\ \ \alpha^6,\ \ \alpha^7.$$

Write each in the basis.

d) *Show that α generates $K^\times$.*

e) *Let $\sigma : K \to K$, $\sigma(a) = a^2$. Show that σ is an automorphism of K and compute*

$$\sigma(\alpha),\ \ \sigma^2(\alpha),\ \ \sigma^3(\alpha).$$

f) *Determine the Galois group*

$$\mathrm{Gal}(K/\mathbb{F}_2).$$

Proof of (a). Since $\deg f \in \{2, 3\}$ we have that f is irreducible if and only if it has no zeros in the ground-field $\mathbb{F}_2$. We quickly see that

$$f(0) = 1 = f(1)$$

so f has no zeros in $\mathbb{F}_2$ and is thus irreducible. $\square$

Solution of (b). Since $\mathbb{F}_2[x]$ is a PID we have that each ideal is generated by some f, and every non-zero prime ideal is a maximal ideal. Thus $\langle f \rangle$ is a maximal ideal if and only if f is irreducible, which we have shown. Since $\langle f \rangle$ is maximal we then have that K is a field.

We have $|K| = |\mathbb{F}_2|^{\deg f} = 8$. This is because since, for $g(x) \in \mathbb{F}_2[x]$, we have $g(x) = f(x)q(x) + r(x)$ where $\deg r < \deg f$. Therefore every element in the quotient can be written as a linear combination of

$$1, x, x^2$$

with scalars in $\mathbb{F}_2$. So we have 2 choices for 1, 2 for x and 2 for x^2. Hence $|K| = 2^3 = 8$.

If $\alpha = x + \langle f(x) \rangle$, we have the standard basis $\{1, \alpha, \alpha^2\}$. $\square$

Solution of (c). We have

$$\alpha^3 + \alpha + 1 = 0$$

so rearranging, and recalling that addition and subtraction are interchangable in $\mathbb{F}_2$ we get

$$\alpha^3 = \alpha + 1.$$

Then

$$\alpha^4 = \alpha^2 + \alpha.$$

Next

$$\alpha^5 = \alpha^3 + \alpha^2 = \alpha^2 + \alpha + 1.$$

Next

$$\alpha^6 = \alpha^3 + \alpha^2 + \alpha = \alpha^2 + 1.$$

Last

$$\alpha^7 = \alpha^3 + \alpha = 1.$$

$\square$

Proof of (d). We have that $|\alpha| = 7$ by (c), so $|\langle \alpha \rangle| = 7$. We have that $|K| = 8$ and since the finite fields are characterized up to isomorphism we have

$$K \cong \mathbb{F}_8$$

which means that $K^\times \cong C_7 \leq \mathbb{F}_8$ (as groups). Since α generates a subgroup of $\mathbb{F}_8$ which has 7 elements and the only subgroup of $\mathbb{F}_8$ with 7 elements is C_7 we conclude that

$$\langle \alpha \rangle \cong C_7 \cong K.$$

$\square$

e. We confirm the homomorphism properties first.

$$\begin{aligned}\sigma(a+b) &= (a+b)^2 \\ &= a^2+2ab+b^2 \\ &= a^2+b^2 \\ &= \sigma(a)+\sigma(b)\end{aligned}$$

since $2 \equiv 0$. Then

$$\begin{aligned}\sigma(ab) &= (ab)^2 \\ &= a^2b^2 \\ &= \sigma(a)\sigma(b).\end{aligned}$$

Let $\phi : \mathbb{F}_8 \to \mathbb{F}_8$ defined by $\phi(a) = a^2$. Then suppose $\phi(a) = 0$. Then $a^2 = 0$ so $a = 0$. This means the map is injective since $\ker \phi = \{0\}$. Furthermore we se see that

$$0 \mapsto 0, \;\; 1 \mapsto 1, \;\; 2 \mapsto 4, \ldots$$

and we can confirm for ourselves in this way that the map is indeed bijective. Now let ψ be the isomorphism from K to $\mathbb{F}_8$. Such a map exists since we have established that $K \cong \mathbb{F}_8$. Then $\sigma : K \to K = \psi \circ \phi \circ \psi^{-1}$, i.e. a composition of isomorphisms and therefore an isormorphism on K. Hence σ is an automorphism of K.

Then

$$\begin{aligned}\sigma(\alpha) &= \alpha^2 \\ \sigma^2(\alpha) &= \alpha^4 = \alpha^2+\alpha \\ \sigma^3(\alpha) &= \alpha^8 = \alpha(\alpha^7) = \alpha.\end{aligned}$$

□

Solution of (f). We use the standard result that for $\mathrm{Gal}(\mathbb{F}_{q^n}/\mathbb{F}_q)$, we have that the galois group is generated by the s-th power of the Frobenius automorphism where s is the characteristic of the field. Thus

$$\mathrm{Gal}(K/\mathbb{F}_2) = \langle \sigma \rangle.$$

As we saw we have $|\sigma^2| = 3$ which is actually inline with $[K : \mathbb{F}_2]$ even though the extension is not automatically Galois as it would be if we we're working over $\mathbb{Q}$ for instance.

In particular we have

$$\mathrm{Gal}(K/\mathbb{F}_2) = \mathbb{Z}/3\mathbb{Z}.$$

□

Problem 26.18. *Let*

$$\phi : \mathbb{Z}[x] \to \mathbb{Z}_6$$

be the ring homomorphism defined by

$$\phi(f(x)) = f(2) \mod 6.$$

a) *Show that ϕ is a ring homomorphism.*

b) *Find $\ker \phi$. Try to describe it as an ideal generated by simple elements of $\mathbb{Z}[x]$.*

c) *Identify*

$$\mathbb{Z}[x]/\ker \phi.$$

d) *Is $\ker \phi$ prime?*

e) *Is $\ker \phi$ maximal?*

f) *Replace $\mathbb{Z}_6$ with $\mathbb{Z}_5$. Define*

$$\psi : \mathbb{Z}[x] \to \mathbb{Z}_5$$

by

$$\psi(f(x)) = f(2) \mod 5.$$

Find the kernel, and decide whether it is prime or maximal.

Solution of (b). $\ker \phi = \{f(x) \in \mathbb{Z}[x] : \phi(f(x)) = 0\}$. This set includes $f(x) = 6n \cdot g(x),\ n \in \mathbb{Z},\ g(x) \in \mathbb{Z}[x]$, $(x-2) \cdot g(x),\ g(x) \in \mathbb{Z}[x]$. It is therefore not hard to see that

$$\ker \phi = \langle x-2, 6 \rangle,$$

since $x - 2, 6 \in \ker \phi$ and for some $f(x) \in \ker \phi$ we have that $f(x) = 0$ for $x = 2$ or $f(x) \equiv 0$ modulo 6. □

Solution of (c). By the first isomorphism theorem we have

$$\mathbb{Z}[x]/\ker \phi \cong \operatorname{im} \phi$$

and since ϕ is trivially surjective (for $n \in \mathbb{Z}_6$ consider the plynomial $f(x) = n$), we have that

$$\mathbb{Z}[x]/\ker \phi \cong \mathbb{Z}_6.$$

□

Proof of (d) and (e). Since $\mathbb{Z}[x]$ is a commutative ring we have that $\mathbb{Z}[x]/\ker \phi$ is an integral domain iff $\ker \phi$ is prime and said quotient ring is a field iff $\ker \phi$ is maximal. As we have identified the quotient ring to be isomorphic to $\mathbb{Z}_6$ we can answer no to both of these questions since $2 \cdot 3 = 6 \equiv 0$ which disqualifies the quotient ring from being an integral domain which disqualifies it from being a field.

Thus $\ker \phi$ is neither maximal nor prime. □

Proof of (f). By the first isomorphism theorem and the same arguments as earlier we have

$$\mathbb{Z}[x]/\ker\psi \cong \mathbb{Z}_5$$

and $\mathbb{Z}_5$ is indeed a field (since $\mathbb{Z}_q$ is a field when q is prime) and thus $\ker\psi$ is maximal which implies that it is also prime. □

Problem 26.19. *Let*

$$f(x) = x^3 - 2 \in \mathbb{Q}[x].$$

Let

$$\alpha = \sqrt[3]{2}$$

be the real cube root of 2*, and let*

$$\omega = e^{2\pi/3}$$

be a primitive third root of unity.

a) *Show that* $f(x)$ *is irreducible over* $\mathbb{Q}$*.*

b) *Find all roots of* f *in* $\mathbb{C}$*, and show that the splitting field is*

$$K = \mathbb{Q}(\alpha, \omega).$$

c) *Compute*

$$[\mathbb{Q}(\alpha) : \mathbb{Q}].$$

d) *Conclude that*

$$[K : \mathbb{Q}] = 6.$$

e) *Show that every* $\mathbb{Q}$*-automorphism of* K *is determined by the images of* α *and* ω*.*

f) *Construct two automorphisms* $\sigma, \tau \in \mathrm{Gal}(K/\mathbb{Q})$ *by*

$$\sigma(\alpha) = \omega\alpha, \;\; \sigma(\omega) = \omega,$$

and

$$\tau(\alpha) = \alpha, \;\; \tau(\omega) = \omega^2.$$

Show that these are valid automorphisms.

g) *Compute*

$$\sigma^3, \;\; \tau^2, \tau\sigma\tau^{-1}.$$

Conclude that

$$\mathrm{Gal}(K/\mathbb{Q}) \cong S_3.$$

Proof of (a). f is a polynomial with integer coefficients so we can use Eisenstein's criterion. We see that for $p = 2$ we have

$$p \mid a_0 = -2$$

and the other terms which are 0 while

$$p \nmid a_3 = 1,$$

and

$$4 = p^2 \nmid a_0 = 2.$$

Thus, by Eisenstein's criterion, f is irreducible over $\mathbb{Q}$. $\square$

Proof of (b). Over $\mathbb{C}$ we have

$$f(\alpha) = f(\sqrt[3]{2}) = (\sqrt[3]{2})^3 - 2 = 0,$$

and

$$f(\omega\alpha) = f(\omega\sqrt[3]{2}) = 0.$$

These are the obvious ones. Next up we see that

$$f(\omega^2\alpha) = 0$$

as well. Then, we have

$$K \subseteq \mathbb{Q}(\alpha, \omega),$$

since $\alpha, \omega\alpha, \omega^2\alpha \in \mathbb{Q}(\alpha, \omega)$.

For the other direction we can see that ω, α generate K when adjoining to $\mathbb{Q}$ so $E \subseteq K$. $\square$

Proof of (c). We have that f is the minimal polynomial of $\sqrt[3]{2}$ over $\mathbb{Q}$ so

$$[\mathbb{Q}(\alpha) : \mathbb{Q}] = \deg f = 3.$$

Now since $\omega \in \mathbb{C}$ we have $\omega \notin \mathbb{Q}(\alpha)$ because $\mathbb{Q}(\alpha)$. We can see that this is the case because

$$\mathbb{Q}(\alpha) = \{a + b\sqrt[3]{2} \mid a, b \in \mathbb{Q}\},$$

and $a, b, \sqrt[3]{2} \in \mathbb{R}$. Since $\mathbb{C} \supset \mathbb{R} \supset \mathbb{Q}(\alpha)$ we conclude that the claim is true.

The minimal polynomial of an n-th root of unity is

$$\prod_{d|n} (x^d - 1)^{\mu(n/d)}$$

which, for prime n works out nicely to

$$\sum_{0 \leq k \leq n-1} x^n.$$

In this case the primitive polynomial is

$$x^2 + x + 1$$

and thus

$$[\mathbb{Q}(\alpha, \omega) : \mathbb{Q}(\alpha)] = 2.$$

Therefore, by the tower law, we have

$$[\mathbb{K} : \mathbb{Q}] = 2 \cdot 3 = 6.$$

$\square$

Proof of (d). A $\mathbb{Q}$-automorphism of K is $\psi : K \to K$ such that

$$\psi[\mathbb{Q}] = \mathbb{Q}.$$

Thus, for every nonidentity $\mathbb{Q}$-automorphism on K we have

$$\psi(\alpha) = \beta$$

where α, β are roots of f.

We have every root is sent to some root so the images of ω must be fixed or sent to ω^2. If $\omega \mapsto \alpha$, for instance we would have $\psi(\omega\alpha) = \psi(\alpha^2) \neq 0$. α can be sent to $\alpha\omega$. □

Proof of (f). We will show $\text{Gal}(K/\mathbb{Q}) \cong S_3$ first. First since K is algebraic over $\mathbb{Q}$ and the splitting field of a $\mathbb{Q}[x]$ polynomial and $\mathbb{Q}$ is characteristic 0 we get that K is a normal- (by the former) and separable (by the latter) extension, since every algebraic axtension of a characteristic 0 field is separable.

Thus $K/\mathbb{Q}$ is Galois and therefore

$$[K : \mathbb{Q}] = |\,\text{Gal}(K/\mathbb{Q})|.$$

Since $\text{Gal}(K/\mathbb{Q}) \leq S_3$ (from K being splitting field over $\deg f = 3$ polynomial) we have

$$\text{Gal}(K/\mathbb{Q}) \cong S_3.$$

□

Problem 26.20. *Let*

$$K = \mathbb{Q}(\sqrt{2}, \sqrt{3}).$$

a) Prove carefully that

$$[\mathbb{Q}(\sqrt{2}) : \mathbb{Q}] = 2.$$

Then prove that

$$\sqrt{3} \notin \mathbb{Q}(\sqrt{2}).$$

Conclude that

$$[K : \mathbb{Q}] = 4.$$

b) Show that

$$K$$

is the splitting field over $\mathbb{Q}$ of

$$f(x) = (x^2 - 2)(x^2 - 3).$$

Conclude that

$$K/\mathbb{Q}$$

is Galois.

c) Show that every $\mathbb{Q}$-automorphism of K is determined by the images of $\sqrt{2}$, and $\sqrt{3}$. Show the possible images and conclude that

$$\mathrm{Gal}(K/\mathbb{Q}) \leq 4.$$

Construct all 4 automorphisms and conclude that

$$\mathrm{Gal}(K/\mathbb{Q}) \cong C_2 \times C_2.$$

Proof of (a). We have that, for $f(x) = x^2 - 2 \in \mathbb{Q}[x]$, $\sqrt{2}$ is a 0. This polynomial is monic, and furthermore no lower degree nonzero polynomial can have $\sqrt{2}$ as a zero. If we suppose that f' is such a polynomial, then

$$f'(x) = x - a$$

such that $\sqrt{2} - a = 0$, meaning $\sqrt{2} = a$, but then $f'(x) \notin \mathbb{Q}[x]$. Thus f is the minimal polynomial of $\sqrt{2}$. Then

$$[\mathbb{Q}(\sqrt{2}) : \mathbb{Q}] = \deg f = 2.$$

Suppose $\sqrt{3} \in \mathbb{Q}(\sqrt{2})$. Then, since $\sqrt{3} \notin \mathbb{Q}$, we have $\sqrt{3} \in \mathbb{Q}(\sqrt{2}) \setminus \mathbb{Q}$. Then $\sqrt{3} = a + b\sqrt{2}$ such that $b \neq 0$. Then

$$3 = a^2 + 2ab\sqrt{2} + b^2.$$

If $a \neq 0$ we have

$$3 - a^2 - b^2 = 2ab\sqrt{2},$$

but the left-hand-side is rational and the right-hand side is not, a contradiction. Suppose $a = 0$. Then

$$3 = b^2$$

so $b = \sqrt{3} \notin \mathbb{Q}$. Thus $\sqrt{3} \notin \mathbb{Q}(\sqrt{2})$. By a similar argument as earlier, the minimal polynomial of $\sqrt{3}$ over $\mathbb{Q}(\sqrt{2})$ is $x^2 - 3$. Thus, by the tower law, we have

$$[\mathbb{Q}(\sqrt{2}, \sqrt{3}) : \mathbb{Q}] = 2 \cdot 2 = 4.$$

□

Proof of (b). Let L be the splitting field of f. It is trivial to see that

$$f(\pm\sqrt{2}) = f(\pm\sqrt{3}) = 0$$

are the zeros of f in $\mathbb{C} \supset \overline{\mathbb{Q}}$. Therefore $L \subseteq K$. The basis of K is $\{1, \sqrt{2}, \sqrt{3}, \sqrt{6}\}$, and so the first 3 are contained in L and furthermore

$$\sqrt{6} = \frac{3\sqrt{2}}{\sqrt{3}} \in L.$$

Therefore $K = L$.

$L = K$ is algebraic over $\mathbb{Q}$ as it contains $\mathbb{Q}$ and elements which are the zeros of $f \in \mathbb{Q}[x]$. Furthermore since $L = K$ is the splitting field of a polynomial in $\mathbb{Q}[x]$ it is a normal extension. Lastly since $\mathbb{Q}$ is characteristic 0, we have that all algebraic extensions are separable.

All three properties give us that $L = K$ is Galois as an extension of $\mathbb{Q}$. □

Proof of (c). The $\mathbb{Q}$-autmorphisms of K are those which fix $\mathbb{Q}$ and preserve

$$f(a) = 0$$

iff

$$f(\psi(a)) = 0.$$

We require

$$(x^2 - 2)(\sqrt{2}) = 0$$

so $\sqrt{2} \mapsto \pm 2$ since

$$(x^2 - 2)(a) = 0$$

if $a \in \{\sqrt{2}, -\sqrt{2}\}$. Similarly for $\sqrt{3} \mapsto \pm\sqrt{3}$.

Then we have a maximum of 4 possible isomorphisms from $K \to K$ which respect this.

There are two choices to make with two options each. For each valid $\mathbb{Q}$-automorphism we have send each of $\sqrt{2}$ and $\sqrt{3}$ to themselves or their additive inverse. Thus $\mathrm{Gal}(K/\mathbb{Q}) \cong C_2 \times C_2$. □

Problem 26.21. *Let G be a finite group and let $H \leq G$ be a subgroup of index n. Let*

$$X = G/H$$

be the set of left cosets of H.
Define an action on X by

$$g \cdot (aH) = (ga)H.$$

a) *Prove carefully that this is a group action.*

b) *Let $\phi : G \to S_x$ be the homomorphism induced by this action. Prove that*

$$\ker \phi = \bigcap_{g \in G} gHg^{-1}.$$

This subgroup is often called the core of H in G.

c) *Prove that $\ker \phi$ is the largest normal subgroup of G contained in H.*

d) *Assume now that*

$$|G| = 21$$

and that $H \leq G$ has order 7.

Use the action to prove that $H \trianglelefteq G$.

e) Use Cauchy's theorem to prove that every group of order 21 has a normal subgroup of order 7.

Proof of (a). We must prove the following:

1. if $aH = bH$, then $(ga)H = (gb)H$,
2. $e \cdot (aH) = aH$,
3. $g_1 \cdot (g_2 \cdot H) = (g_1 g_2) \cdot H$.

We begin with the first.

Suppose $aH = bH$. We defined the action to be $g \cdot (aH) = (ga)H$ so

$$\begin{aligned} aH &= bH \\ g(aH) &= g(bH) \\ (ga)H &= (gb)H. \end{aligned}$$

Next the second property:

$$\begin{aligned} e \cdot aH &= (ea)H \\ &= (a)H \\ &= aH. \end{aligned}$$

The last property is also evident from the definition

$$g_1(g_2 H) \underset{\text{def}}{=} (g_1 g_2)H.$$

$\square$

Proof of (b). The homomorphism induced by G acting on X is $\psi : G \to \mathrm{Sym}(X)$, defined by

$$\psi(g)(aH) = g \cdot aH = (ga)H.$$

We want to find

$$\ker \phi = \{g \in G : \psi(g) = e\}.$$

That is, we want to find all $g \in G$ such that $\psi(g)(aH) = aH$ for every $aH \in X$.

We remark that

$$aH = bH \Leftrightarrow b^{-1}a \in H,$$

and we need to find $g \in G$ such that

$$gxH = xH, \ \forall x \in G.$$

Then we are looking for $g \in G$ such that

$$x^{-1}gx \in H$$

for every $x \in G$.

This is equivalent to

$$g \in xHx^{-1}$$

for every $x \in G$.

Summing up all these equivalences we get

$$\ker \phi = \bigcap_{g \in G} gHg^{-1}.$$

□

Proof of (c). The map ψ is a group homomorphism. Then $\ker \psi$ is a normal subgroup of the domain $\ker \psi \trianglelefteq G$.

We have $\ker \psi = \bigcap_{g \in G} gHg^{-1}$ from (b). For $g = e$ we have

$$\bigcap_{g \in G} \subseteq eHe^{-1} = H.$$

Let N be some normal subgroup of G such that $N \subseteq H$. We must show that $N \subseteq \ker \psi$. Since N is invariant under conjugation we have

$$N = gNg^{-1}$$

for every $g \in G$. Since $N \subseteq H$, we have

$$N = gNg^{-1} \subseteq gHg^{-1}, \ \forall g \in G.$$

Thus

$$N \subseteq \bigcap_{g \in G} gHg^{-1} = \ker \psi.$$

□

Proof of (d). From Lagrange's theorem we have

$$|X| = \left|\frac{G}{H}\right| = \frac{|G|}{|H|} = 3.$$

The action induces the homomorphism $\psi : G \to S_3$.

Thus by the first isomorphism theorem we have that

$$\frac{G}{\ker \psi} \cong \operatorname{im} \phi \leq S_3.$$

Thus
$$\left|\frac{G}{\ker\psi}\right| \leq 6.$$
We have $\ker\psi \subseteq H$, so $|\ker\psi| \leq 7$. We must also have that the order of the kernel divides $|G| = 21$. We have $\ker\psi \geq 3$ by the constraint of the quotient group having ≤ 6 elements. Then, $4, 5, 6$ do not divide 21 so we must have that $|\ker\psi| = 7$. By (c), $\ker\psi$ is the largest normal subgroup of G contained entirely in H, but $|\ker\psi| = |H|$, so $H = \ker\psi$. Thus H is normal. □

Proof of (e). Cauchy's theorem tells us that if G is a finite group and p is prime such that p divides $|G|$. Then there exists an element in G with order p. Let g be that element. Then $\langle g\rangle \leq G$. Apply this for $p = 7$ so that we get $|\langle g\rangle| = 7$. Apply (d) to see that the kernel of the homomorphism induced by the group action on the set of cosets of $\langle g\rangle$ to see that $\langle g\rangle$ is normal. □

Problem 26.22. *Let*
$$R = \mathbb{Z} \times \mathbb{Z}$$
with componentwise addition and multiplication.

a) *Find all units in R.*

b) *Find all zero divisors in R.*

c) *Prove that every ideal $I \trianglelefteq R$ has the form*
$$I = m\mathbb{Z} \times n\mathbb{Z}$$
for some integers $m, n \geq 0$.

d) *Let*
$$J = 2\mathbb{Z} \times 3\mathbb{Z}.$$
Identify the quotient
$$\frac{R}{J}$$
up to isomorphism.

e) *Is J prime, maximal?*

f) *Find all pairs (m, n) such that*
$$m\mathbb{Z} \times n\mathbb{Z}$$
is a prime ideal of R.

Solution of (a). We want to find $(a, b) \in R$ for which there exists $(d, c) \in R$ such that
$$(a, b)(d, c) = (1, 1),$$

meaning

$$ad = 1, \text{ and } bc = 1.$$

Then $a^{-1} = d$ and $b^{-1} = c$ in $\mathbb{Z}$. The only units in $\mathbb{Z}$ are 1 and -1 both having themselves as inverses. Thus the units are $(a, b) = (1, 1), (a, b) = (1, -1), (a, b) = (-1, 1), (a, b) = (-1, -1)$ with themselves as the multiplicative inverse $(a, b)^{-1} = (c, d) = (a, b)$. □

Solution of (b). We want to find nonzero (a, b) for which there exists nonzero (c, d) such that

$$(a, b)(c, d) = (0, 0).$$

Therefore $(0, 0)$ is out of the question. We then want to find $a, b, c, d \in \mathbb{Z}$ such that

$$ab = 0 \text{ and } cd = 0,$$

but we forbid $a = b = 0$ or $c = d = 0$ similtaneously. Thus we can choose e.g. $a, d \in \mathbb{Z} \setminus \{0\}$, $b = c = 0$ and vice-versa.

Then the zero-divsors are all pairs

$$(a, b), (c, d) \text{ s.t. } a, d \in \mathbb{Z}\setminus\{0\} \text{ and } b = c = 0 \text{ or } a = d = 0 \text{ and } b, c \in \mathbb{Z}\setminus\{0\}.$$

□

Proof of (c). It is obvious from the problem statement that we are talking about nontrivial proper ideals.

We will first prove, for arbitrary $m, n \in \mathbb{Z}$, that $m\mathbb{Z} \times n\mathbb{Z}$ is an ideal (two-sided, though all ideals of R are two-sided since it is commutative) of R.

$m\mathbb{Z} \times n\mathbb{Z}$ is the direct product of the additive groups $m\mathbb{Z}, n\mathbb{Z}$ and is therefore itself an additive group. It is a subgroup of $\mathbb{Z} \times \mathbb{Z}$.

Suppose $(a, b) \in m\mathbb{Z} \times n\mathbb{Z}$ and that $(c, d) \in R$. Then $(a, b) = (m\alpha, n\beta)$ for some $\alpha, \beta \in \mathbb{Z}$. Then

$$(c, d)(a, b) = (c \cdot m\alpha, d \cdot n\beta) = (m(c\alpha), n(d\beta)) \in m\mathbb{Z} \times n\mathbb{Z},$$

and likewise for right-multiplication. Thus $m\mathbb{Z} \times n\mathbb{Z}$ is an ideal of R for arbitary m, n.

Suppose I is an ideal of R. Then I is an additive subgroup of R and absorbs multiplication from R.

The additive group $(R, +)$ is $(\mathbb{Z} \times \mathbb{Z}, +)$, a (finitely generated) abelian group. Thus every subgroup is a finitely generated abelian group, meaning that for every subgroup $I \leq R$, we have

$$I \cong \prod_{i=1}^{k} \mathbb{Z}_{p_i^{e_i}} \times \prod_{i=1}^{l} \mathbb{Z},$$

by the fundamental theorem of finitely generated abelian groups. From this it is clear that $I \cong m\mathbb{Z} \times n\mathbb{Z}$ or $I \cong m\mathbb{Z} \times \mathbb{Z}$. □

Solution of (d). We recall that $\mathbb{Z}/n\mathbb{Z} \cong \mathbb{Z}_n$ and thus immediately suspect

$$\frac{\mathbb{Z} \times \mathbb{Z}}{2\mathbb{Z} \times 3\mathbb{Z}} \cong \mathbb{Z}_2 \times \mathbb{Z}_3.$$

Define the homomorphism $\phi : R \to \mathbb{Z}_2 \times \mathbb{Z}_3$ by

$$\phi(a, b) = (a \mod 2, b \mod 3).$$

Clearly R is a homomorphism and surjective. Thus $\operatorname{im}\phi = \mathbb{Z}_2 \times \mathbb{Z}_3$. Furthermore $\phi(a, b) = 0$ if and only if $a \cong 0 \mod 2$ and $b \cong 0 \mod 3$, meaning $a = 2x, b = 3y$ so

$$\ker\phi = 2\mathbb{Z} \times 3\mathbb{Z}.$$

By the first isomorphism theorem we have

$$\frac{\mathbb{Z} \times \mathbb{Z}}{2\mathbb{Z} \times 3\mathbb{Z}} = \frac{R}{\ker\phi} \cong \operatorname{im}\phi = \mathbb{Z}_2 \times \mathbb{Z}_3.$$

□

Proof of (e). R is a commutative ring so the quotient ring is a field if and only if the ideal is maximal. Similarly the quotient is an integral domain if and only if the quotient is a prime ideal.

Thus we can see that the quotient is neither since $(1, 0)(0, 1) = (0, 0)$ in $\mathbb{Z}_2 \times \mathbb{Z}_3$. □

Proof of (e). Only $(m, n) \in \{(1, 0), (1, p), (0, 1), (p, 1)\}$ where p are prime give us prime ideals. Then the maximal ideals are a subset of this and clearly they are those $(m, n) \in \{(p, 1), (1, p)\}$. Up to isomorphism these sets are cut in half. □

Problem 26.23. *Let G be a finite abelian group with $|G| = 72 = 2^3 \cdot 3^2$.*

a) *List all abelian groups of order $|G|$ up to ismorphism.*

b) *Which of the groups are cyclic?*

c) *Let*
$$A = (\mathbb{Z}_2)^3 \times (\mathbb{Z}_3)^2.$$
How many elements of order exactly 6 exist?

d) *Let*
$$B = \mathbb{Z}_4 \times \mathbb{Z}_2 \times \mathbb{Z}_9.$$
Find an element of order 36 or prove that no such elements exists.

e) *Suppose G is an abelian group of order 72 and contains an element of order 72. Prove $G \cong \mathbb{Z}_{72}$. Suppose G contains an* *element of order 36. Is G cyclic or can you find a counter-example?*

Solution of (a). We use a special case of the fundamental theorem of finitely generated abelian groups, that being the classification theorem for finite abelian groups. We have

$$G \cong \prod_i \mathbb{Z}_{p_i^{e_i}}$$

such that

$$\prod_i p_i^{e_i} = 72.$$

We must then find the unique ways of doing this. We have $72 = 2^3 \cdot 3^2$. Then we get the following groups

$$\begin{aligned}(\mathbb{Z}_2)^3 \times (\mathbb{Z}_3)^2 \\ \mathbb{Z}_2 \times \mathbb{Z}_2^2 \times (\mathbb{Z}_3)^2 \\ \mathbb{Z}_2^3 \times (\mathbb{Z}_3)^2 \\ \mathbb{Z}_2^3 \times \mathbb{Z}_3^2 \\ \mathbb{Z} \times \mathbb{Z}_2^2 \times \mathbb{Z}_3^2 \\ (\mathbb{Z}_2)^3 \times \mathbb{Z}_3^2.\end{aligned}$$

□

Solution of (b). Each $\mathbb{Z}_n$ is cyclic and the direct product of such groups is cyclic if and only if their orders are relatively prime.

That leaves only

$$\mathbb{Z}_8 \times \mathbb{Z}_9.$$

This group is isomorphic to $\mathbb{Z}_{72}$ which can be shown with the first isomorphism theorem using the surjective $\phi : \mathbb{Z} \to \mathbb{Z}_8 \times \mathbb{Z}_9$,

$$\phi(x) = (x \mod 8, x \mod 9),$$

and identifying that $\ker \phi = \mathbb{Z}_{72}$. □

Proof of (c). We notice that 1 has order 2 in $\mathbb{Z}_2$ and 1 and 2 have order 3 in $\mathbb{Z}_3$. Then in the direct product the order is the least common multiple of the entries. Thus

$$(a, b, c, d, e)$$

has order 6 when at least one of a, b, c is non-zero and at least one of d, e is non-zero. Then we can make 2 choices for a, b, c since one must be non-zero and 1 choice for d, e for the same reason. Thus there are $2^2 \times 3 = 12$ such elements. □

Proof of (d). We have 1 having order n in $\mathbb{Z}_n$. Therefore $(1,1,1)$ has order $\operatorname{lcm}(2,4,9) = 36$ in B. □

Proof of (e). We begin with the second claim. B as defined in (d) has an element of order 36 and is not cyclic.

It is evident that $\mathbb{Z}_{72}$ has an element of order 72, namely 1.

If $|G| = 72$ and there exists $g \in G$ such that $|g| = 72$ then $\langle g \rangle$ is a subgroup of G with order 72, meaning

$$\langle g \rangle \cong C_{72} \cong \mathbb{Z}_{72}.$$

The only subgroup of G with order $|G|$, is G so $\langle g \rangle = G$. Thus

$$G \cong \mathbb{Z}_{72}.$$

□

Problem 26.24. *Let $E/\mathbb{Q}$ be a finite Galois extension such that*

$$\mathtt{Gal}(E/\mathbb{Q}) \cong S_5.$$

Let

$$G = \mathtt{Gal}(E/\mathbb{Q}).$$

1. *Compute $[E : \mathbb{Q}]$.*
2. *Prove that there exists an intermediate field K so that*
$$[K : \mathbb{Q}] = 24.$$
3. *Prove that there exists an intermediate field L such that*
$$[L : \mathbb{Q}] = 10.$$
4. *Does there necessarily exist an intermediate field M such that*
$$[M : \mathbb{Q}] = 7?$$
5. *Let $A_5 \leq S_5$. Let $K = E^{A_5}$, the fixed field of A_5. Compute*
$$[K : \mathbb{Q}]$$
and
$$[E : K].$$
Decide whether $K/\mathbb{Q}$ is Galois.
6. *Let*
$$H = (\langle 1\ \ 2\ \ 3\ \ 4\ \ 5) \rangle \leq S_5.$$
Let $K = E^H$. Compute
$$[K : \mathbb{Q}]$$
and
$$[E : K].$$
Is $K/\mathbb{Q}$ necessarily Galois?

Solution of (a). The extension $E/\mathbb{Q}$ is Galois. Then

$$|\,\mathtt{Gal}(E/\mathbb{Q})| = [E:\mathbb{Q}].$$

Thus

$$[E:\mathbb{Q}] = |S_5| = 120.$$

□

Proof of (b). We have $H := \langle (1\ \ 2\ \ 3\ \ 4\ \ 5)\rangle \leq S_5$ and $|H| = 5$. By the tower law:

$$[K:\mathbb{Q}] = 24$$

□

Proof of (c). We need a subgroup of order 12. Take $H := A_4$.

Let $L = E^{A_4}$. Then

$$[L:\mathbb{Q}] = [S_5 : S_4] = 10.$$

□

Proof of (d). Suppose there exists $M \subset E$ so that $[M:\mathbb{Q}] = 7$. Then $M/\mathbb{Q}$ is Galois and the Galois group H satisfies $H \leq G$. Then

$$120 = |G| = [G:H]\cdot 7,$$

meaning that 7 would divide 120, which it does not. Therefore there does not exist such an intermediary field. □

Problem 26.25. *Let*

$$E = \mathbb{Q}(\sqrt{2}, \sqrt{3}, \sqrt{5}).$$

For $a, b, c \in \{0, 1\}$, every element of E can be written uniquely as a $\mathbb{Q}$-linear combination of $(\sqrt{2})^a, (\sqrt{3})^b, (\sqrt{5})^c$. The basis is

$$\{1, \sqrt{2}, \sqrt{3}, \sqrt{5}, \sqrt{6}, \sqrt{10}, \sqrt{15}, \sqrt{30}\}.$$

Define automorphisms $\sigma_2, \sigma_3, \sigma_5 \in \mathtt{Gal}(E/\mathbb{Q})$ by

$$\begin{aligned} \sigma_2(\sqrt{2}) &= -\sqrt{2}, & \sigma_2(\sqrt{3}) &= \sqrt{3}, & \sigma_2(\sqrt{5}) &= \sqrt{5},\\ \sigma_3(\sqrt{2}) &= \sqrt{2}, & \sigma_3(\sqrt{3}) &= -\sqrt{3}, & \sigma_3(\sqrt{5}) &= \sqrt{5},\\ \sigma_5(\sqrt{2}) &= \sqrt{2}, & \sigma_5(\sqrt{3}) &= \sqrt{3}, & \sigma_5(\sqrt{5}) &= -\sqrt{5}. \end{aligned}$$

a) Show that $[E:\mathbb{Q}] = 8$. Then explain why

$$\mathtt{Gal}(E/\mathbb{Q}) \cong C_2 \times \cong C_2 \times \cong C_2.$$

b) Let

$$H = \langle \sigma_2 \rangle.$$

Find the fixed field

$$E^H.$$

c) Let
$$H = \langle \sigma_2, \sigma_3 \rangle.$$
Find E^H.

d) Let
$$H = \langle \sigma_2 \sigma_3 \rangle.$$
Find E^H.

e) Let
$$H = \langle \sigma_2 \sigma_3, \sigma_5 \rangle.$$
Find E^H. Compute the degrees
$$[E : E^H], [E^H : \mathbb{Q}].$$

f) Let $K = \mathbb{Q}(\sqrt{30})$. Find $\mathtt{Gal}(E/K)$.

g) Let $K = \mathbb{Q}(\sqrt{2} + \sqrt{3})$. Find $\mathtt{Gal}(E/K)$.

Proof of (a). The problem states that every element of E can be uniquely written as a $\mathbb{Q}$-linear combination of the 8-element basis
$$\{1, \sqrt{2}, \sqrt{3}, \sqrt{5}, \sqrt{6}, \sqrt{10}, \sqrt{15}, \sqrt{30}\}.$$
Since the dimension of E as a vector space over $\mathbb{Q}$ is equal to the cardinality of its basis, we immediately obtain $[E : \mathbb{Q}] = 8$.

Because E is the splitting field of the separable polynomial $(x^2 - 2)(x^2 - 3)(x^2 - 5)$ over $\mathbb{Q}$, the extension is Galois, which implies $|\mathtt{Gal}(E/\mathbb{Q})| = [E : \mathbb{Q}] = 8$.

The given automorphisms $\sigma_2, \sigma_3, \sigma_5$ satisfy $\sigma_i^2 = \mathrm{id}$ for each $i \in \{2, 3, 5\}$ because applying any of them twice negates the respective radical twice (e.g., $\sigma_2(\sigma_2(\sqrt{2})) = \sigma_2(-\sqrt{2}) = \sqrt{2}$). Furthermore, these automorphisms commute with each other. Since $\mathtt{Gal}(E/\mathbb{Q})$ is an abelian group of order 8 where every non-identity element has order 2, it must be isomorphic to the elementary abelian group of order 8:
$$\mathtt{Gal}(E/\mathbb{Q}) \cong C_2 \times C_2 \times C_2.$$
□

Solution to (b). The subgroup H is generated by σ_2, which sends $\sqrt{2} \mapsto -\sqrt{2}$ while fixing $\sqrt{3}$ and $\sqrt{5}$. An element $x \in E$ written in the standard basis is fixed by σ_2 if and only if the coefficients of all basis elements containing $\sqrt{2}$ (namely $\sqrt{2}, \sqrt{6}, \sqrt{10}, \sqrt{30}$) are zero. The remaining elements are generated by $\sqrt{3}$ and $\sqrt{5}$. Thus, the fixed field is:
$$E^H = \mathbb{Q}(\sqrt{3}, \sqrt{5}).$$
□

Solution to (c). An element is fixed by H if it is simultaneously fixed by σ_2 and σ_3. Being fixed by σ_2 eliminates coefficients for base components containing $\sqrt{2}$, while being fixed by σ_3 eliminates coefficients for base components containing $\sqrt{3}$. The only remaining basis elements unaffected by both negations are 1 and $\sqrt{5}$. Thus, the fixed field is:

$$E^H = \mathbb{Q}(\sqrt{5}).$$

□

Solution to (d). The generator acts on the primary radicals as follows:

$$\sigma_2\sigma_3(\sqrt{2}) = -\sqrt{2}, \quad \sigma_2\sigma_3(\sqrt{3}) = -\sqrt{3}, \quad \sigma_2\sigma_3(\sqrt{5}) = \sqrt{5}.$$

Evaluating its effect on the product elements of the basis yields:

$$\sigma_2\sigma_3(\sqrt{6}) = (-\sqrt{2})(-\sqrt{3}) = \sqrt{6}.$$

Therefore, the basis elements explicitly fixed by $\sigma_2\sigma_3$ are 1, $\sqrt{5}$, $\sqrt{6}$, and $\sqrt{30}$ (since $\sqrt{30} = \sqrt{5}\sqrt{6}$). Thus, the fixed field is:

$$E^H = \mathbb{Q}(\sqrt{6}, \sqrt{5}).$$

□

Solution to (e). From part (d), we know $\sigma_2\sigma_3$ fixes $\mathbb{Q}(\sqrt{6}, \sqrt{5})$. Now we apply the second generator σ_5, which maps $\sqrt{5} \mapsto -\sqrt{5}$ and fixes $\sqrt{6}$. The only basis components from part (d) fixed by σ_5 are 1 and $\sqrt{6}$. Thus, the fixed field is:

$$E^H = \mathbb{Q}(\sqrt{6}).$$

By the Fundamental Theorem of Galois Theory, the degree of the extension over the fixed field equals the order of the subgroup:

$$[E : E^H] = |H| = 4.$$

By the tower law property of extension degrees ($[E : \mathbb{Q}] = [E : E^H][E^H : \mathbb{Q}]$):

$$[E^H : \mathbb{Q}] = \frac{8}{4} = 2.$$

□

Solution to (f). By Galois correspondence, Gal(E/K) consists of the automorphisms in Gal$(E/\mathbb{Q})$ that fix $K = \mathbb{Q}(\sqrt{30})$ elementwise. We test how the basic automorphisms act on $\sqrt{30}$:

$$\sigma_2(\sqrt{30}) = (-\sqrt{2})\sqrt{3}\sqrt{5} = -\sqrt{30},$$
$$\sigma_3(\sqrt{30}) = \sqrt{2}(-\sqrt{3})\sqrt{5} = -\sqrt{30},$$
$$\sigma_5(\sqrt{30}) = \sqrt{2}\sqrt{3}(-\sqrt{5}) = -\sqrt{30}.$$

Products of pairs of these maps will resolve the double negation: $\sigma_2\sigma_3(\sqrt{30}) = \sqrt{30}$, $\sigma_2\sigma_5(\sqrt{30}) = \sqrt{30}$, and $\sigma_3\sigma_5(\sqrt{30}) = \sqrt{30}$. Since $[E : K] = 4$, the subgroup must have order 4. Thus:

$$\mathtt{Gal}(E/K) = \{\mathtt{id}, \sigma_2\sigma_3, \sigma_2\sigma_5, \sigma_3\sigma_5\}.$$

□

Solution to (g). We find which non-identity automorphisms fix $\sqrt{2}+\sqrt{3}$:

$$\sigma_2(\sqrt{2}+\sqrt{3}) = -\sqrt{2}+\sqrt{3} \neq \sqrt{2}+\sqrt{3},$$
$$\sigma_3(\sqrt{2}+\sqrt{3}) = \sqrt{2}-\sqrt{3} \neq \sqrt{2}+\sqrt{3},$$
$$\sigma_5(\sqrt{2}+\sqrt{3}) = \sqrt{2}+\sqrt{3}.$$

Since σ_5 fixes the generator of K, and $[E : K] = 2$ (as $K = \mathbb{Q}(\sqrt{2}, \sqrt{3})$ has degree 4 over $\mathbb{Q}$), the Galois group must have order 2. Therefore:

$$\mathtt{Gal}(E/K) = \langle \sigma_5 \rangle = \{\mathtt{id}, \sigma_5\}.$$

□

Problem 26.26. *Let*

$$K = \mathbb{Q}(\sqrt{2}, \sqrt{5}).$$

1. *Show that* K *is the splitting field of*

$$f(x) = (x^2-2)(x^2-5)$$

over $\mathbb{Q}$*, and prove that*

$$[K : \mathbb{Q}] = 4.$$

2. *Find* $G = \mathtt{Gal}(K/\mathbb{Q})$*. Describe it's elements explicitly by their action on* $\sqrt{2}$ *and* $\sqrt{5}$*, and prove that*

$$G \cong C_2 \times C_2.$$

3. *List the subgroups of* G*.*
4. *For each* $H \leq G$*, find the fixed field* K^H*.*
5. *Explain which intermediate extensions are Galois over* $\mathbb{Q}$*.*

Proof of (a). We begin by finding the roots of $f(x)$ in $\overline{\mathbb{Q}}$.

x^2-2 has zeros $\pm\sqrt{2}$ and x^2-5 has zeros $\pm\sqrt{5}$. These are contained in K. Thus the splitting field L is a subset $L \subseteq K$.

Now, K has $\{1, \sqrt{2}, \sqrt{5}, \sqrt{10}\}$ as the basis as a vector space over $\mathbb{Q}$. The first 3 elements are in the splitting field L and $\sqrt{10} = \sqrt{2}\sqrt{5} \in L$ so since every elements of K is a linear combination of these elements which are indeed contained in the splitting field we have $K \subseteq L$. Thus $K = L$.

We claimed that $B = \{1, \sqrt{2}, \sqrt{5}, \sqrt{10}\}$ is a basis for the extension and then

$$|B| = [K : \mathbb{Q}],$$

but we did this without proof. We proceed with the tower law instead and observe that as $x^2 - 2$ is the minimal polynomial of $\sqrt{2}$ over $\mathbb{Q}$ we have

$$[\mathbb{Q}(\sqrt{2}) : \mathbb{Q}] = \deg(x^2 - 2) = 2.$$

We observe that $\sqrt{5} \notin \mathbb{Q}(\sqrt{2})$, as if this were the case we would have $\sqrt{5} = a + b\sqrt{2}$. If $b = 0$ then we would have $\sqrt{5} \in \mathbb{Q}$ which we do not. Working out the rest of the cases we get contradictions where $\sqrt{2}$ is asserted to be rational which it is not. Then $x^2 - 5$ is the minimal polynomial of $\sqrt{5}$ over $\mathbb{Q}(\sqrt{2})$ meaning

$$[K : \mathbb{Q}(\sqrt{2})] = 2$$

as before and using the tower law we get

$$[K : \mathbb{Q}] = 4.$$

□

Solution and proof (b). $G \leq S_4$ since K is a splitting field of some polynomial of degree 4.

Specifically G is the group of automorphisms on K which fix $\mathbb{Q}$ and preserve

$$\psi(\sqrt{2})^2 - 2 = 0$$

as with the minimal polynomial of $\sqrt{5}$. Thus we get $\psi(\sqrt{2}) = \pm\sqrt{2}$ and $\psi(\sqrt{5}) = \pm\sqrt{5}$ for every $\psi \in G$. Thus, for each $\psi \in G$ we make the choice of whether to flip the sign of each root or not, meaning $G \cong C_2 \times C_2$. □

Solution of (c). The nontrivial subgroups of G are those which one root is fixed, i.e.

$$\{0\} \times C_2, \;\; C_2 \times \{0\}.$$

□

Solution of (d). For each $\psi \in G$ we have

$$\psi[\mathbb{Q}] = \mathbb{Q}.$$

If ψ fixes a basiselement α of K, then ψ will fix $\mathbb{Q}(\alpha)$. Suppose α is a basiselement and $\psi(\alpha) = \alpha$. Then for each $a + b\alpha \in \mathbb{Q}(\alpha)$, we have (since ψ is a homomorphism),

$$\psi(a + b\alpha) = \psi(a) + \psi(b)\psi(\alpha) = a + b\alpha.$$

Thus ψ fixes $\mathbb{Q}(\alpha)$.

Then the fixed fields are

$$K, \ \mathbb{Q}(\sqrt{2}), \ \mathbb{Q}(\sqrt{5}), \ \mathbb{Q}(\sqrt{10}), \ \mathbb{Q}.$$

□

Solution of (e). $G \cong C_2 \times C_2$ is abelian, hence every subgroup is normal. Thus every intermediate extension is Galois. □

Problem 26.27. *Work over $\mathbb{F}_2$. Let*

$$f(x) = x^3 + x + 1, g(x) = x^3 + x^2 + 1$$

in $\mathbb{F}_2[x]$.
Let

$$K = \mathbb{F}_2[x]/\langle f\rangle, \ \ L = \mathbb{F}_2[x]/\langle g\rangle.$$

Write

$$\alpha = x + \langle f\rangle \in K, \ \ \beta = x + \langle g\rangle \in L.$$

a) *Show that f and g are irreducible over $\mathbb{F}_2$. Conclude that they are fields.*

b) *Show that $K \cong L$ as fields.*

c) *Does the assignment $\alpha \mapsto \beta$ necessarily define a $\mathbb{F}_2$-field homomorphism*

$$K \to L?$$

Decide in this specific case. That is, check whether there exists an $\mathbb{F}_2$-field homomorphism $\phi : K \to L$ such that

$$\phi(\alpha) = \beta.$$

d) *Find an explicit element of L to which α can be sent. In other words, find $\gamma \in L$ such that*

$$f(\gamma) = 0.$$

Then explain why $\alpha \mapsto \gamma$ does define a $\mathbb{F}_2$-field homomorphism $K \to L$.

e) *Let*

$$\sigma : K \to K, \ \ \sigma(a) = a^2.$$

Compute

$$\sigma(\alpha), \ \ \sigma^2(\alpha), \ \ \sigma^3(\alpha).$$

Then determine Gal$(K/\mathbb{F}_2)$.

Proof of (a). Both f, g have degree $3 \in \{2, 3\}$ and are thus irreducible over $\mathbb{F}_2$ if and only if they have no zeros in $\mathbb{F}_2$.

We quickly check that

$$f(0) = f(1) = g(0) = g(1) = 1.$$

Since $\mathbb{F}_2$ is a field, the polynomial ring (in one variable) is a PID and thus a commutative ring. Therefore the quotient ring is a field if and only if the ideal we quotient by is maximal. As $\mathbb{F}_2[x]$ is a PID we have that the ideals are maximal if and only if they are prime. Thus irreducibility implies the ideal is maximal which in turn implies the quotient ring is a field.

Therefore we conclude that $\mathbb{F}_2[x]/\langle f\rangle$ and $\mathbb{F}_2[x]/\langle g\rangle$ are fields. □

Proof of (b). We have that f, g are irreducible and there is a standard result which says that in this case

$$\left|\frac{\mathbb{F}_2[x]}{\langle f\rangle}\right| = |\mathbb{F}_2|^{\deg f} = |\mathbb{F}_2|^{\deg g} = \left|\frac{\mathbb{F}_2[x]}{\langle g\rangle}\right|$$

and from here we can apply the classification of finite fields to conclude that both quotient rings are then isomorphic.

For the $|\mathbb{F}_2|^{\deg f}$ part we can elucidate the claim as follows. We will do this for the quotient with $\langle f\rangle$, and the approach is the same for g.

Suppose $l(x) \in \mathbb{F}_2[x]$. Then $l(x) = f(x)q(x) + r(x)$ where $0 \leq \deg r(x) < \deg f$. Therefore each element in the quotient can be written as a linear combination of $1, x, x^2$ with scalars in $\mathbb{F}_2$, since the $f(x)q(x)$ term vanishes. Thus there are 2^3 choices. □

Proof of (c). Suppose $\phi : K \to L$ is a map such that $\phi(\alpha) = \beta$. The question is then if

$$f(\phi(\alpha)) = 0.$$

We have

$$f(\beta) = \beta^3 + \beta + 1,$$

and

$$\beta^3 = \beta^2 + 1$$

so the question is whether

$$\beta(\beta + 1) = 0.$$

This cannot be since L is a field and this would imply $\beta = 1$ or $\beta = 0$, neither of which can be the case.

Thus $\alpha \mapsto \beta$ does not constitute a $\mathbb{F}_2$-fixing homomorphism from $K \to L$. □

Solution of (d). We need $\gamma \in L$ such that $f(\gamma) = 0$.

Let $\gamma := \beta^3$. Then

$$f(\gamma) = f(\beta^3) = \beta^2 + (\beta^2 + 1) + 1 = 0.$$

Define $\Phi : K \to L$ by

$$\Phi(h(\alpha)) = h(\gamma)$$

for $h \in \mathbb{F}_2[x]$. $\square$

Solution of (e). We have

$$\sigma(\alpha) = \alpha^2.$$

Then

$$\sigma^2(\alpha) = \sigma(\alpha^2) = \alpha^4 = \alpha^2 + \alpha.$$

Lastly

$$\sigma^3(\alpha) = \sigma(\alpha^2 + \alpha) = \alpha^4 + \alpha^2 = \alpha.$$

We always have that $\mathbb{F}_{q^n}/\mathbb{F}_q$ is Galois with $[\mathbb{F}_{q^n} : \mathbb{F}_q] = n$, and furthermore the Galois group is always the cyclic group of order n generated by the Frobenious automorphism $a \mapsto a^q$. Thus

$$\mathrm{Gal}(K/\mathbb{F}_2) = C_3 = \langle \sigma \rangle.$$

$\square$

Problem 26.28. *Let*

$$G = \mathbb{Z}_{12} \times \mathbb{Z}_8.$$

Let

$$H = \langle (4, 2) \rangle \leq G.$$

We want to determine G/H up to isomorphism.

a) *Find the order of $(4, 2) \in \mathbb{Z}_{12} \times \mathbb{Z}_8$. Conclude $|H|$ and compute*

$$|G/H|.$$

b) *Show that in G/H, we have the relation*

$$4\overline{e_1} + 2\overline{e_2} = 0,$$

where

$$\overline{e_1} = (1, 0) + H, \ \ \overline{e_2} = (0, 1) + H.$$

Record the original relations so G/H is generated by $\overline{e_1}, \overline{e_2}$ with $12\overline{e_1} = 0, 8\overline{e_2} = 0$ and the above relation.

c) *Prove that G/H is not cyclic.*

d) *Conclude that $G/H \cong \mathbb{Z}_4 \times \mathbb{Z}_2$.*

e) *Construct a surjective homomorphism*

$$\phi : \mathbb{Z}_{12} \times \mathbb{Z}_8 \to \mathbb{Z}_4 \times \mathbb{Z}_2$$

such that

$$\ker \phi = H.$$

Use the first isomorphism theorem to conclude the previous result.

Solution of (a). We have that 4 has order 3 in $\mathbb{Z}_{12}$ and 2 has the order 4 in $\mathbb{Z}_8$. Then $|(4,2)| = \text{lcm}(3,4) = 12$. Thus $|H| = 12$.

Then, by Lagrange's theorem we have

$$|G/H| = 12 \cdot 8/12 = 8.$$

□

Proof of (b). Because LaTeX is annoying we will denote $\overline{e_1} =: e_1$ and likewise for $\overline{e_2}$. We have $H = \langle 4,2 \rangle$ so $(4,2)H = H$, equivalently $(4,2) \in H$, equivalently $(4,2)H = 0$ in the quotient. Then, since $(4,2)H = 4e_1 + 2e_2$ we see that the relation holds. □

Proof of (c). If G/H was cyclic it must contain some element of order $|G/H| = 8$. Suppose $(a,b)H \in G/H$. Then $(a,b) = ae_1 + be_2$. Then it is evident that (a,b) cannot have order 8 since we already have the relation given in (b). □

Proof of (d). By the classification theorem for finite abelian groups we have that the groups of order $|G/H| = 8$, up to isomorphism are

$$\prod_i \mathbb{Z}_{p_i^{e_i}}$$

such that

$$\prod_i p_i^{e_i} = 8.$$

These are $\mathbb{Z}_2^3,\ \mathbb{Z}_2 \times \mathbb{Z}_4,\ \mathbb{Z}_8$.

We have that G/H has no element of order 8 hence $\mathbb{Z}_8$ is out of the question. Furthermore, $(1,1)$ has order 4 so $\mathbb{Z}_2^3$ is also out of the question. Therefore

$$G \cong \mathbb{Z}_2 \times \mathbb{Z}_4.$$

□

Proof of (e). Let $\phi : \mathbb{Z}_{12} \times \mathbb{Z}_8 \to \mathbb{Z}_4 \times \mathbb{Z}_2$, by

$$\phi(a,b) = (a \mod 4, b \mod 2).$$

Then this map is clearly surjective and

$$\ker \phi = H.$$

Thus, by the first isomorphism theorem we have the desired result. □

Problem 26.29. *Let $G = D_4$ be the symmetry group of the square. It has 8 elements: four rotations and four reflections. We color the four vertices using black and white. Two colorings are considered equivalent if you can obtain one from the other by*

a symmetry of the square.
Let X be the set of all such colorings. Then $|X| = 16$.
The group D_4 acts on X by permuting the vertices.
We want to count the number of different colorings up to the equivalence class created by this action.

a) *State Burnside's formula.*

b) *Label the vertices $1, 2, 3, 4$ cyclically. List the cycle of each symmetry of the square as the permutation across vertices.*

c) *For a permutation g of the vertices prove that the number of colorings fixed by g is $2^{c(g)}$ where $c(g)$ is the number of disjoint cycles in the cycle decomposition of g.*

d) *Use Burnside's formula to compute the number of essentially different colorings of the square.*

e) *Now color the square using m colors instead of 2. Find a formula for the number of essentially different colorings in terms of m.*

Solution of (a). Burnside's formula is

$$\#(X/G) = \frac{1}{|G|}\sum_{g\in G}|X^g|,$$

where

$$X^g = \{x \in X : g \cdot x = x\}.$$

X^g is the set of $x \in X$ such that g fixes x, meaning it is like the opposite of $\text{Stab}_G(x)$. Instead of the $g \in G$ which leave x untouched, it is the set of x which $g \in G$ leave untouched. □

Solution of (b). Denote r_{90} as rotating 90 degrees, r_{180} for 180, etc. Then $r_{90}(1, 2, 3, 4) = (4, 1, 2, 3)$ and $r_{180}(1, 2, 3, 4) = (3, 4, 1, 2)$ and $r_{270}(1, 2, 3, 4) = (2, 3, 4, 1)$.

For reflections we denote $s_\uparrow$ as a reflection along vertical center-line, $s_\rightarrow$ as a reflection along the horizontal, $s_\nearrow$ as one diagonal and $s_\searrow$ as the other. Then $s_\uparrow(1, 2, 3, 4) = (3, 4, 2, 1)$, $s_\rightarrow(1, 2, 3, 4) = (2, 1, 4, 3)$, $s_\nearrow(1, 2, 3, 4) = (1, 4, 3, 2)$ and $s_\searrow(1, 2, 3, 4) = (3, 2, 1, 4)$.

In cycle notation we get $r_{90} = (1\ 2\ 3\ 4)$ and then $r_{180} = r_{90}^2 = (1\ 3)(2\ 4)$ and similarly for 270 we get $r_{270} = r_{90} \circ r_{180} = (1\ 4\ 3\ 2)$. Then reflections are $s_\uparrow = (1\ 4\ 2\ 3)$, $s_\rightarrow = (1\ 2)(3\ 4)$, $s_\nearrow = (2\ 4)$ and $s_\searrow = (1\ 3)$. □

Proof of (c). It is evident that this works for a single 4-cycle as no vertex is fixed hence only the colorings where every vertex are the same color works (2 such colorings exist), but the claim does not hold in general. We have that the reflection along the bottom-left to top-right diagonal is $(2\ 4)$ as it swaps the vertex in the top-left and bottom-right. However there exists much more than two colorings which are fixed by this map. Notice that there are similtaneously 2 choices for vertex 1, 2 for 3, and 2 choices for $2, 4$ which must be the same color. $\square$